

# Emergent Communities in Decentralised Autonomous Organisations



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## **Declaration**

I declare that this thesis is entirely my own work. Where the work of others is referenced, clear citations have been made to credit the original authors, including within figures and diagrams. The contents of this thesis have not been submitted for any other degree or qualification.

Thomas Lloyd  
April 2026

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## Abstract

Decentralised Autonomous Organisations (DAOs) collectively govern billions of dollars in digital assets across blockchain networks. The governance of DAOs is typically analysed at the address level, on the assumption that each address represents a single actor. In practice, an actor may control several addresses, and a single address may act for several actors. Address-level analysis therefore obscures concentrated influence among aligned voting blocs, so decisions over substantial shared capital can rest with far fewer actors than the appearance of decentralised participation suggests.

This thesis re-frames DAO governance through the lens of communities. Three problems motivate the work: the *detection* of internal and external communities from on-chain and off-chain governance data; the *distortion* that external communities introduce when they build layered governance on top of an existing DAO; and the *division* of a DAO when partisan communities consolidate to the point of fragmentation.

We *detect* internal communities through pairwise dissimilarity analysis of voting records, and external communities through a heuristic that identifies governance smart contracts and extracts their voters. We measure *distortion* through a case study of the Curve, Convex, and Votium ecosystem, correlating off-chain incentives with on-chain voting outcomes. We measure *division* through partisan cluster formation in DAOs that have undergone forks.

We identify 72 metagovernance relationships across 61 DAOs. Within the Curve ecosystem, bribes paid at higher tiers correlate with subsequent vote outcomes ( $r = 0.99$ ), demonstrating that layered governance can fundamentally alter the rules of engagement of an underlying DAO. In Nouns DAO, addresses that later participated in a fork increasingly cluster in voting behaviour as the fork approaches, with over 90% aligned in a single cluster in the final proposals before fragmentation.

Our findings demonstrate that community-level analysis can support governance monitoring by identifying concentrations of power, locating instances of metagovernance, and detecting early signs of fragmentation within DAOs. For DAO design, they show that governance mechanisms should be evaluated in terms of the emergent communities they enable, not only the behaviour of individual addresses.

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# List of acronyms

**ABI** Application Binary Interface. The specification of a smart contract's functions and events, used to encode and decode interactions with it.

**AMM** Automated Market Maker. A decentralised exchange model in which liquidity pools and a pricing formula replace order books.

**API** Application Programming Interface. An interface through which programs request data or services from another system.

**DAO** Decentralised Autonomous Organisation. An organisation governed collectively by members, in which the rules and participation are implemented in whole or in part by smart contracts.

**DeFi** Decentralised Finance. Financial services implemented as composable smart contracts on public blockchains.

**DEX** Decentralised Exchange. A smart contract protocol for trading tokens without a centralised intermediary.

**ENS** Ethereum Name Service. A naming system that maps human-readable names to Ethereum addresses.

**EOA** Externally Owned Account. A blockchain account controlled directly by a private key rather than by smart contract code.

**ERC-20** Ethereum Request for Comments 20. The Ethereum standard interface for fungible tokens.

**EVM** Ethereum Virtual Machine. A virtual computer and execution environment for smart contracts, initially used by Ethereum.

**IPFS** InterPlanetary File System. A peer-to-peer protocol for storing and sharing content-addressed data.

- LLM** Large Language Model. A machine learning model trained on text that can parse and generate natural language.
- MDS** Multidimensional Scaling. A statistical technique that projects pairwise dissimilarities between items into a low-dimensional spatial representation.
- MP** Member of Parliament. An elected member of the UK House of Commons.
- NFT** Non-Fungible Token. A token that is unique within its token contract, in contrast to interchangeable fungible tokens.
- RPC** Remote Procedure Call. The interface through which clients query a blockchain node, such as Ethereum’s JSON-RPC interface.
- RQ** Research Question. One of the three overarching questions, RQ1 to RQ3, that this thesis addresses.
- SRQ** Specific Research Question. A scoped form of a research question that defines the boundaries of the corresponding empirical study; SRQ1 to SRQ3 scope RQ1 to RQ3 respectively.
- TVL** Total Value Locked. The total value of assets deposited in a protocol’s smart contracts.

# Glossary

**DAO community** An analytical term describing forms of collective participation in DAO governance, identified through behavioural patterns or governance structures.

**DAO fork** The splitting of a DAO in which a subset of members creates a new instance of the organisation, typically dividing the membership and the treasury.

**DAO metagovernance** The process in which one collective governance entity participates in the governance of another DAO.

**delegation** The transfer of voting power from one address to another without transferring the underlying tokens.

**Ethereum** A public blockchain network that supports smart contracts through the EVM, and the empirical setting for the studies in this thesis.

**external DAO community** A collective governance actor external to a DAO that participates in its governance through a contract account, such as a multi-signature address.

**gauge** A Curve Finance mechanism that measures the liquidity provided to each pool; gauge weights, set by gauge proposal votes, direct the emission of new CRV tokens.

**governance token** A token whose balance confers voting power in a DAO.

**internal DAO community** A group of addresses within a DAO inferred from sustained alignment in their observed voting behaviour.

**multi-signature wallet** A smart contract account that requires a predefined number of authorised signers to approve a transaction before it is executed.

**Nouns DAO** An NFT-based DAO in which Nouns NFTs are auctioned with the proceeds directed to a treasury governed by Nouns NFT holders.

**off-chain** Actions or data that occur outside of the blockchain and are therefore not directly executed or recorded as blockchain transactions.

**on-chain** Actions or data that are executed or recorded directly on a blockchain.

**smart contract** A program deployed to and executed within a blockchain network, controlling the account at its contract address.

**Snapshot** The most popular off-chain governance platform used by DAOs to conduct governance procedures.

**token graph** A directed graph in which each vertex is a token and each edge indicates that the source token is tokenised by the target token.

**tokenisation** The process by which an existing token is deposited into a smart contract and represented by a newly minted token.

**tokenised external DAO community** An external DAO community that accepts deposits of a DAO's governance token, issues a new token in return, and operates its own governance process to determine how the deposited voting power is exercised.

**tokenised metagovernance** A form of metagovernance in which an external governance entity acquires governance power in another DAO through the tokenisation of its governance token.

**tokenising meta-event** A heuristic-defined pair of transfer events within a single transaction indicating that one token is tokenised by another, through either a deposit & mint or a withdraw & burn.

**veToken** A governance token model in which users lock tokens for a fixed period in exchange for voting weight, where longer lockup periods equate to a larger voting weight.



# Chapter 1

## Introduction

### 1.1 Background

A Decentralised Autonomous Organisation, or DAO, is a governance system implemented as a set of smart contracts on a public blockchain. DAOs allow pseudonymous participants to create proposals, vote on them, and execute the outcomes without relying on a central authority. The concept was first realised on the Ethereum blockchain in 2016 [37], and DAOs have since become the predominant organisational structure for blockchain-based protocols. At the time of writing, thousands of DAOs operate across multiple blockchain networks, collectively managing treasuries worth more than twenty billion US dollars [31].

The dominant governance model is token-based voting: each governance token held by an address confers one unit of voting power, and tokens are freely tradable on decentralised exchanges. A typical governance process consists of a proposal phase, a voting period in which token-weighted votes are cast, and an execution phase in which the outcome is enacted, either automatically through smart contract code or manually by a trusted set of signers. Governance may take place entirely on-chain, where every action is a blockchain transaction, or off-chain through protocols such as Snapshot [102], where votes are signed messages verified against on-chain token balances. Regardless of the venue, voting power is ultimately derived from on-chain token holdings.

Two well-documented properties of this model shape DAO governance. First, voting power tends to concentrate in a small number of addresses. Studies of major DAOs consistently find that a handful of addresses control the majority of voting power [48, 121], a pattern analogous to large-shareholder dominance in corporate governance. Second, voter participation is low: most token holders never vote, and those who do participate sporadically [45]. Together, concentration and apathy mean that governance outcomes are routinely determined by a small fraction of eligible participants.

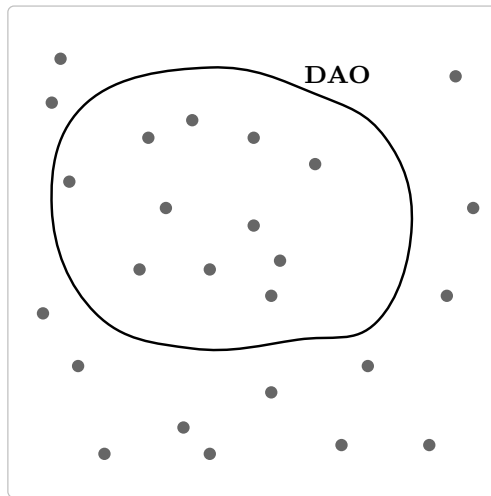
A further property of public blockchains is that smart contracts are composable: any contract can call functions on any other contract within the same network. In decentralised finance, this composability has produced layered ecosystems in which protocols build on top of one another, often described as financial building blocks [99]. The same composability applies to governance contracts. New protocols can acquire governance tokens, lock them in smart contracts, issue derivative tokens, and construct entirely new governance processes on top of an existing DAO. This creates the conditions for the multi-layered governance structures examined in this thesis.

The first DAO deployed on Ethereum, simply named “The DAO”, was exploited through a re-entrancy attack that drained approximately a third of its funds and led to a hard fork of the Ethereum network [37]. In the decade since, DAOs have suffered a range of governance attacks, including flash loan exploits, market manipulation, and vote-buying schemes [45]. These incidents have driven substantial improvements in smart contract security through better coding languages, formal verification tools, and professional security audits. The technical attack surface has narrowed as a result. The social attack surface, however, remains comparatively understudied: the dynamics of how participants organise, coordinate, and exert influence within and across DAOs have received far less attention than the smart contracts through which they act.

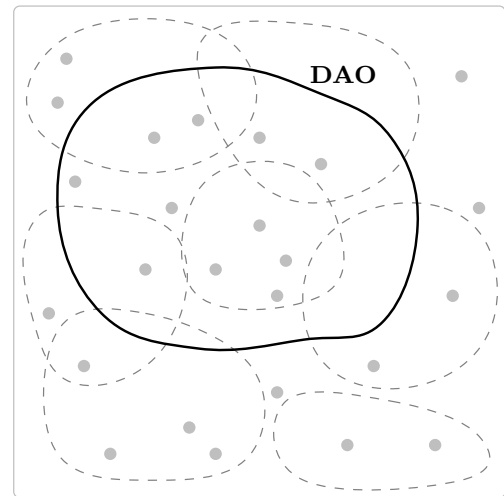
## 1.2 Problem Statement

Research into DAO governance has overwhelmingly adopted the individual address as the unit of analysis. Studies measure token concentration across addresses, compute decentralisation indices such as the Nakamoto coefficient, and track the voting records of individual participants [48, 121]. These analyses have revealed important properties of DAO governance, including the dominance of a small number of high-balance addresses, known as whales, and persistently low voter turnout. However, the individual address is an unreliable proxy for the individual participant. A single actor can distribute tokens across many addresses to create an appearance of decentralisation [11]. Conversely, many independent actors can align in their voting behaviour, forming a coordinated bloc that is invisible when each address is examined in isolation. Governance power does not reside in addresses alone; it resides in the communities that form among them. Figure 1.1 illustrates this distinction: the same set of addresses can be viewed as independent individuals or as members of overlapping communities that span the boundary of the DAO.

We use the term *community* as a broad analytical category for forms of collective participation in DAO governance. Rather than referring to a single type of governance actor, the term covers related forms of collective participation that differ in how they are identified



(a) Addresses treated as independent individuals.



(b) The same addresses viewed as members of overlapping communities.

Fig. 1.1 Two perspectives on DAO governance participants. In (a), each address is analysed in isolation. In (b), community structure reveals groups of addresses that cross the DAO boundary, including external organisations that participate from outside.

and how they participate in governance. For example *Internal communities* are groups of addresses whose observable voting records show sustained behavioural alignment across proposals. They are not formally constituted; rather, they are inferred solely from observed governance behaviour and do not imply deliberate coordination between participants. *External communities* are organisations that participate in a DAO's governance through a smart contract account, typically a multi-signature address controlled by several signers. An external community may be a university research group, a venture fund, or another DAO. From the perspective of the underlying DAO, the external community appears as a single voting address, yet the decision of how to vote is made elsewhere. *Tokenised external communities* extend this pattern further: they accept deposits of the underlying DAO's governance token, issue a new token in return, and operate their own governance process to determine how the deposited voting power is exercised. The result is a layered governance structure in which the rules of engagement at the higher tier may differ from those of the underlying DAO.

Three gaps in the existing literature motivate this thesis. We summarise them here; Chapter 3 reviews the literature in detail.

The first gap concerns *detection*. No systematic method exists to identify communities that participate in DAO governance. Internal communities must be inferred from voting records, a task complicated by pseudonymous addresses, voter churn, and low participation rates. External communities must be distinguished from ordinary contract accounts among a DAO's voters. Prior work has identified voter coalitions in individual DAOs [106] and

measured the concentration of voting power [48], but has not provided a general framework for detecting and characterising the range of community forms that exist in practice.

The second gap concerns *distortion*. DAO governance mechanisms encode specific rules and incentive structures. For example, the veToken model requires voters to lock tokens for up to four years in order to receive voting weight, a design intended to align voter interests with the long-term health of the protocol. Because governance contracts are composable, external communities can build new governance layers on top of an existing DAO without its permission. A tokenised external community can wrap the underlying token, shorten the lockup period, and introduce bribe markets that redirect voting behaviour. The effects of this layered governance on the mechanisms of the underlying DAO are not well understood. A scoping review of metagovernance research found only seven studies that directly address governance interactions between DAOs [124], and the consequences of governance composability for the intended rules of engagement remain largely unexplored.

The third gap concerns *division*. DAO governance analysis typically examines individual proposals or static snapshots of token distributions. It does not track how communities form, consolidate, and fracture over time. When disagreements within a DAO become irreconcilable, the community may split through a governance fork, dividing both the membership and the treasury. While a fork can represent the successful use of decentralised mechanics, forking can also represent fragmentation. In these cases, forking is costly: it undermines the legitimacy of the original organisation and fragments the resources it controls [94]. Yet there is little research on whether the voting behaviour of communities provides observable warning signs before a fragmentation event occurs. If partisan alignment could be detected early, it might be possible to intervene before a fork becomes inevitable by explicitly notifying DAO members of emerging behavioural divisions, and encouraging deliberation within the DAO forums or Discord servers that go beyond binary proposal voting.

## 1.3 Research Questions

The three gaps identified above give rise to three research questions, referred to hereafter as RQ 1, RQ 2 and RQ 3. The research questions are investigated through a series of empirical studies. From each research question, we have derived a *specific research question* (SRQ) that defines the scope of the individual investigation we conducted, while still contributing to answering the overarching research question. Each SRQ is numbered after the research question it scopes and is stated beneath it below: SRQ1 scopes RQ 1, SRQ2 scopes RQ 2, and SRQ3 scopes RQ 3.

**Research Question 1.** *How can communities that participate in DAO governance be detected and characterised from Ethereum governance data?*

This question addresses the detection gap. It asks how internal communities can be inferred from voting records and how external communities, including tokenised metagovernors, can be identified among a DAO's governing addresses.

**SRQ1** How can historical voting behaviour and multi-signature governance mechanisms be used to identify groups participating in Ethereum DAO governance?

**Research Question 2.** *How do communities that participate in DAO governance distort the intended governance mechanisms?*

This question addresses the distortion gap. It asks what happens to a DAO's rules of engagement when external communities build layered governance on top of it, and whether the resulting incentive structures diverge from those the underlying DAO was designed to enforce.

**SRQ2** How does metagovernance distort the intended governance mechanisms of the veToken model within the Curve ecosystem?

**Research Question 3.** *How do community voting patterns provide early-warning signals of partisan division and organisational fragmentation in DAOs?*

This question addresses the division gap. It asks whether the voting behaviour of internal communities contains observable patterns that precede governance failures such as DAO forks.

**SRQ3** How does retrospective analysis of Nouns DAO community voting patterns approaching a documented fork offer insights into organisational fragmentation?

## 1.4 Contributions

The research questions are addressed across four chapters, each based on a published paper. Table 1.1 shows which chapters contribute to each question.

For RQ 1, Chapter 5 develops heuristics to detect governance contracts and extract smart contract voters from both on-chain and off-chain DAOs. Chapter 4 identifies internal partisan communities through pairwise dissimilarity analysis of voting records. Chapter 6 characterises tokenised external communities within the Curve governance ecosystem. Chapter 7 extends this work to the Ethereum blockchain as a whole, presenting a heuristic that detects tokenising meta-events from EVM logs and a token graph that captures composition patterns at network scale.

Table 1.1 Research questions and the chapters that address them. A checkmark (✓) indicates the chapter contributes to the question.

|      | Ch. 4: Internal Communities | Ch. 5: Meta-governance | Ch. 6: The veToken Model | Ch. 7: Token Composition |
|------|-----------------------------|------------------------|--------------------------|--------------------------|
| RQ 1 | ✓                           | ✓                      | ✓                        | ✓                        |
| RQ 2 |                             |                        | ✓                        |                          |
| RQ 3 | ✓                           |                        |                          |                          |

For RQ 2, Chapter 6 shows that tokenised metagovernance can change the rules of engagement established by an underlying DAO, with bribes at higher tiers of governance exhibiting a near-perfect correlation with voting outcomes.

For RQ 3, Chapter 4 shows that addresses which later participate in a DAO fork increasingly cluster together in voting behaviour as the fork approaches, with over 90% aligned in a single cluster in the final proposals before fragmentation.

Chapter 7 contributes to RQ 1 at a different level of abstraction from the other three chapters: it presents a general detection technique rather than an analysis of a specific DAO or ecosystem. It is a co-authored paper that grew out of the work on the veToken model in Chapter 6, generalising the token composition patterns observed in the Curve ecosystem to the Ethereum blockchain as a whole.

## 1.5 Methodology

This thesis adopts an empirical, quantitative approach to the study of DAO governance. Public blockchains record every transaction, token transfer, and governance action in an immutable ledger. This makes blockchain data a naturally occurring dataset of governance activity: it is comprehensive, publicly accessible, and independently verifiable. We treat this data as the primary source of evidence for all four studies in the thesis.

Our data falls into three categories. The first is *on-chain data* from the Ethereum blockchain, including transaction records, smart contract event logs, and internal execution traces. The second is *off-chain governance data* from platforms such as Snapshot, where voting takes place outside the blockchain but voting power is still derived from on-chain token holdings. The third is *supplementary metadata* from services that index, label, or aggregate blockchain data, such as block explorers, token registries, and market data providers. Together, these sources provide empirical coverage of both on-chain and off-chain DAO governance. The specific data sources and collection procedures for each study are described in the chapter where they are used.

Four analytical methods recur across the studies. First, we develop *heuristics* to identify governance structures and community relationships from raw blockchain data, such as matching smart contract event signatures against known governance patterns to detect DAO contracts, or identifying pairs of token transfers within a single transaction to detect tokenisation events. Second, we construct *graphs and networks* to represent relationships: a metagovernance network connecting DAOs through governance participation, and a token graph connecting tokens through composition. Third, we use *clustering and dimensionality reduction* to detect community structure in voting behaviour, projecting pairwise dissimilarity scores into spatial representations and applying cluster analysis. Fourth, we use *statistical analysis* to measure relationships and validate results, including correlation analysis between bribe distributions and voting outcomes, and comparison of genuine voting data against randomised baselines.

Each study applies these methods to specific DAOs or sets of DAOs chosen for properties relevant to the research question under investigation and all studies are conducted on the Ethereum network and therefore share several methodological assumptions. All studies analyse observable blockchain activity associated with pseudonymous Ethereum addresses and smart contracts. While this preserves public immutable governance data, allowing accurate data collection within this thesis, it also obscures the true motive of voting behaviour. For example, behavioural patterns are inferred from observable blockchain activity, such as voting for or against in a binary DAO voting system; however, the motivations underlying an actor's voting decisions cannot be determined from the empirical data collected in this thesis and would require additional qualitative data to be gathered from the actor in control of the address. Moreover, inferring behaviour in a pseudonymous setting requires the use of third-party, supplementary indexers and labellers, where information is gathered, assigned to addresses, and presented outside of the blockchain itself, for example, by block explorers such as Etherscan. These challenges present themselves across all four of the studies described in this thesis. For example, RQ 1 primarily seeks to infer relationships between addresses from voting records alone for internal communities, as the true motivations of pseudonymous addresses are not explicitly stated in the governance records. This same principle applies to the inference of voting division in RQ 3. Moreover, when detecting external communities, we rely on labelling systems from Etherscan, whereas RQ 2 relies on the off-chain governance infrastructure to gather off-chain voting records of addresses.

Where a DAO has documented governance events, such as forks, that provide a ground truth for validation, we use them. The detailed method for each study, including data collection procedures, parameter choices, and validation protocols, is presented in Chapters 4–7.

## 1.6 Thesis-level Argument

In this thesis, we argue that DAO governance can be understood more completely by viewing DAOs through a community lens, rather than individual addresses, as the primary unit of analysis. This perspective reveals governance structures and relationships that are hidden in address-level analyses, enables layered governance mechanisms and the outcomes they produce to be analysed more transparently, and provides insights into voting behaviour and organisational fragmentation. In this way, the three empirical strands of the thesis support a layered, community-level account of detection, distortion and division in DAO governance.

The foundation of the argument in this thesis is built upon communities. Within Chapters 4–7 we detect communities by developing methods to identify and characterise them from observable governance activity. This community-level perspective enables further analysis beyond individual addresses, and provides a basis for quantifying the impact of communities on DAO governance.

Our results show that internal community analysis can provide insights into voting behaviour, including early signs of partisan division (Chapter 4). Moreover, external communities have an impact on voting outcomes through the distortion of DAO mechanics, whether through tokenised metagovernance (Chapter 6) or token wrappings (Chapter 7). These insights are unavailable when DAOs are examined at the level of the single address; community analysis makes the relationships between addresses apparent, offering new insights into DAO dynamics.

## 1.7 Publications

The results of this work have been published in four peer-reviewed conference papers, three as first author and one as a contributing author:

- Thomas Lloyd, Daire Ó Broin, and Martin Harrigan. “Emergent Outcomes of the veToken Model”. In *The IEEE International Conference on Omni-Layer Intelligent Systems (COINS)*. IEEE, 2023.
- Thomas Lloyd, Daire Ó Broin, and Martin Harrigan. “The On-Chain and Off-Chain Mechanisms of DAO-to-DAO Voting”. In *The IEEE International Conference on Blockchain (BLOCKCHAIN)*, pages 649–655. IEEE, 2024.
- Thomas Lloyd, Daire Ó Broin, and Martin Harrigan. “Mapping Partisan Fault Lines Within DAOs”. In *The IEEE International Conference on Blockchain (BLOCKCHAIN)*, pages 38–45. IEEE, 2025.



- Martin Harrigan, Thomas Lloyd, and Daire Ó Broin. “Token Composition: A Graph Based on EVM Logs”. In *The IEEE International Conference on Omni-Layer Intelligent Systems (COINS)*. IEEE, 2024.

The candidate’s contributions and responsibilities for the first three publications in which he is the first author are: conceptualisation, methodology design, programming, validation, formal analysis, investigation, data curation, writing the original draft and editing and reviewing. The candidate held a more limited role in the fourth paper, where he contributed to: conceptualisation, methodology design, validation and writing.

## 1.8 Organisation

The remainder of this thesis is organised as follows.

Chapter 2 introduces the foundational concepts that underpin DAO governance, including Ethereum and its account types, smart contracts and their composition, blockchain data analysis, tokens and token wrapping, decentralised finance, and the mechanics of on-chain and off-chain DAOs. It defines DAO communities, metagovernance, and DAO forks, and concludes with the analytical framework adopted throughout the thesis.

Chapter 3 reviews the relevant literature on smart contract identification and composition, internal DAO governance mechanisms and voting behaviour, external DAO governance, and governance exploits.

Chapter 4 presents a method for detecting internal partisan communities within DAOs by analysing on-chain voting behaviour. Using Nouns DAO as a case study, it shows that addresses which later participate in a fork increasingly cluster together in the proposals before fragmentation.

Chapter 5 develops heuristics to automate the detection of metagovernance relationships between DAOs on the Ethereum network. It identifies 72 external community relationships across 61 DAOs and examines their governance dynamics through three case studies.

Chapter 6 analyses the veToken governance model as implemented by Curve Finance and the ecosystem of higher-level protocols that has emerged on top of it. It shows that bribes at higher tiers of governance exhibit a near-perfect correlation with voting outcomes.

Chapter 7 presents a token graph constructed from EVM logs that represents the composition of tokens on the Ethereum blockchain. It analyses the macro- and micro-topological structure of the graph and visualises the dependencies of composed tokens. The graph provides empirical evidence of the composability that enables the layered governance structures examined in earlier chapters.

Chapter 8 discusses the findings from the preceding chapters in relation to the three research questions, considers their broader implications for DAO governance, and identifies directions for future work.

## Chapter 2

# Foundational Concepts

In this chapter, we describe the foundational concepts that underpin the research in this thesis. We begin with the blockchain infrastructure on which Decentralised Autonomous Organisations operate, then introduce smart contracts, the data analysis techniques used to study them, the token systems that determine voting power, and the decentralised finance primitives that recur in our case studies. We then define DAOs and the community structures that form within and across them, including DAO forks and metagovernance. The chapter concludes with a visual grammar for the governance diagrams used throughout the thesis. Together, these concepts provide the foundation for the empirical chapters that address RQ 1 to RQ 3, and the community structures defined here underpin the thesis-level argument that DAO governance is understood more completely through a community lens than through individual addresses.

### 2.1 Blockchain and Ethereum

The concept of a blockchain was first introduced in 2008 by Nakamoto, the pseudonymous author of the Bitcoin whitepaper [87]. A blockchain is a distributed ledger maintained by a network of nodes that agree on its contents through a consensus protocol. Once a block of transactions is appended to the chain, it is computationally infeasible to alter, giving the ledger its property of immutability.

Public blockchains are decentralised, permissionless networks: any actor with a blockchain address and a sufficient balance of the network's native currency can read from and write to the network. Since Bitcoin's inception, many blockchain networks have emerged, building upon and evolving Nakamoto's original concept.

### 2.1.1 Ethereum and the EVM

In 2013, Buterin published the Ethereum whitepaper, in which he introduced the Ethereum Virtual Machine (EVM), an execution environment where programs can be deployed and executed within the blockchain network [22]. The EVM is Turing-complete in principle, but each operation consumes gas (Section 2.1.2), so execution is bounded by the gas limit of each transaction. These programs are referred to as smart contracts and are described in Section 2.2. The EVM provides a shared execution environment: every smart contract on the network can interact with every other smart contract, permissionlessly. This property, known as composability, is central to the governance structures examined in this thesis.

Throughout Chapters 4, 5, 6, and 7 our analysis takes place within the Ethereum network. We selected this network because Ethereum provides the longest and most mature history of DAO governance within blockchain networks. It has served as the birthplace of the ERC-20 token standard, introduced DAO governance frameworks such as Compound Bravo and innovated upon governance mechanisms such as Curve’s veToken. The Ethereum network also dominates off-chain DAO governance, with 65.4% of the most prevalent DAOs within the off-chain governance protocol Snapshot being on Ethereum [122].

While this thesis is concerned with the Ethereum network, many of the concepts exist in public blockchains that support smart contracts. In the case of other EVM blockchains, such as Arbitrum, the terminology and approaches of our method are directly transferable by updating the chain ID when gathering data through the associated blockchain node. Simply put, the following sections, aside from Section 2.3.1, apply to all EVM-based blockchain networks. However, blockchains that support smart contracts but are not based on the EVM architecture, such as Solana, largely share the same high-level concepts described in the following sections, although the technical details may vary.

### 2.1.2 Transactions and Gas

All state changes on Ethereum are initiated by transactions. A transaction is a cryptographically signed instruction submitted by an account. Executing a transaction consumes computational resources on the network; the sender pays for these resources in Ethereum’s native currency, ether, through a fee mechanism called *gas*. More complex operations, such as deploying or interacting with smart contracts, require more gas than simple transfers of ether.

### 2.1.3 Account Types

The Ethereum network has two types of accounts, both identified by a 20-byte hexadecimal address [128]. An *externally owned account* (EOA) is controlled by a private key held by a person or organisation. EOAs can initiate transactions but contain no code. A *contract account* is controlled by the code of a smart contract deployed at that address. Contract accounts cannot initiate transactions on their own; they execute code in response to transactions or calls from other contracts. The distinction between EOAs and contract accounts is important throughout this thesis: when a contract account participates in DAO governance, it may represent a multi-signature wallet, a yield aggregator, or an entire external organisation, rather than a single individual.

## 2.2 Smart Contracts

The concept of a smart contract was initially described by Szabo before the advent of blockchains in 1996, where he defined smart contracts as contractual clauses embedded in hardware and software designed to make the breach of the contract expensive or prohibitive for the breaching party [108]. In the context of Ethereum, smart contracts are algorithmically enforced agreements deployed to and executed within the EVM [128]. They are typically written in a high-level language such as Solidity or Vyper, compiled into bytecode, and deployed to the network through a transaction.

### 2.2.1 Contract Addresses

Smart contracts are referred to by their unique 20-byte hexadecimal address within the Ethereum blockchain network. The contract address is determined by the deployer's address and the number of transactions the deployer's address has made (the nonce) when calling the CREATE opcode [128]. The opcode CREATE2 replaces the nonce with a user-defined salt and the hash of the smart contract's initialisation code, making the resulting address deterministic and independent of the deployer's transaction count. Blockchain explorers such as Etherscan index deployed smart contracts and attach metadata to contextualise the hexadecimal address. For example, the Uniswap token smart contract exists at address 0x1F9840A85D5AF5BF1D1762F925BDADDC4201F984 within the Ethereum network. When searching for this address on Etherscan, additional data such as the name, ticker symbol, and icon are displayed.

### 2.2.2 Bytecode and Source Code

Smart contract bytecode is the compiled code executed by the EVM. Once a contract has been deployed, the bytecode is stored on the blockchain so that functions within the contract may be invoked through transactions. Bytecode is not human-readable. Third-party services such as the block explorer Etherscan allow developers to verify and publish smart contract source code. Etherscan compiles the submitted source code and matches the result against the existing bytecode on the blockchain; if they match, the source code is published alongside the contract data. At the time of this research, Etherscan had detected over 93 million smart contracts within the Ethereum blockchain, 840 thousand of which had been verified [43].

### 2.2.3 Event Logs

Smart contracts may emit logs within the EVM. Logs are emitted through opcodes LOG 0 through LOG 4 [128]. Logs do not change the storage within a smart contract but instead emit a receipt of the events that have taken place during execution. This allows off-chain analysis tools to retrieve important data directly without processing every transaction for a given contract. For example, the ERC-20 token standard [117] specifies the transfer event `TRANSFER(ADDRESS INDEXED _FROM, ADDRESS INDEXED _TO, UINT256 _VALUE)`, which records the sender, the recipient, and the number of tokens transferred.

Event logs are identified through their event signature, which is derived from the Keccak-256 hash of the event name and its parameter datatypes. For example, the event `TRANSFER(ADDRESS,ADDRESS,UINT256)` produces the signature `DDF252AD...`, which is prefixed with `0X` and stored in `topic0` of the log. Third-party tools such as the 4byte directory index event signatures [1]. Functions within smart contracts are identified by a related mechanism: the function selector is the first four bytes of the Keccak-256 hash of the function name and its parameter datatypes.

### 2.2.4 Smart Contract Composition

Smart contracts can interact with other smart contracts within the same blockchain network by executing an external call. The called contract runs its code and returns a result to the caller. This inter-contract interaction is the basis of *smart contract composition*: new functionality can be built by combining existing contracts rather than reimplementing their logic.

Composition is prevalent on Ethereum. Over a four-year period between 2020 and 2024, over 76% of Ethereum transactions involved at least two inter-contract calls [135]. In decentralised finance, composition is often described using the metaphor of building

blocks or financial legos [7]: multiple protocols can be combined by letting their smart contracts interact with one another. Schar [99] proposes a multi-layered framework to analyse these building blocks and describes composability as one of the most promising features in the DeFi ecosystem. The same composability applies to governance contracts and is the mechanism through which the multi-layered governance structures examined in this thesis are constructed.

## 2.3 Blockchain Data and Analysis

### 2.3.1 Ethereum Nodes

An Ethereum node is a computer connected to the Ethereum network that maintains a copy of the blockchain's history. Nodes can be configured in several ways. A *full node* keeps the entire history of the blockchain. An *archive node* also keeps the entire history but additionally stores the state of the blockchain at every block, allowing historical queries such as the balance of an account at a specific block height.

### 2.3.2 On-chain Data

Collecting blockchain data requires either a node for the associated blockchain or access to third-party indexers that store and format blockchain data. Within this research, we predominantly used our own instance of an Ethereum node to retrieve on-chain data. For certain aspects of this research, such as quick look-ups of blockchain data, third-party indexers such as *Dune Analytics* and *The Graph* were used.

A full node keeps a record of every transaction that has taken place within the Ethereum network. Transactions can be recalled through their transaction hash. Event logs may also be retrieved through event signature look-ups. Specifying the smart contract address allows the analyst to recover the entire history of a smart contract's interactions. These queries are made through the Ethereum JSON-RPC interface, commonly referred to as RPC calls.

### 2.3.3 Trace Data

When a transaction triggers a smart contract that in turn calls other smart contracts, the resulting chain of internal calls is recorded as *trace data* (also called internal transactions). Trace data captures the from and to addresses of each inter-contract call, the function invoked, and any value transferred. Unlike event logs, which are explicitly emitted by contract code, trace data is a by-product of EVM execution and is available only from archive nodes or specialised indexing services.

Trace data is important for this thesis because it reveals relationships between contracts that are not visible from transaction-level data alone. For example, a governance contract that checks a voter's token balance will generate a trace call to the token contract. By analysing these trace calls, we can identify which contracts interact with a given token or governance contract, a technique used in Chapter 5 to detect metagovernance relationships.

### **2.3.4 Third-Party Data Sources**

Several third-party services supplement raw blockchain data. Block explorers such as Etherscan index transactions, contracts, and token transfers, and provide public name tags that attach human-readable identifiers to addresses. Token registries such as CoinMarketCap and CoinGecko aggregate market data including prices and market capitalisation. Governance-specific platforms such as Snapshot provide APIs for retrieving off-chain voting data. We use these sources throughout the thesis and describe the specific data collection procedures in each chapter.

## **2.4 Tokens**

Buterin [22] describes token systems as having a range of uses, from sub-currencies representing real-world assets like USD or gold, to company stocks. Tokens are themselves smart contracts, and are typically categorised by their fungibility. Within decentralised governance on Ethereum, tokens play a central role: token-based voting systems are employed in most DAOs [54], with voting power typically proportional to the number of governance tokens an address holds. Governance tokens share similarities with shares in traditional corporate governance; however, DAO governance tokens do not require the formalities of a broker and are traded through decentralised exchanges permissionlessly and pseudonymously.

### **2.4.1 Fungible Tokens**

Fungible tokens are tokens in which any one token within a smart contract is directly equal to another token within that same smart contract. For example, each USDC token created by the financial technology company Circle is equivalent to any other USDC.

Fungible tokens are more commonly referred to by their implementation standard from the Ethereum Request for Comment standards. A fungible token smart contract typically follows the ERC-20 standard [117], and is, as a result, commonly referred to as an ERC-20 token. The ERC-20 standard includes important functionality: the token's name, number of decimals, total supply, transfer capability, and minting. The token contract keeps a



record of the balance of each address that owns the token, with the transfer functionality deducting value from the sender's balance and increasing the recipient's balance.

### 2.4.2 Non-Fungible Tokens

Non-fungible tokens (NFTs) are not interchangeable: each token within the token contract is unique in some way. The standard implementation for NFT token smart contracts is ERC-721 [40]. ERC-721 shares most of the functionality of ERC-20, such as name, balance, and transfer, but also includes a token ID to uniquely identify each token within the contract and a token URI (Uniform Resource Identifier) which typically resolves to a JSON metadata document.

### 2.4.3 Token Wrapping

Token wrapping is a process in which tokens of one type are deposited into a smart contract and new tokens of a different type are minted in return. The deposited tokens are held by the contract, while the newly minted tokens represent a claim on the underlying deposit. The process can be reversed: the holder burns the wrapper token and withdraws the underlying token from the contract.

Token wrapping is widespread on Ethereum. For example, up to 84% of DAI-related transactions involved wrapped or composed derivatives of DAI [118]. Schar [99] illustrates the risks of deep wrapping chains through the scenario of a “token on top of a token on top of a token”: if any contract in the chain fails, tokens at higher tiers could become worthless.

Token wrapping has direct implications for governance. When a protocol wraps a DAO's governance token and issues its own token in return, the voting power associated with the deposited tokens is transferred to the wrapping protocol. This is the mechanism through which tokenised metagovernance operates, as described in Section 2.8. The composition of tokens through wrapping is examined in detail in Chapter 7.

## 2.5 Decentralised Finance

Decentralised Finance (DeFi) refers to financial services implemented as smart contracts on public blockchains, operating without traditional intermediaries such as banks or brokerages. DeFi protocols compose with one another through smart contract interactions, creating an ecosystem often described as financial building blocks [99] or money legos [7]. Several DeFi concepts recur throughout this thesis and are defined below.

### 2.5.1 Decentralised Exchanges and Liquidity Pools

A *decentralised exchange* (DEX) is a smart contract protocol that enables permissionless trading of tokens without a centralised intermediary. Unlike centralised exchanges, where an order book matches buyers and sellers, many DEXs use an automated market maker (AMM) model. In an AMM, users called *liquidity providers* deposit pairs of tokens into a smart contract known as a *liquidity pool*. The pool uses a mathematical formula to determine the exchange rate between the two tokens. In return for providing liquidity, providers receive a share of the trading fees generated by the pool, often represented by a liquidity pool token.

Liquidity pools are relevant to this thesis because several governance mechanisms, including Curve Finance’s gauge system (Chapter 6), direct token emissions to liquidity providers based on governance votes.

### 2.5.2 Stablecoins

A *stablecoin* is a token designed to maintain a stable value relative to a reference asset, typically the US dollar. Stablecoins achieve their peg through various mechanisms: fiat-backed stablecoins such as USDC hold reserves of the reference currency, while collateralised stablecoins such as DAI are minted against overcollateralised deposits managed by lending protocols. Stablecoins are relevant to this thesis because they are among the most composed tokens on Ethereum (Chapter 7) and because several of the DAOs examined, including Curve Finance and Frax Finance, are stablecoin issuers or stablecoin-focused trading platforms.

## 2.6 Decentralised Autonomous Organisations

Decentralised Autonomous Organisations (DAOs) are governance protocols built on public blockchains. DAOs have become a common way to organise for blockchain-based protocols, as blockchain networks enable decentralised governance through smart contracts that are secured by the underlying network. DAOs provide typical governance functions such as proposal submission, voting power determination, and voting.

A DAO’s governance can take place entirely on-chain, where every action is a blockchain transaction, or off-chain through third-party protocols, where votes are signed messages verified against on-chain token balances. Regardless of the venue, voting power is ultimately derived from on-chain token holdings.

The analysis of both on-chain and off-chain DAOs within this thesis is based on observable governance records such as addresses, token balances, votes, token wrappings, smart

contracts and voting power delegations. Governance processes that are not empirically observable, such as private deliberation and individual motivations, are outside the scope of this research.

### 2.6.1 On-chain DAOs

On-chain DAOs operate entirely within a blockchain network. The governance protocol itself is a series of smart contracts that implement governance through on-chain transactions. Common frameworks for DAO governance have emerged from protocols such as Compound Finance's Governor Alpha and Governor Bravo. OpenZeppelin offers modular templates based on Compound's framework [91].

A DAO's on-chain governance ecosystem typically consists of several composed smart contracts, each with its own functionality. For example, the timelock smart contract within the OpenZeppelin governance suite requires successful proposals to wait a mandatory period before execution. On-chain governance is final in nature: the rules of engagement are auditable through the smart contract code itself, from the initial proposal to the execution of the results. Proposals can include executable code that is triggered automatically once a vote has passed. This removes the need to trust any particular actor to enact the will of the DAO. The cost of on-chain governance is that every participant must spend gas to submit proposals and cast votes.

Figure 2.1 shows Compound Governor Bravo's governance proposal flow. An address with at least 25 000 delegated COMP tokens may make a DAO proposal. The proposal enters a two-day review period. Once this review period has elapsed, voting weights per address are recorded at the time the review period ended and voting opens. Voting lasts for three days, at the end of which all votes are tallied. Compound requires 400 000 votes to have been cast for a vote to be valid. If the proposal is successful and has met the criteria, it is queued in the timelock contract. Once the timelock's waiting period of two days has been met, the proposal may be executed.

### 2.6.2 Off-chain DAOs

Off-chain DAOs do not rely on blockchain transactions for their governance process. They operate predominantly outside a blockchain network; however, components of governance such as voting power are still determined through on-chain means.

Off-chain DAOs typically use third-party governance protocols such as Snapshot. Snapshot hosts over 5000 DAOs, has held 41 000 proposals, and recorded 31 million votes. To create a Snapshot Space, a DAO must register its governance address using an Ethereum Name Service domain. A voting power source may then be defined, such as an Ethereum

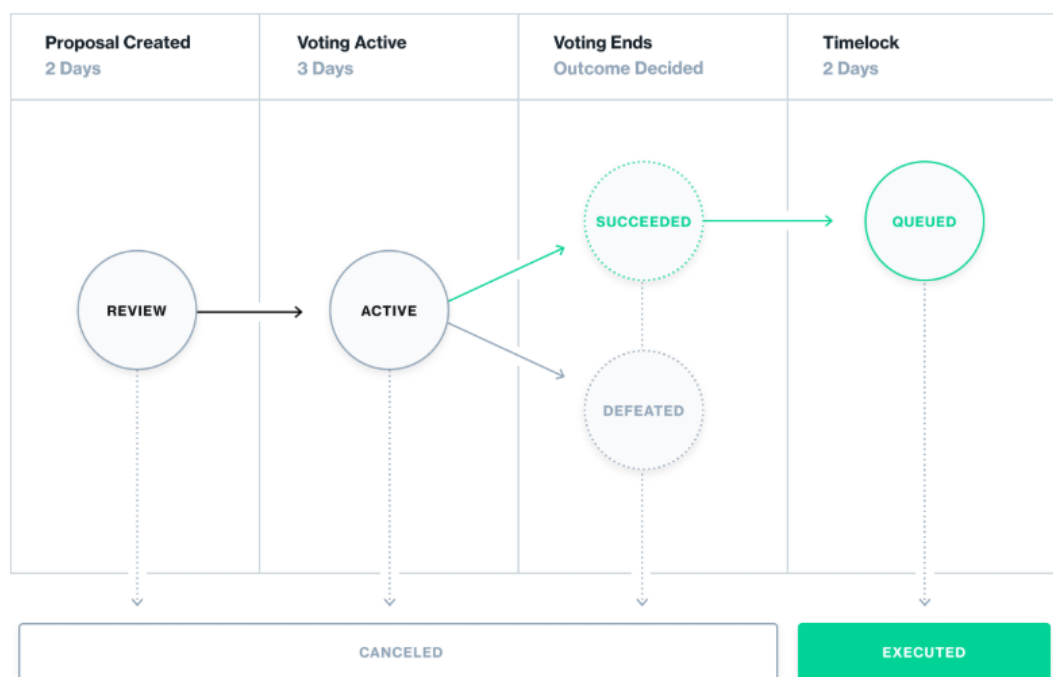


Fig. 2.1 The Compound Governor Bravo proposal lifecycle. A proposal enters a two-day review period, followed by three days of active voting. If the proposal succeeds, it is queued in a timelock for a further two days before it can be executed. A proposal may be cancelled at any stage. Adapted from the Compound Finance documentation.

token smart contract, from which the voting weight of participants is read. Once a proposal has been initiated, addresses with voting power may participate. Votes are recorded by the signing of messages off-chain: a user signs a message supporting a proposal with the private key of their blockchain address, but this message is not placed on-chain via a transaction. Instead, the message is verified by the Snapshot protocol and subsequently stored on IPFS.

Off-chain governance typically requires an intermediary, such as a multi-signature wallet controlled by trusted community members, to execute the will of the DAO once a proposal has passed. This creates a centralised risk. However, participants benefit from free voting without incurring transaction fees, and smart contract vulnerability risks are reduced as malicious proposals that pass can be vetoed or simply not implemented by the controllers of the DAO.

### 2.6.3 Voting Power and Delegation

While tokens issued by DAOs play a central role in determining voting power, the mechanisms by which voting power is realised are not always as simple as holding a token to receive a vote. For example, the Governor Bravo framework implemented by Compound Finance requires holders of the COMP token to delegate the token's voting power to an address for that power to be recorded during a snapshot before a proposal opens for voting. The off-chain governance protocol Snapshot also has bespoke smart contracts to manage delegation for off-chain governance.

While it is common for token-holding addresses to delegate voting power to themselves, delegation also enables the transfer of voting power to other addresses. This can create metagovernance relationships, as described in Section 2.8.

### 2.6.4 Multi-Signature Wallets

A *multi-signature wallet* is a smart contract that requires a predefined number of authorised signers to approve a transaction before it can be executed. A multi-signature wallet configured as *m-of-n* requires at least *m* out of *n* designated signers to provide their cryptographic signatures. For example, a 3-of-5 multi-signature wallet requires three of its five signers to approve each transaction.

Multi-signature wallets serve two primary roles in DAO governance. First, they are used as the execution mechanism for off-chain DAOs: trusted community members controlling a multi-signature wallet enact the results of Snapshot votes on-chain. Second, multi-signature wallets are used by external organisations, including other DAOs, to participate in governance. In both cases, a governance vote that appears as a single address in a

Table 2.1 Comparison between corporate governance and DAO governance. While both rest on transferable units of ownership, they differ in decision making, execution, transparency, identity, regulation, proposal scope, and transferability.

|                      | Corporate                                                 | DAO                                                                                |
|----------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|
| Decision making      | Delegated to managers and board of directors              | Direct participation by token holders                                              |
| Execution            | Enacted through organisational hierarchy                  | Executed automatically by smart contracts                                          |
| Voting transparency  | Votes are typically private                               | All votes are recorded on a public ledger                                          |
| Identity             | Shareholders are identified through regulated registries  | Participants are pseudonymous addresses                                            |
| Regulatory oversight | Subject to established legal frameworks                   | Limited legal and regulatory structures                                            |
| Proposal scope       | Infrequent, high-level decisions such as board elections  | Frequent, operational decisions such as treasury allocations and protocol upgrades |
| Transferability      | Share transfers require brokers and regulatory compliance | Tokens are freely tradable on decentralised exchanges                              |

DAO's voting records may represent the collective decision of multiple actors. Identifying multi-signature voters among a DAO's governing addresses is a central task in Chapter 5.

### 2.6.5 Corporate and DAO Governance

DAOs offer a mechanism for maintaining decentralisation within blockchain protocols. While parallels can be drawn between traditional corporate governance and DAO governance, such as voting power being denominated in transferable units of ownership (shares in corporate governance, tokens in DAO governance), the governance mechanisms required to maintain decentralisation differentiate DAOs from corporate governance in fundamental ways [54]. Table 2.1 outlines these differences.

As voting power in DAOs grants those governing direct access to DAO resources, including treasury funds and protocol parameters, the stakes of governance are high. DAOs are exposed to both technical exploits, where a smart contract's logic is used in a way not intended, and social exploits, where actors manipulate the governance process while operating within its rules. We review the literature on DAO governance attacks in Section 3.4.

## 2.7 DAO Communities

In general discourse, referring to a DAO community typically covers the entirety of a DAO's members. Membership of a DAO is a fuzzy term and could include developers, token holders, and users of the protocol. For example, if one refers to the Uniswap community, it is unclear whether this includes those holding the UNI token, those working on the Uniswap protocol, those using Uniswap's decentralised exchange, or all three. Even if membership were more clearly defined, this definition implies that the individuals making up the community are a single entity.

We therefore define community in more specific terms: a *DAO community* is a cohesive group of actors that collectively exerts influence within a DAO's governance. This definition allows for multiple communities to be distinguished within a single DAO and introduces a tangible aspect that permits an analytical process to identify groups based on cohesion.

We further distinguish between two types of DAO communities: internal and external.

### 2.7.1 Internal Communities

*Internal DAO communities* are groups of voting addresses that demonstrate persistent alignment across a DAO's proposals. Cohesion is inferred from the similarity of their voting behaviour: addresses that consistently vote the same way across multiple proposals are considered to belong to the same internal community.

While voting records of each address are public, voter identities are pseudonymous, limiting our ability to determine community alignment through identity alone. This is in contrast to traditional forms of governance such as corporate or political governance systems, where governance strategies and party affiliations are often explicitly stated. In this thesis, we explore the accuracy with which internal DAO communities can be inferred from past governance data and examine whether the historical voting behaviour of internal communities can provide early warning signals for future governance behaviour (Chapter 4).

### 2.7.2 External Communities

*External DAO communities* are defined by their mechanism of participation: governance is exercised through a smart contract address, typically a multi-signature address. This mechanism allows a decision executed by a single address to represent the collective decision of multiple actors. It relocates the decision-making process outside of the DAO's governance smart contract to a predefined group of addresses.

We also define a sub-category of external community governance: *tokenised external community governance*. While this is still typically enacted through a multi-signature address within the underlying DAO's governance, external to the underlying DAO, a new governance protocol is built. The external community protocol is itself a DAO, complete with on-chain or off-chain governance mechanisms and a token. Token acquisition for the underlying DAO is achieved through token wrapping (Section 2.4.3), in which the token of the underlying DAO is deposited in the external community's protocol and a new token is minted, native to the external community. We identify external governance communities in Chapter 5 and examine tokenised external communities in Chapter 6.

### 2.7.3 Forks

A core component of DAO governance is resolving differences between members of a DAO. Purely relying on the internal governance process may not be sufficient for extreme disagreements that are deemed unresolvable. Beck et al. [13] describe forking as a drastic resolution for governance disagreements, whereby participants copy the existing code and set up an alternative project. Within their study, they analyse the fork from Arcade City to Swarm City, where a token exchange between the ARC token and the SWT token was implemented at a one-to-one ratio.

Further examples of forking include automated forking mechanisms, in which forking is programmed into the governance mechanisms of the DAO. Nouns DAO, the case study examined in Chapter 4, implements such a mechanism: those wishing to create a new instance of their current DAO can do so through the governance system, with tokens automatically minted within the new DAO to retain voting power.

Forking is an extreme solution for resolving governance differences. While it may be necessary, it undermines the legitimacy of the original organisation and fractures the community and the resources the DAO controls. The relatively low costs may encourage participants to fork preemptively before all other options have been exhausted [94].

## 2.8 Metagovernance

Metagovernance is not a concept unique to blockchain governance. The term has been previously defined by Jessop [59] when discussing governance relating to governments. When discussing governance networks, Bortel and Mullins [115] describe metagovernance as an “attempt by politicians, administrators or other governance networks to construct, structure and influence the game-like interaction within governance networks”. Simply put, metagovernance is the governance of governance.



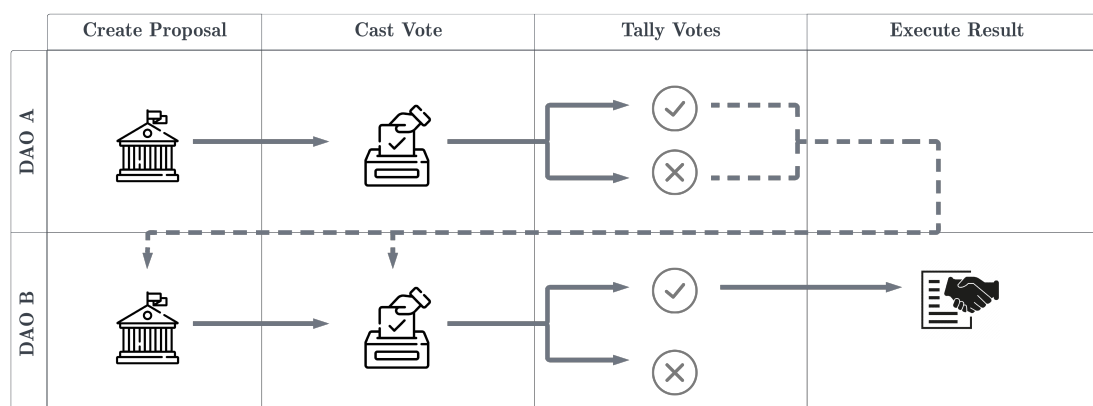


Fig. 2.2 Metagovernance occurs between two DAOs, DAO A and DAO B, when the result of a proposal in DAO A either creates or votes on a proposal in DAO B. The solid lines show the progression between the stages in a typical proposal process within DAO A and DAO B; the dotted lines show the output of the process in DAO A becoming an input to the process in DAO B.

DAO metagovernance involves an organised external community participating in the governance of a DAO. This community can take different forms, such as another DAO, a protocol, or a collective of actors. Metagovernance therefore requires governance to be exercised by a community rather than a single participant. Metagovernance between DAOs is an attempt by one DAO to govern another. Figure 2.2 illustrates one form of metagovernance between two DAOs, DAO A and DAO B. A proposal in DAO A progresses through the typical stages: creation, voting, tallying, and execution. The outcome of the proposal either creates or votes on a proposal in DAO B. The output of the former proposal is an input to the latter; this is represented by dotted lines. The execution of metagovernance proposals is typically conducted through a multi-signature address (Section 2.6.4).

In Figure 2.2, DAO A holds voting power in DAO B. As described in Section 2.4, tokens typically determine voting power within DAOs. Metagovernance is therefore conducted through the acquisition of tokens in DAO B by DAO A. DAO A can obtain DAO B's governance token in a multitude of ways: on the open market, through token swaps, as a reward for contributions, or through liquidity provision.

More complex forms of metagovernance voting power acquisition also exist. One such mechanism is through *token wrapping* (Section 2.4.3). Users deposit their tokens from DAO B into the protocol operated by DAO A. The DAO B tokens are held within DAO A's smart contract, while DAO A issues a new token representing the deposited assets. This mechanism enables DAO A to accumulate voting power in DAO B through the wrapped tokens. We refer to this model of metagovernance as *tokenised metagovernance*.

As described in Section 2.6.3, delegation enables an address with voting rights to delegate those rights to another address. This can also create metagovernance relationships, as DAO A could receive a delegation of voting power within DAO B, creating a metagovernance relationship without DAO A ever having to hold DAO B's governance token.

## 2.9 Visual Grammar for Governance Diagrams

Having defined the foundational concepts, we now describe the visual grammar adopted in this thesis. The standard approach to DAO governance analysis treats each voting address as an independent actor and examines individual voting records in isolation. This thesis takes a different approach: we analyse governance through the lens of communities. The following subsections describe the progression from individual-level analysis to community-level analysis and outline how the chapters map to this grammar.

To describe these governance structures precisely, we define a visual grammar (Table 2.2) comprising five vertex types, five edge types, and one annotation convention. We use this grammar throughout the thesis to explain and visualise the governance scenarios in our analysis. The grammar is introduced in Figures 2.3–2.6 and provides a shared vocabulary for the chapters that follow. Where multiple instances of the same type appear in a diagram, they are distinguished by labels (e.g., Token A and Token B, or DAO A and DAO B).

Table 2.2 Visual grammar used in Figures 2.3–2.6 to illustrate governance structures at increasing levels of complexity. The grammar is divided into vertices (the entities), edges (the relationships between entities), and an annotation convention.

| (a) Vertices                                                                        |       |                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Actor | A person or entity that participates in governance, possibly controlling one or more blockchain addresses.                                        |
|  | EOA   | An externally owned account: a blockchain address controlled by a single private key.                                                             |
|  | Token | A governance token. Holding the token confers voting power in the associated DAO. Labelled per instance (A, B, etc.) when multiple tokens appear. |

*Continued on next page*

Table 2.2 continued

|                                                                                     |                        |                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Multi-Signature Wallet | A smart contract address requiring $m$ -of- $n$ signers to approve a transaction.                                                                                                     |
|    | DAO                    | A governance contract: the hub where proposals are submitted and votes are cast. Labelled per instance (A, B, etc.) when multiple DAOs appear.                                        |
| <b>(b) Edges</b>                                                                    |                        |                                                                                                                                                                                       |
|    | Controls               | A double-lined arrow indicating that the source actor holds the private key for the target address. For a multi-signature wallet, multiple control edges converge on a single vertex. |
|   | Holds                  | A solid arrow indicating that the source address holds the target token, conferring voting power.                                                                                     |
|  | Votes yes              | A dashed green arrow indicating a vote cast in support of a proposal.                                                                                                                 |
|  | Votes no               | A dashed red arrow indicating a vote cast in opposition to a proposal.                                                                                                                |
|  | Tokenised by           | A dash-dot arrow indicating a tokenisation relationship: the source token is deposited into a contract that mints the target token.                                                   |
| <b>(c) Conventions</b>                                                              |                        |                                                                                                                                                                                       |
|  | Community boundary     | A grey rounded rectangle grouping actors that share governance alignment or organisational membership.                                                                                |

The five vertex types capture the distinct roles that entities play in a governance system. An *actor* is a person or entity that participates in governance. We use “actor” rather than “owner” because not all governance participants own tokens: some receive delegated voting power, and others act through multi-signature wallets they share with other actors.

Actors interact with the blockchain through addresses. An *externally owned account* (EOA) is a standard blockchain address controlled by a single private key. A *multi-signature*

*wallet* is a smart contract address that requires  $m$ -of- $n$  designated signers to approve a transaction before it can execute. The distinction between these two address types is important: in Chapter 5, we show that many governance voters are contract accounts rather than EOAs, and automated detection of multi-signature wallets is a key step in identifying external communities.

A *token* is a governance token whose balance determines voting power in a DAO. A *DAO* is the governance system itself: the contract (or off-chain space) where proposals are submitted and votes are cast. Where multiple tokens or DAOs appear in the same diagram, they are distinguished by labels such as Token A, Token B, DAO A, and DAO B.

The five edge types represent the relationships between vertices. A *controls* edge connects an actor to an address, indicating that the actor holds the private key for that EOA or is one of the designated signers of that multi-signature wallet. A *holds* edge connects an address to a token, indicating that the address has a token balance and therefore voting power.

Two vote edges capture governance actions. A *yes* edge (dashed green arrow) connects an address to a DAO, indicating a vote cast in support of a proposal. A *no* edge (dashed red arrow) indicates opposition. In both cases, the voting power exercised by the address derives from the tokens it holds.

A *tokenised by* edge connects one token to another, indicating a tokenisation relationship: holders of the source token deposit it into a smart contract that mints the target token. This edge is directed from the underlying token to the derivative. Token wrapping is the mechanism behind the veToken model analysed in Chapter 6 and is the sole relationship type in the token composition graph of Chapter 7.

One annotation convention completes the grammar. A *community boundary* is a rounded rectangle drawn around a group of actors that share governance alignment or organisational membership. Community boundaries are the visual counterpart of our core contribution: detecting and characterising groups of aligned actors within and across DAOs.

The grammar enforces constraints on which edges may connect which vertex types. Actors connect only to addresses (EOAs or multi-signature wallets) via controls edges; they never connect directly to tokens or DAOs. This reflects the blockchain's structure: actors interact with the governance system through their addresses. Addresses connect to tokens via holds edges and to DAOs via yes or no edges. Tokenised by edges connect one token to another. Tokens never vote; only addresses do.

A community boundary may also serve as the source or target of an edge, indicating that the action is performed collectively by the community rather than by any single actor. For example, a holds edge originating from a community boundary means that the

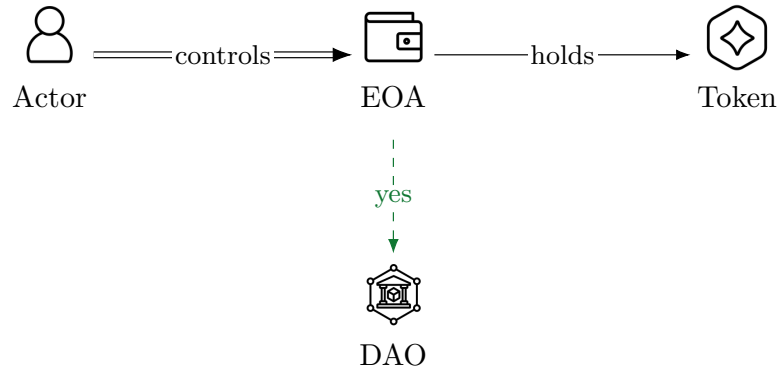


Fig. 2.3 Governance analysis in its simplest form. An actor controls an EOA that holds a governance token and casts a yes vote in the associated DAO. Grammar elements: Actor, EOA, Token, and DAO vertices; Controls, Holds, and Yes edges.

community’s members collectively hold the token, and a yes or no edge originating from a community boundary means that the community’s members vote in the same direction.

These constraints mean that every governance action in the grammar passes through an address. An actor who wishes to vote must first control an address that holds the relevant token. An actor who wishes to participate in tokenised metagovernance must hold a derivative token (connected to the underlying token by a tokenised by edge) through an address that can then vote in the higher-tier DAO. The grammar thus captures the layered, composable nature of DAO governance as a directed graph of typed vertices and edges.

### 2.9.1 Individual-Level Analysis

Figure 2.3 shows governance analysis in its simplest form using four vertex types and three edge types from the grammar. An actor controls an externally owned account (EOA); the EOA holds a governance token, and the token confers voting power that the EOA exercises in the associated DAO. The path from actor, to EOA, to token, to cast vote is conceptually simple. This level of analysis assumes the controller of each address is a single individual and treats each voter interaction independently, overlooking relationships and behavioural alignment between voters.

### 2.9.2 Community-Level Analysis

We approached the analysis of governance dynamics by addressing the limitations of individual-address analysis. In Chapter 4, we go beyond analysing voters and proposals in isolation, instead examining the similarity between voters across multiple proposals to identify shared voting alignment, grouping addresses into larger communities. Figure 2.4 introduces the community boundary convention: actors whose voting behaviour is aligned

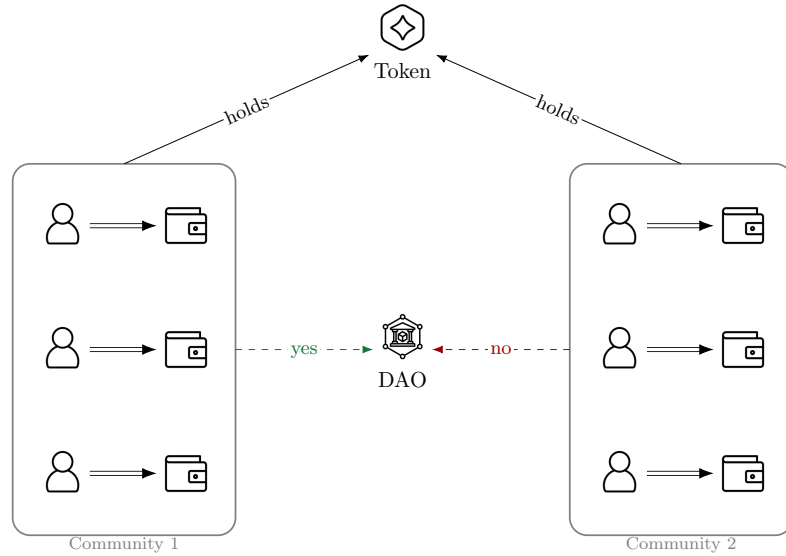


Fig. 2.4 Community-level governance analysis. Three actors in Community 1 exercise yes votes, while three actors in Community 2 exercise no votes. All actors hold a single shared token. Community boundaries (rounded rectangles) group actors with aligned behaviour.

are enclosed in a rounded rectangle. Community 1 exercises yes votes and Community 2 exercises no votes. The token is shown as a single shared vertex outside the community boundaries, since all actors in both communities hold the same governance token.

An often overlooked aspect of DAO governance is that governing addresses can be smart contracts, such as multi-signature wallets, resulting in multiple actors controlling a single voting address. Figure 2.5 introduces the multi-signature vertex type: three actors control a single 2-of-3 multi-signature address, which requires two of the three to sign before its voting power can be exercised. In Chapter 5, we automate multi-signature detection, broadening the analysis of governance dynamics to the external context.

In Chapter 6, we expand upon multi-signature identification to investigate tokenised metagovernance, a governance phenomenon that operates a DAO on top of an existing DAO to create a new layer of governance. Figure 2.6 introduces the tokenised by edge: Token B is tokenised by Token A, meaning that holders of Token B deposit it into a contract that mints Token A. Token A confers voting power in DAO A, whose two communities vote yes or no on proposals. DAO A controls a multi-signature wallet that holds Token B and exercises votes in DAO B. Where tokenised metagovernance relationships exist, analysis is important because the locus of governance can shift to the higher tiers. This broadening of a DAO's context also introduces a recursive structure of analysis, where participants in the higher tiers of governance can be investigated for internal community dynamics. Chapter 7 extends this by constructing a token composition graph in which every vertex is

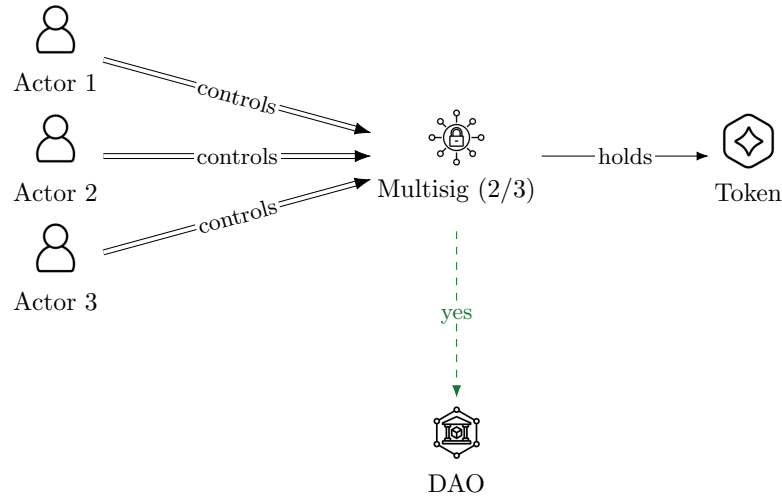


Fig. 2.5 Multi-signature governance. Three actors control a single 2-of-3 multi-signature address that requires two of three signers to approve a transaction.

a token and every edge is a tokenised by relationship, providing a system-level view of tokenisation across the Ethereum ecosystem.

### 2.9.3 Extensions

The grammar defined above is deliberately minimal: it includes only the vertex and edge types needed to describe the governance structures analysed in this thesis. Additional edge types could be introduced to capture relationships that the grammar currently represents implicitly.

The most natural candidate is a *delegates* edge connecting one address to another, representing the delegation of voting power described in Section 2.6.3. Such an edge would be straightforward to define (for example, a dotted arrow from the delegating address to the delegate). Delegation is common in on-chain governance and can itself constitute a form of metagovernance when one organisation delegates to another (Section 2.8).

We omit this edge because delegation introduces an additional level of indirection into every diagram in which it appears. A delegating address must still hold the token, and the delegate must still cast the vote, so the path from token to ballot lengthens from one hop to two. When multiple addresses delegate to the same target, the resulting fan-in pattern dominates the diagram and obscures the governance relationships that are the focus of our analysis. In practice, delegation is most visible in its effect (concentrated voting power at the delegate) rather than in the individual delegation acts themselves, and the grammar already captures that effect through the holds and vote edges.

The visual grammar described is also limited by the scope of governance data considered in this thesis, which is restricted to empirically observable governance activities.

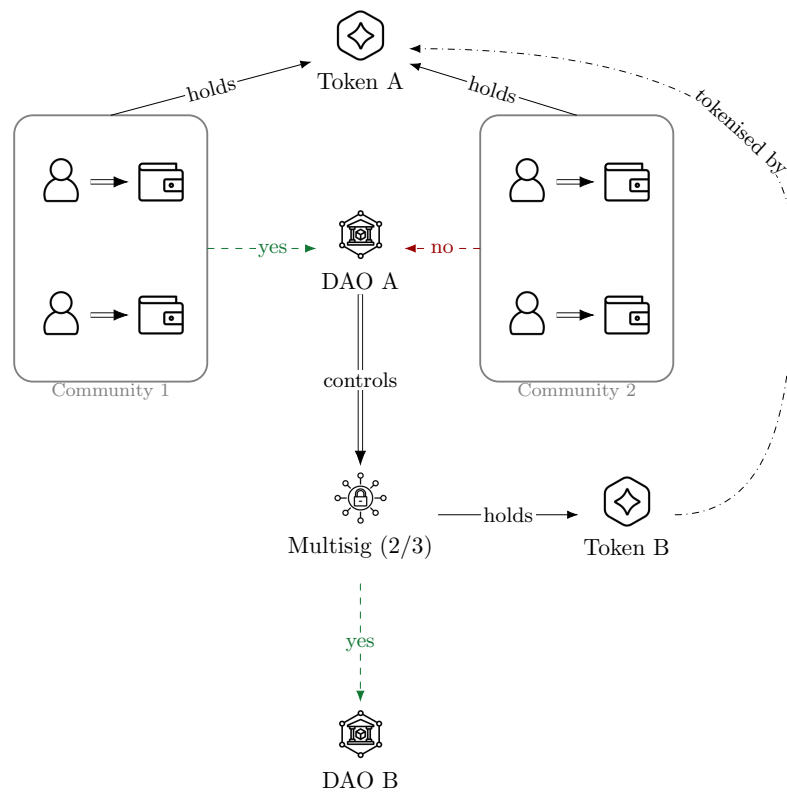


Fig. 2.6 Tokenised metagovernance. Token B is tokenised by Token A, creating a new governance layer (DAO A) on top of the underlying DAO B. Communities within DAO A vote on replicated proposals, and a multi-signature wallet controlled by DAO A enacts the outcome in DAO B. This figure uses nearly all grammar elements: five vertex types, four edge types, and the community boundary convention.



## 2.9 Visual Grammar for Governance Diagrams

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While the visual grammar represents actors controlling addresses that cast votes, it does not capture private deliberation, individual motivations, informal coordination, or other off-chain social processes that may have influenced a governance decision.

We prioritise diagram readability over completeness. The five edge types retain enough expressiveness to describe the governance structures in Chapters 4–7, while keeping diagrams compact enough to compare across chapters.

## Chapter 3

# Literature Review

In this chapter, we review the literature that motivates the research questions posed in Chapter 1. We examine four topical areas, each tied to one or more of the research questions. Smart contract analysis (Section 3.1) underpins the detection of governance contracts and the composition mechanisms through which external communities form, informing RQ 1. Internal DAO governance (Section 3.2) covers governance mechanisms, voting behaviour, dimensionality reduction techniques borrowed from political-science research, and forking, informing RQ 1 and RQ 3. External DAO governance (Section 3.3) addresses metagovernance and DAO-to-DAO interactions, informing RQ 1 and RQ 2. DAO exploits (Section 3.4) describe the threat model that motivates community-level analysis. We close with a synthesis of the gaps in knowledge (Section 3.5) that the thesis addresses.

### 3.1 Smart Contract Analysis

#### 3.1.1 Identifying Smart Contracts

Smart contracts are identified within blockchain data as hexadecimal addresses that do not describe the functionality or purpose of the contract. This makes searching for specific smart contracts difficult. However, extensive research has produced a range of approaches to illuminate smart contract functionality and purpose through their interfaces, bytecode, and behaviour.

Tokens are a core component within a smart contract economy and the primary mechanism for determining voting power within DAOs. Identifying and monitoring token contracts is essential for representing market data and governance security. The ERC-20 standard provides a standardised interface for fungible token smart contracts. Frowis et al. [49] use this standardisation to identify tokens within Ethereum. They provide two

methods to identify token smart contracts and find most success through function-signature matching against the ERC-20 standard. They note that this method, while yielding nearly complete identification of the contracts they investigated, could be evaded by deviating from the standard. Norvill et al. [89] cluster smart contracts by function selectors, using clusters of ERC-20 contracts to validate their results. Conformance to the ERC-20 interface does not guarantee conforming behaviour: Moon and Park [84] examine the top hundred ERC-20 tokens and find variation between specified interfaces and implementations.

Di Angelo et al. [34] address the identification of token contracts that do not conform to the ERC-20 standard. In addition to structural identification through function and event signatures, they use log event data to detect behavioural aspects such as token transfers. This combined approach identified 19% additional non-conforming ERC-20 tokens that interface-only methods miss.

Smart contract code is heavily reused across deployers, partly because of well-established templates and frameworks (such as those provided by OpenZeppelin) and the increase in development that follows when such frameworks are provided [50]. He et al. [57] categorise Ethereum smart contracts by hashing the structural logic of each contract and combining the results into a single fingerprint. Pairs of fingerprints are compared for similarity, producing clusters of similar contracts; the authors deem contracts with a similarity of 70% or more to be clones. Of the ten million contracts examined, 96% were identified as having a clone, with token contracts being the most cloned. Liu et al. [73] detect smart contract clones by symbolically executing contracts to observe their behaviour, such as reads from and writes to storage; these actions are transformed into vectors for comparison. They later expanded this work to consider the conditions preceding each action [72]. Zhu et al. [136] identify contract clones through opcode patterns within the control flow graph.

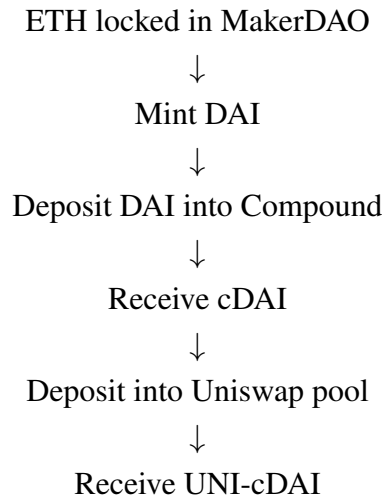
Bytecode-based identification is harder to evade than interface matching but is not infallible. Yu et al. [133] introduce BOSC, a bytecode obfuscation approach that complicates the identification of contracts whose authors deliberately avoid detection, such as honeypot contracts [75] and Ponzi contracts [126]. Di Angelo and Salzer [33] analyse the Ethereum smart contract ecosystem and categorise contracts both by interface and by behaviour derived from historical trace data. One category they describe is *prolific* contracts, which create other contracts; they note that DAOs warrant further study of governance and usage.

Smart contract identification combines structural cues (interfaces, bytecode) with behavioural cues (transfers, opcode traces). Standard contracts such as ERC-20 tokens are well covered by existing methods, but DAO governance contracts, which use heterogeneous frameworks and configurable modules, remain a comparatively open problem. Common smart contract interfaces enable interoperability, the topic of the next subsection.

### 3.1.2 Smart Contract Composition

Smart contracts can interact with other smart contracts within the same blockchain network through external calls. Over a four-year period between 2020 and 2024, over 76% of Ethereum transactions involved at least two inter-contract calls [135], making smart contract composition pervasive in the Ethereum ecosystem. Composition is a fundamental aspect of Decentralised Finance (DeFi). In DeFi, multiple products can be combined by allowing their smart contracts to interact, often described as financial legos [7] or DeFi building blocks. Composition can produce entirely new protocols built on top of existing ones.

Schar [99] proposes a multi-layered framework for analysing these building blocks and describes composability as one of the most promising features in the DeFi ecosystem, as new services emerge from combining existing ones. However, an issue with one smart contract may affect the wider ecosystem, illustrated through the “token on top of a token on top of a token” scenario:



In this scenario, the risks of bugs accumulate with each layer; if any contract within the chain fails, tokens at higher tiers could become worthless. The theoretical transparency of the blockchain is intact, but practical transparency is obscured: holding UNI-cDAI requires knowledge of every dependency in the stack. Token wrapping is prevalent on Ethereum, with up to 84% of DAI-related transactions involving wrapped or composed derivatives of DAI [118]. Composability also enables liquid staking [129], in which tokens representing the staked underlying token are themselves liquid. Composition between Lido, Aave, and Curve enabled approximately USD\$877 million worth of ETH volume over 963 days.

Kitzler et al. [64] examine smart contract composition empirically. They identify building blocks within the DeFi ecosystem through repeated internal transaction traces and map them to their associated protocol using a ground-truth dataset. They show that

nested building blocks belonging to different protocols are detected frequently within single transactions, evidence of protocol composition alongside contract composition.

Composition introduces new security risks, as a calling contract relinquishes control to the composed contract. Almakhour et al. [6] use finite state machines to verify expected interactions between smart contracts; Wang et al. [123] construct cross-contract control flow graphs to analyse the execution behaviour between interacting contracts. Knecht and Stiller [65] propose a different approach: smart contract *traits*, identified by hashes of their bytecode, allow the calling contract to verify the specific functionality of the trait before invocation.

While smart contract composition occurs within a single blockchain network, Wilson et al. [127] explore general interoperability across blockchain networks, highlighting its potential to enable more complex applications. The authors note that interoperable systems may inherit vulnerabilities from connected networks. This raises questions regarding DAO governance composition across multiple blockchains. For example, Aave Governance v3 enables users to vote on proposals through networks such as Polygon and Avalanche, while their governance tokens remain on the Ethereum mainnet [2]. Cross-chain DAO governance is outside the scope of this thesis.

Smart contract composition enables decentralised contract interactions within the Ethereum network. It produces new on-chain assets through token wrapping, drives the layered structure of the DeFi ecosystem, and underpins the layered governance structures examined in this thesis. Composition is largely viewed as a positive within DeFi but, as the following sections show, its consequences for governance are less settled.

## 3.2 Internal DAO Governance

Decentralised governance on public blockchains relies on two main mechanisms: on-chain governance and off-chain governance. This section reviews the governance mechanisms that DAOs use, the voting behaviour their participants exhibit, the legislative-voting analysis methods on which our internal-community detection is built, and the resolution of irreconcilable governance disputes through forking.

### 3.2.1 DAO Governance Mechanisms

A core part of any governance system is the mechanism by which participants vote and how voting power is determined. Within DAO governance, this takes many forms. Ding et al. [35] explore the mechanisms typically used in DAO governance. Token-based quorum voting, in which token contracts keep a record of balances and a minimum number of participants or voting power must be met for the vote to be approved, is the most

common. Weighted voting assigns each voter a number of votes that need not be equal; reputation-based voting is similar but uses a participant's reputation within the community to determine voting weight. Multi-signature voting requires multiple addresses to agree on a voting action before it is executed; this is the primary mechanism for external governance interactions covered in Section 3.3. The configurability of DAO governance frameworks such as Aragon [29] and OpenZeppelin's governance suite [91] allows DAOs to adopt these mechanisms in different combinations.

Rikken et al. [95] examine how DAO design influences the longevity of 220 DAOs. They find that one-account-one-vote systems significantly and positively impact long-term viability. DAOs without explicit governance rewards were also more likely to remain active, although the authors do not establish causation.

Berg et al. [18] describe a conceptual governance model in which *commitment voting* grants voting power according to both the quantity of tokens under a voter's control and the time the voter is willing to lock those tokens. Voters' tokens remain illiquid for the duration of the lockup, aligning their economic interest with the long-term success of the protocol. Curve Finance implements a *voter-escrowed token* (veToken) model that operationalises this idea: locking the CRV token grants voting power proportional to both the duration of the lock and the number of tokens locked [39]. Tamai and Kasahara [109] combine the veToken model with quadratic voting in an attempt to align voter and protocol incentives while reducing whale dominance; they acknowledge that splitting tokens across many addresses (a Sybil attack) could exacerbate centralisation. Voshmgir [119] provides broader context for token-based governance, surveying the range of token systems on public blockchains. Lockup-based mechanisms introduce illiquidity costs that the literature treats as an alignment device; whether they hold up in the presence of external communities that wrap the locked token is a question the literature has not yet addressed and which Chapter 6 takes up.

The initial distribution of tokens shapes governance dynamics before a DAO has held its first proposal. Bhambhwani [19] examines the top fifty protocols and categorises tokenomics by insider control. In 39 of 50 protocols, tokens were distributed to team members as a form of compensation, averaging 15% of the total distribution. Tokens were typically distributed via vesting smart contracts to prevent contributors from selling immediately. Kitzler et al. [63] examine the governance behaviour of contributors within DAOs and find that contributors held majority voting power in 7.5% of DAOs and singularly determined a governance proposal in a fifth of DAOs. They also find similar voting patterns among contributors, suggesting an internal community with aligned incentives.

Voting power within DAOs is rarely one-address-one-vote: it is more commonly determined by token balance, with addresses that acquire disproportionate voting power referred to as governance whales. Fritsch et al. [48] examine the distribution of voting

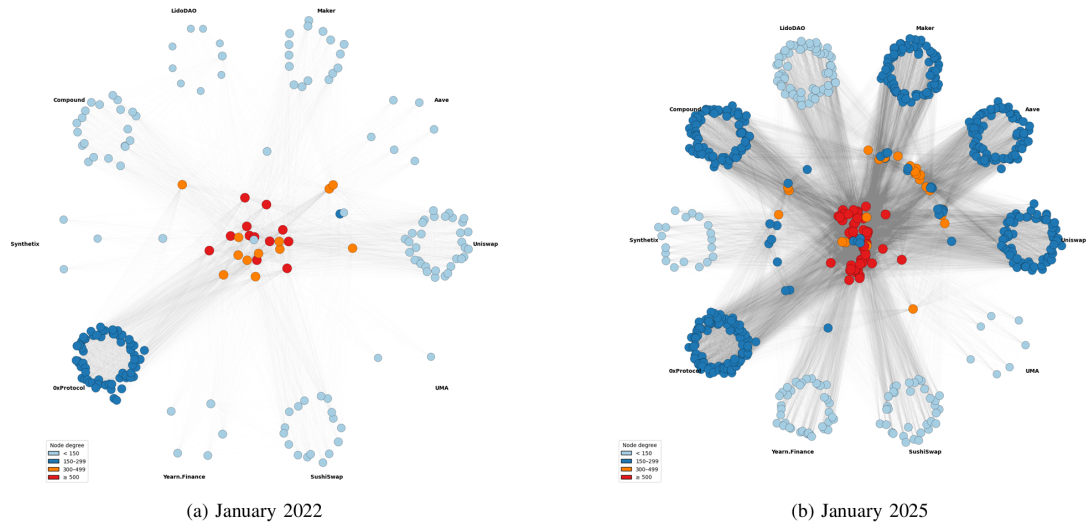


Fig. 3.1 Comparison of DAO holder-overlap networks over time by Satybaldy et al. [98]. Nodes represent addresses, edges represent co-holdings, and colour represents degree of centrality.

power within three token-based DAOs and describe the delegation mechanism by which token holders may transfer voting power to another address. They distinguish between single-holding delegates, who receive the majority of their delegation from a single address, and community delegates. In most cases, single-address delegates vote in the same way as community delegates. They conclude that, given the small number of addresses controlling over 50% of voting power, DAO control is centralised. Wang et al. [121] reach a similar conclusion through a large-scale study of 581 DAOs on the off-chain governance protocol Snapshot.

Satybaldy et al. [98] examine systemic risk in DAO networks by examining overlapping governance token holdings across addresses. They analyse three years of token-holding records (2022 to 2025) for the top ten Ethereum-based DeFi governance tokens, retaining only the smallest set of addresses whose cumulative balances account for 80% of the total token supply. This filters the dataset from over 1.3 million addresses to just 667. They find that the number of multi-DAO token holders grew by 42% over the period, indicating cross-DAO engagement, and that the holder-overlap network became more connected and more central (Figure 3.1), with the upper-degree categories (addresses connected to 500 or more others) growing dramatically. This work shows interconnectivity at the token-holding level but does not measure governance participation directly.

Messias et al. [82] examine delegation within DAO governance, gathering on-chain data from five DAOs and discussion posts from 14 forums. They map Ethereum Name Service domains and forum identifiers to blockchain addresses and analyse the resulting

delegation networks. Many low-degree delegators connect to a few high-degree delegates: a16z accumulates an in-degree of 344 and 245 across two addresses, while Nouns DAO predominantly self-delegates. Delegation networks are typically shaped by a small number of dominant delegates, centralising control even when token distribution is broad.

Weidener et al. [125] conduct a systematic review of 13 delegation publications and identify shared insights across five categories: efficiency, centralisation, inclusivity, transparency, and fairness. Delegation can improve efficiency when exploitative behaviours such as vote buying are limited, but it typically centralises governance power in a few key stakeholders, and limited transparency in off-chain delegations can undermine governance outcomes.

Buterin [23] argues that decentralised governance is necessary for the growth of decentralised protocols, but echoes the concerns of voter apathy, misaligned incentives, and centralisation discussed above. He suggests alternative mechanisms such as quadratic voting and proof of personhood. Quadratic voting reduces voting-power centralisation but introduces new attack surfaces, particularly Sybil attacks; Dupont [38] provides a theoretical framework that applies deep learning to voting networks to identify suspicious patterns indicative of Sybil activity.

The mechanisms catalogued above are heterogeneous in detail but converge on a small set of structural concerns: low participation, concentration of voting power, and the influence of contributors and delegates. Where the literature treats these as properties of individual addresses or single-DAO ecosystems, this thesis treats them as community-level phenomena that span multiple DAOs and multiple governance tiers.

### 3.2.2 DAO Voting Behaviour

Sun et al. [106] analyse past voting behaviour of participants in MakerDAO and discover three voter coalitions of participants who vote consistently with one another. They examine each coalition's effect on the DAI stablecoin and suggest that voter coalitions can affect a DAO's economic standing. The authors note the unique position that public blockchains offer for empirically analysing voter coalitions when compared to corporate shareholder governance, as all governance data is public.

Austgen et al. [11] propose a different way to interpret centralisation within DAOs. Instead of measuring token holdings, they examine the similarity of voters' utility functions and introduce a metric called Voting-Bloc Entropy (VBE). They illustrate the idea through a Dark DAO that participates in another DAO's governance by opaquely buying votes; introducing bribes within a DAO's governance ecosystem aligns economically incentivised voters with one another.



Goldberg et al. [52] investigate governance within the metaverse protocol Decentraland. They find a heavily centralised DAO and observe that addresses with large voting power vote strategically, delaying their votes until late in the proposal period to observe earlier votes. In one case a proposal had a ‘No’ vote leading by two million voting power until just before the proposal closed, when a single vote of 4.4 million voting power flipped the outcome. In 45% of grant proposals, the decision was made unilaterally by a single governing address. Pseudonymity makes it impossible to determine whether the proposer is the same person who benefits from the grant.

Saesen et al. [96] examine how on-chain governance design choices affect token performance. They analyse signalling theory in DAO contexts: a signaller (a developer) sends a signal (such as a white paper) and potential token holders assess merit through the signal. The mechanisms a DAO chooses also signal to investors. The authors investigate the minimum token quorum (MTQ) as such a signal and find that, of 204 DAOs, over half had an MTQ, only 15% had voting incentives, and 19% had a measurable founder’s stake. MTQ is positively associated with token performance; voting incentives are negatively associated. A founder’s stake negates the effect of voting incentives but does not significantly change the effect of MTQ.

Esposito et al. [41] examine DAO governance through the lens of Elinor Ostrom’s principles for governing digital commons [105], including defining boundaries, participatory rule-making, monitoring and enforcement, sanctions, and structuring governance across nested layers. They identify governance models that align with Ostrom’s principles: quadratic governance in Bitcoin grants, reputation-based voting that reduces wealth-based concentration, Soul Bound tokens that bind responsibility to non-transferable holders, and rotating governance councils that change periodically to oversee decisions. They contrast MakerDAO’s focus on financial stability, Commons Stack’s focus on funding public goods, and Optimism’s bicameral Token-House and Citizen-House model, concluding that DAO success depends on how the community structures power and accountability.

Alamsyah et al. [4] apply transformer-based text classification to 10 840 user-generated posts across X, Reddit, YouTube, and the ENS DAO forum, classifying each post by Web3 industry domain, emotional tone, and communicative intent. Their results show that classification of Web3 discourse is feasible and that signals extracted from forum activity could provide foresight into future DAO governance behaviour.

Voting behaviour in DAOs is shaped by coalitions, by economic incentives, and by strategic delays among large holders. The literature has begun to identify these behaviours but has not connected them to the formation, consolidation, and fracture of communities that this thesis treats as the central object of analysis.

### 3.2.3 Political Voting Analysis

Detecting community alignment from voting records is a well-studied problem in the analysis of legislative bodies, where voters' party affiliations are typically known and provide a ground truth for validating clustering methods. Mickevicius et al. [83] analyse voting behaviour in the Lithuanian Parliament using cluster analysis and multidimensional scaling (MDS). They find that MDS visualises opposing parties almost perfectly and that clusters correctly assigned to political parties with 68% accuracy.

Parliamentary voting differs from DAO voting in two respects that matter for transferring methods between domains. First, parliamentary data segments naturally into epochs between general elections; DAOs do not have an equivalent. Voters may enter and exit DAO governance at any time, complicating dissimilarity comparisons between addresses that have never voted on the same proposal. Second, party affiliation provides a ground truth for cluster validation in legislative settings; DAOs are pseudonymous and rarely provide such labels. Chapter 4 adapts the dissimilarity-and-MDS approach for DAOs that do supply a ground truth through documented forks, allowing internal-community detection to be validated despite the pseudonymity of voter addresses.

### 3.2.4 DAO Forks

Forking is the most extreme form of conflict resolution available to a DAO. Beck et al. [13] describe forking as a drastic resolution for governance disagreements, in which participants copy the existing code and set up an alternative project. Their case study of the fork from Arcade City to Swarm City shows a token exchange between ARC and SWT at a one-to-one ratio.

Forking has since been operationalised within DAO governance mechanisms themselves. Nouns DAO implements an automated fork mechanism: members who wish to leave the DAO can trigger the creation of a new instance through the governance system, with tokens minted in the new DAO to retain voting power and a proportional share of the treasury transferred [67, 56]. The first Nouns DAO fork in September 2023 saw 472 Nouns leave the DAO with USD\$27 million from the treasury [56].

Riggins and Fosso Wamba [94] analyse governance adaptation in DAOs and observe that the comparatively low cost of forking can encourage participants to fork preemptively before all other options have been exhausted. Forking undermines the legitimacy of the original organisation and fragments both the membership and the resources the DAO controls.

The literature documents what a fork is and what it costs but does not provide warning signals. No prior study identifies the patterns of voting alignment that precede a fork; the comparison between “forkers” and “stayers” as ground-truth communities, used in

Chapter 4 and our prior work [78], has not previously been applied as a validation method for community detection.

## 3.3 External DAO Governance

### 3.3.1 Metagovernance

In Section 2.8 we defined metagovernance and described how it functions between DAOs within blockchain networks. Weidener et al. [124] provide an overview of metagovernance research and identify only seven papers that directly address DAO-to-DAO governance, indicating an under-researched area despite the prevalence of metagovernance in practice.

Allen et al. [5] examine decentralised governance through the analysis of three DAOs (Curve Finance, Lido, and Metagov). They contrast a ‘romantic view’ of DAO governance, characterised by consensus through community voting, with an ‘exchange view’, characterised by an entrepreneurial process of bargaining and exchanging votes under uncertainty. They argue the exchange view is more appropriate. Their analysis of Curve introduces what they label metagovernance: holding one DAO’s token to influence decisions in another, with players such as Convex Finance, Votium, and Redacted participating in Curve governance. Their analysis of Lido focuses on its accumulation of 30% of the Ether in the Ethereum network and the failed governance proposals that sought to limit Lido’s influence. Their treatment of Metagov frames it as a research collective lowering the costs of political bargaining and exchange in governance systems. This conceptualisation goes beyond individual-address analysis and treats governance as a system of exchange; the present thesis is methodologically aligned with this move beyond individual-level analysis but uses the community lens rather than the exchange lens.

Kaal [61] introduces the concept of a ‘DAO of DAOs’, a metagovernance framework designed to enhance coordination across the DAO ecosystem. He argues that centralisation, regulatory uncertainty, and suboptimal incentive mechanisms limit existing DAOs and proposes a dual governance structure combining on-chain reputation-based voting with an off-chain legal wrapper.

Bennett [17] examines the Hypha governance platform, which enables decentralised governance through modular ‘spaces’. The author conceptualises ‘DAO 3.0’, a third-generation governance architecture that combines technical automation with human interaction. The model incorporates multi-layered voting and dynamic capital flow mechanisms intended to allow governance systems to evolve in response to social and organisational complexity. Such human-centric approaches face the obstacle of high voter apathy in existing DAOs.

Xue et al. [130] examine DAO members who participate in multiple DAOs, naming this type of actor a *member interlock*. They draw comparisons with corporate interlocking directors who serve on multiple boards, creating an information flow. They use the Optimism token airdrop as a case study and analyse voting data from all addresses participating in governance within the Tally and Snapshot governance protocols. Addresses with prior Snapshot governance experience were more likely to claim the OP token airdrop and more likely to participate in Optimism’s governance. This work analyses governance behaviour at the address level for actors that span multiple DAOs; this thesis takes the complementary DAO-centric view, examining each DAO and the addresses participating within it.

Metagovernance is not confined to a single blockchain. Notland et al. [90] investigate cross-chain DAOs (XDAOs), which extend the typical security risks of metagovernance to include the bridge contracts that connect chains.

Metagovernance research is in its infancy. The literature confirms that DAO-to-DAO governance is taking place and identifies its broad shapes, but the mechanisms by which it is detected from blockchain data, the prevalence of these relationships in the wider Ethereum ecosystem, and the consequences for the governance mechanisms of the underlying DAOs are largely unstudied.

## 3.4 DAO Exploits

DAOs are nascent forms of decentralised organisation that manage substantial capital, making them targets for exploitation. As DAOs are both technical and social systems, they are open to two primary forms of attack. Technical exploits have been the most prominent risk since the earliest DAOs were deployed [85, 69]; they are well researched and outside the scope of this thesis. Risks at the governance layer present a more subtle avenue of attack and are reviewed below. Figure 3.2 summarises documented governance attacks between 2020 and 2024.

Feichtinger et al. [45] systematically analyse security threats to DAOs by investigating past exploits. They categorise attack vectors into four categories: bribes, token control, human-computer interaction, and code and protocol vulnerability.

The bribes category covers both informal bribes at the holder/delegate level and formal vote-buying protocols. Most delegates with substantial voting power are publicly or pseudonymously known, which deters them from accepting bribes. Vote-buying protocols use smart contracts to enable the purchase and sale of votes without the briber being exposed to the risk of holding the governing token. The authors present Paladin Lending as a case study; while its liquidity is too low to enable an attack on its own, the existence

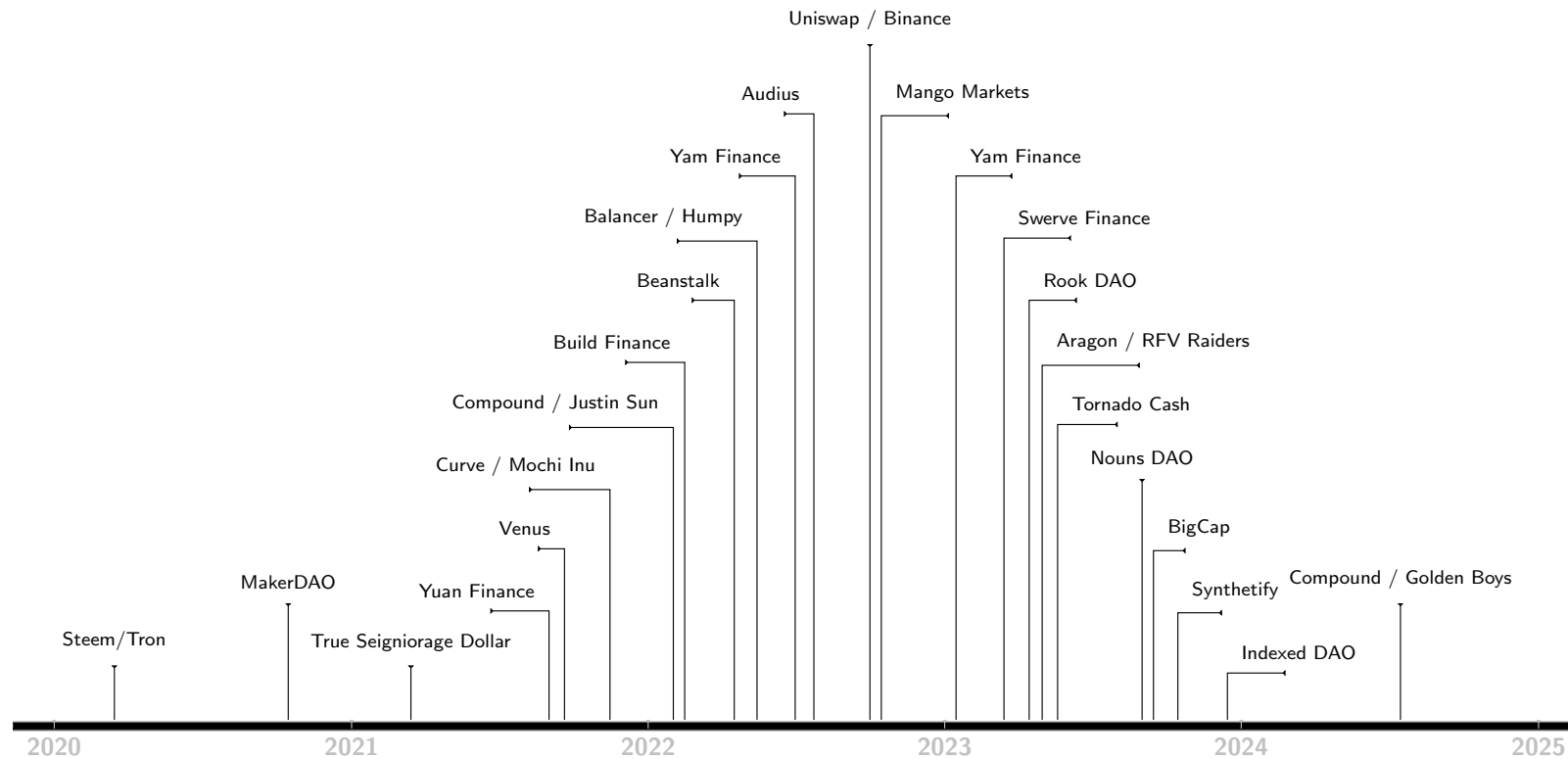


Fig. 3.2 Timeline of governance attacks and incidents targeting DAOs between 2020 and 2024, adapted from the survey by Feichtinger et al. [45]. Appendix A provides details and citations for each incident.

of better-capitalised vote-buying protocols, or attacks combining them with other sources of voting power, raises broader concerns.

The token control category includes flash loan attacks [79, 44, 46], where the attacker acquires tokens, uses them for governance, and returns them in a single transaction; it also includes majority coalitions that can pass any proposal they choose. This risk is acute when the treasury of a DAO is substantial.

In the human-computer interaction category the authors highlight not only technical attacks but also user-interface issues. They use Tally as a case study and describe a UI bug that incorrectly assigned ‘no’ votes as ‘yes’ votes. The bug went unnoticed, suggesting that typical DAO users do not analyse the exact data the governance process produces. The code and protocol vulnerability category largely concerns technical attacks outside the scope of this thesis.

The authors suggest mitigations and safeguards including, for less time-sensitive attacks, governance forks: forking allows participants who wish to leave during an attack an avenue to do so while taking a proportional share of treasury funds. Forking thus deserves practical review as a mechanism to prevent coalition takeovers.

Ma et al. [80] conduct an analysis of 3348 DAOs across nine blockchains. They assess three properties of each DAO’s governance contract: correctness (whether the contract structurally implements decentralised governance), self-governance (whether privileged functions are controlled by the contract itself rather than by an external address), and immutability (whether the contract’s code can be altered after deployment). All DAOs satisfy the first property, but 176 DAOs allow external entities to control privileged functions in their governance contracts, and one DAO permits the developer to alter the contract’s logic directly. They also examine consistency between the descriptions of governance proposals and the code within them; inconsistencies can be a sign of a governance attack via a proposal. Although the target address of the proposal had verified source code in over 90% of cases, making the contract logic auditable in principle, 91% of the proposals investigated nevertheless failed to provide a description consistent with the corresponding code.

Tsai et al. [114] examine vulnerabilities in DAOs by investigating 54 real-world events from 2016 to 2023. They identify technical vulnerabilities (flash-loan attacks, reentrancy attacks, oracle manipulation) and governance-takeover vulnerabilities, including majority attacks (where an attacker acquires majority voting power, exemplified by the Mango Markets and Audius exploits) and malicious proposals (where on-chain proposals execute attacker code, exemplified by the Beanstalk and Tornado Cash exploits). They suggest mitigations including higher voting thresholds, execution delays, improved voting mechanisms such as quadratic voting (which itself necessitates Sybil identification), and proposal-content analysis.

Liu et al. [74] conduct a literature review of DAOs and identify forking culture as a result of the original DAO hack. They highlight the infancy of blockchain laws, particularly given that blockchain-based systems become vulnerable when private off-chain interest groups force on-chain governance decisions [92].

The literature on DAO exploits is substantial for technical attacks but thinner for governance-layer risks. Where governance-layer risks have been studied, the unit of analysis is typically the individual attacker or the single proposal; the role of communities, both internal and external, in either enabling or defending against these attacks is largely unexplored.

## 3.5 Gaps in Knowledge

Chapter 1 (Section 1.2) introduces three gaps in the existing literature on DAO governance: detection, distortion, and division. In this section we tie each gap to the work reviewed above and to the auxiliary problem of identifying governance data on which the detection methods depend.

### 3.5.1 Identifying Governance Data

Smart contract identification is well studied for tokens, where ERC-20 conformance enables interface-based detection [49, 89] and bytecode-based detection catches non-conforming variants [34]. Smart contract code is widely reused [57, 50], and even functionally cloned contracts can be identified through symbolic execution [73]. These methods do not transfer straightforwardly to DAO governance contracts, which use heterogeneous frameworks [121] with configurable modules. OpenZeppelin templates [91] support some out-of-the-box deployments, but DAOs do not uniformly implement the same governance contracts. No prior work has produced a framework-agnostic heuristic for identifying governance contracts on Ethereum. Chapter 5 closes this gap by combining trace-data analysis, function- and log-signature matching, and similarity-based identification of known governance contracts.

### 3.5.2 Community Detection

Prior research analyses DAO governance behaviour at the address level, identifying governance whales [48], documenting centralisation [121], and characterising contributor influence [63]. Coalitions of co-voting participants have been identified within individual DAOs [106] and overlap in token holdings has been mapped across DAOs [98], but no

general method exists for inferring communities from observed behaviour or for identifying external communities through the smart contracts they use to participate. Delegation networks add a further layer of indirection that obfuscates true voting power and is itself a form of community [82, 125]. Methods from political-voting analysis [83] provide a template for inferring partisan communities from voting records, but have not been adapted and applied to pseudonymous DAO data. Adapting these methods to DAOs requires accounting for the unique dynamics within DAO governance, such as the pseudonymous nature of blockchain addresses, voter apathy resulting in sparse participation across governance proposals, voting power swings due to the ease with which delegations and token acquisitions can occur, and the lack of defined party labels. Chapters 4 and 5 close this gap by adapting dissimilarity-and-clustering methods for internal communities and by detecting external communities through the multi-signature contracts they use to vote.

### 3.5.3 Distorting Governance Mechanics

DAO governance mechanisms are heterogeneous: token voting is dominant [35], but variants such as quadratic voting, one-token-one-vote, weighted voting, and lockup-based voting all see use, and design choices materially affect DAO viability [95] and token performance [96]. These mechanisms are not isolated: public blockchains permit smart contract composition, which enables new protocols to build on top of existing governance contracts [99, 64]. Composition between governance contracts produces metagovernance [124], through which one DAO governs another. Examples of metagovernance protocols accumulating significant voting power are emerging [5], but the consequences of layered governance for the rules of engagement of the underlying DAO are not well understood. Chapter 6 closes this gap by analysing the Curve, Convex, and Votium ecosystem, where stringent lockup rules at the underlying tier coexist with shorter lockups and bribe markets at higher tiers.

### 3.5.4 Warning Signals of Division

Pseudonymity makes it difficult to anticipate the behaviour of blockchain addresses. Attempts to link off-chain forum identities to on-chain addresses through ENS-name matching [82, 4] provide some signal but require off-chain data that the majority of governing addresses do not generate. On-chain behavioural patterns can reveal strategic delays among whales [52] and changes in participation following airdrops [130], but these patterns indicate when an address will vote rather than how. Forking is well documented as an outcome [13, 94, 67, 56], but no prior work identifies the voting-behaviour patterns that precede a fork. Chapter 4 closes this gap by showing that addresses which later participate



in a fork increasingly cluster together in voting behaviour as the fork approaches, providing an observable warning signal of impending fragmentation.

### 3.5.5 Community-Level Analysis

Across the gaps identified in the above subsections, past DAO governance research has predominantly examined governance through individual addresses, including token concentration, governance whales, voter participation, delegation, token-holder overlap, and individual proposal outcomes. While these approaches provide important insights into the distribution and exercise of governance power within DAOs, they do not fully capture the collective and layered structures through which governance participation can occur.

This thesis takes a community-level approach, examining both patterns of collective behaviour between voting addresses and the governance structures represented by individual addresses. This approach identifies additional ways to interpret DAO governance data, with repercussions for past and future DAO analysis research. For example, historical voting behaviour can reveal groups of addresses that exhibit sustained voting alignment, potentially increasing centralisation scores compared to single-address analysis alone. Moreover, a single voting address may itself represent a collective actor, such as a DAO participating through a multi-signature contract. This moves the decision-making process outside of the governance smart contract and increases the number of addresses governing that decision. What a single-address analysis would record as one address casting a vote may, in fact, be the result of a collective decision from hundreds or thousands of addresses through tokenised metagovernance.

Importantly, the community-level approach provides an additional layer of analysis that contextualises the observations made at the address level. Governance is inherently a collective decision-making process, yet past DAO research analyses this process as actions performed by individual addresses. Analysing relationships between these actions can reveal patterns consistent with collective governance behaviour, such as sustained voting alignment, while examining the mechanisms behind voting addresses can reveal decision-making processes that occur outside the primary governance venue. Community-level analysis can therefore provide context for collective behaviour that is not apparent when governance addresses are considered independently.

## Chapter 4

# Internal DAO Communities

### 4.1 Introduction

In Section 3.2 we reviewed prior research on internal DAO dynamics and the mechanisms by which DAOs function. That literature documents a heterogeneous range of governance frameworks and voting models. Within these frameworks, studies have characterised the well-documented accrual of voting power by whales and, more recently, the lesser-understood formation of voter coalitions whose role in DAO outcomes has yet to be examined systematically.

Section 3.4 reviewed the catalogue of exploits that DAOs have suffered, and the failures that have followed. Technical exploits remain the most documented category, but tooling to mitigate them is improving rapidly. DAO failures rooted in social behaviour, by contrast, remain comparatively understudied. In this chapter, we examine these less-understood aspects: the internal communities within DAOs and the social dynamics that lead to governance failures. We pursue three objectives. First, we identify internal communities within DAOs from their governance records alone. Second, we examine the relationships between co-existing communities within the same DAO. Third, we ask whether community voting patterns provide an observable signal of impending DAO fragmentation. In doing so, this chapter contributes to RQ 1 by detecting internal communities from voting records, and primarily addresses RQ 3 by asking whether the voting patterns of those communities contain early-warning signals of fragmentation. It provides the internal-community strand of the thesis-level argument that DAO governance is shaped by communities rather than independent addresses.

We present a method for identifying and visualising partisan communities within DAOs by analysing on-chain voting behaviour. The method transforms blockchain governance data into spatial representations that reveal behavioural alignment patterns among participants. By examining the voting history of addresses across multiple proposals, we can

detect the emergence of cohesive sub-communities before they manifest as organisational splits.

The method consists of several stages. We extract voting events from on-chain governance smart contracts and construct voter matrices that encode participation and support patterns across all proposals. We apply pairwise dissimilarity analysis to quantify voting divergence between addresses based on their shared voting history. The resulting dissimilarity scores are projected into two dimensions using multidimensional scaling (MDS), creating spatial representations in which proximity indicates voting alignment. Finally, we apply  $k$ -means clustering with silhouette score optimisation to identify distinct partisan communities, and we validate the results against a randomised baseline that isolates genuine alignment from coincidental agreement.

We demonstrate the method on Nouns DAO, an NFT-based governance system on Ethereum. Section 4.3 describes Nouns DAO in more detail and explains why we chose it as the case study: its three documented forks provide a behavioural ground truth against which our clusters can be validated. Section 4.2 first applies an initial version of the method to UK parliamentary voting data, where party affiliation supplies a similar ground truth in a non-pseudonymous setting and serves as a sanity check on the dissimilarity-and-MDS pipeline before we apply it to pseudonymous blockchain data. Section 4.4 describes the steps required to transform raw blockchain data into a uniform voter matrix, compute pairwise dissimilarity between addresses, and identify and visualise clusters of aligned voter addresses. Section 4.5 discusses the results, and Section 4.6 concludes.

## 4.2 Preliminary Political Analysis

Detecting community alignment from voting records is a well-studied problem in legislative analysis. As reviewed in Section 3.2.3, Mickevicius et al. [83] analysed voting behaviour in the Lithuanian Parliament and found that multidimensional scaling visualised opposing parties almost perfectly, with cluster assignments matching political parties at 68% accuracy. The appeal of legislative data for this kind of analysis is that the governance record is part of the public parliamentary record, and party members are typically expected to vote in alignment with the party’s policy objectives, providing a known ground truth against which community-detection heuristics can be evaluated.

A clear example is the UK parliamentary system, where political parties formalise cohesive voting through the party whip: members are instructed on how to vote in each parliamentary proposal. The UK government also releases all governance data to the public. Because the major parties operate the whip system and the data is freely available, we used a subset of the voting records from the House of Commons to apply an initial

version of our heuristic to detect similarities between voters and visualise the result in an MDS graph.

Unlike the Nouns DAO study we present later in this chapter, which evaluates the complete internal-community identification framework, the UK parliamentary analysis described in this section applies only the pairwise dissimilarity measure and the MDS visualisation. This preliminary analysis serves as a sanity check demonstrating that meaningful visualisations of governance communities can be generated directly from voting history, where the known party affiliations of MPs provide a ground truth by which the resulting MDS visualisation can be assessed.

House of Commons governance data is released per proposal and includes each MP's name, party affiliation, constituency, vote, and proxy member. Our data collection began on 8 May 2024 and continued to 9 October 2024. We unified the proposals into a single voter matrix and determined voter alignment scores through pairwise dissimilarity comparisons of all voters; the method is described in detail in Section 4.4. Once the dissimilarity matrix had been computed, we projected the data into two dimensions using MDS. During the period of analysis the UK held a general election on 4 July 2024; consequently, the set of MPs eligible to participate in governance changed. We therefore generated MDS graphs both for the entirety of our data and for the post-election period alone. Figure 4.1 shows the post-election projection of MP voting alignment. Figure 4.2 projects dissimilarity across both epochs, pre-election and post-election.

In Figure 4.1, each vertex is coloured by its political-party affiliation, with proximity indicating greater voting alignment and separation indicating misalignment. Conservatives, Labour, and Liberal Democrats occupy distinct regions of the graph. This is consistent with our expectation that the whip system aligns voting behaviour within each party, and validates the dissimilarity and MDS pipeline on data with a known ground truth. The seemingly random placement of the *Other* category reflects its composition: a mix of smaller parties and independents whose voting behaviour is not tied to a single party line and may align with either the left-leaning or right-leaning blocs.

Figure 4.2 is harder to interpret. Conservative and Labour vertices each split into two distinct groups within the chart. This is an artefact of the way we determine dissimilarity for voters who have never participated in the same governance proposal: when there is no overlap between two voters' activity, we treat them as maximally dissimilar. Parliamentary data segments naturally into epochs between general elections, with limited cross-overs through by-elections, and MPs who sat only before the election therefore never share a proposal with those who sat only after. DAO governance has no equivalent boundary: voters may enter or exit the system at any time and without a formal process, so the same dissimilarity convention will produce noise rather than coherent groupings unless the active voters are filtered, which we do in Section 4.4.4.

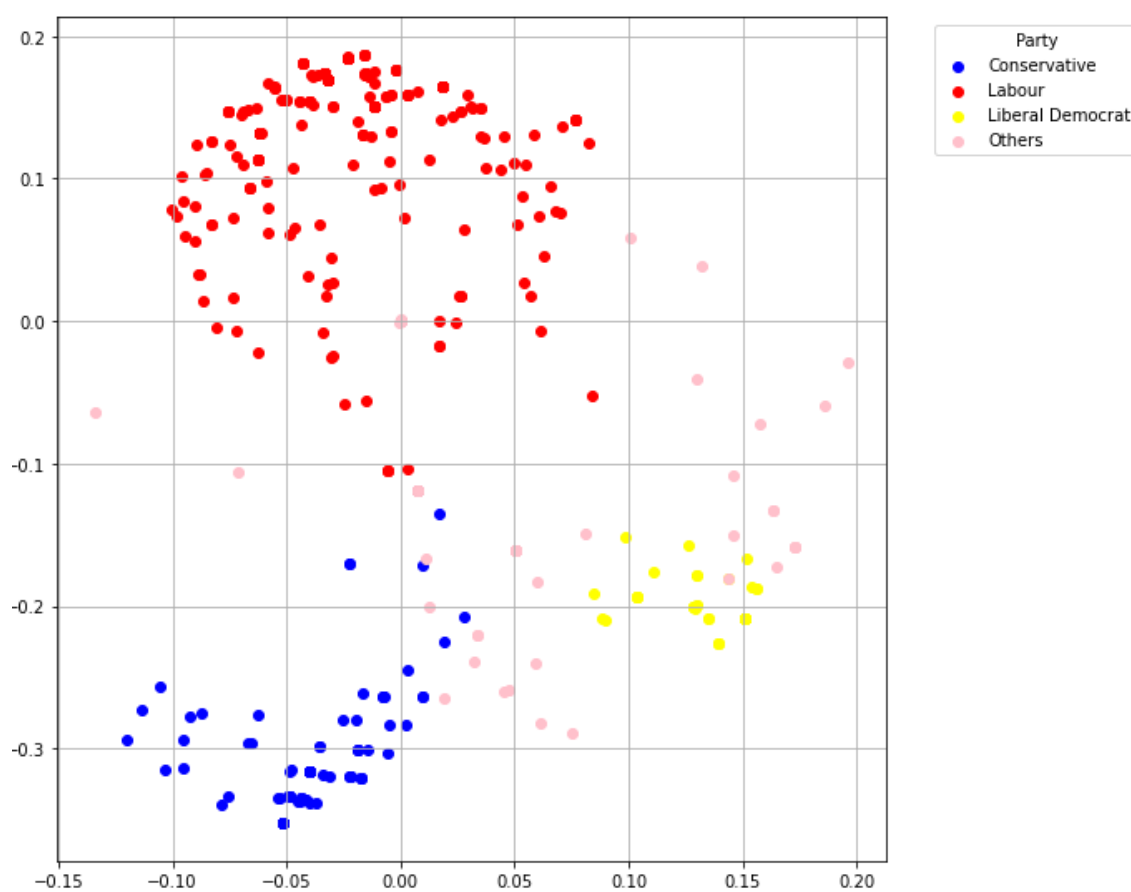


Fig. 4.1 Parliamentary MDS graph between proposals 1827 and 1844. Each vertex is a Member of Parliament, coloured by political party affiliation. The distance between vertices reflects dissimilarity in voting behaviour: greater distance indicates greater dissimilarity.

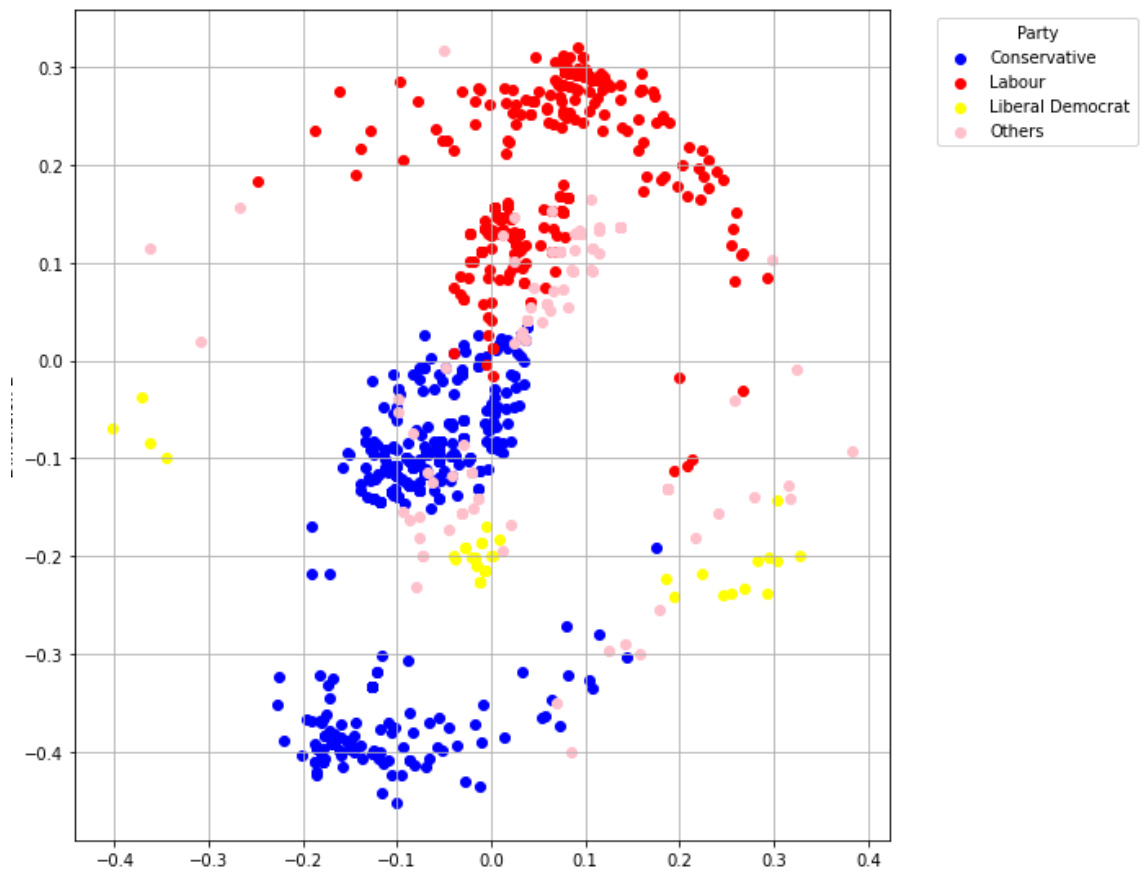


Fig. 4.2 Parliamentary MDS graph across all proposals. The distinct grouping of same-party vertices reveals two separate epochs, before and after the election.

The preliminary analysis confirms that our approach can visualise partisan communities in a setting with a known ground truth. Translating the method to DAO governance introduces challenges that are absent in parliamentary data. The chief difficulty is the inability to verify community detection directly: blockchain addresses are pseudonymous, and no equivalent of party affiliation is available against which clusters can be validated. We therefore restrict our subsequent analysis to a single DAO that provides a behavioural ground truth. We selected Nouns DAO as our test case because it has previously undergone DAO forks, in which a subset of the membership separated from the main DAO to create a new instance. The forks identify, after the fact, which addresses left, providing a proxy for partisan-community membership.

## 4.3 Nouns DAO

Nouns DAO describes itself as an experimental attempt to improve the formation of on-chain avatar communities. Noun avatars are represented as on-chain NFTs, each with a distinct combination of visual traits. The DAO generates revenue by auctioning a new Noun NFT every 24 hours on Ethereum: bidders pay in ether, and after 24 hours the highest bidder receives the NFT and the proceeds are transferred to the Nouns DAO treasury.

The treasury is operated by the DAO itself. Nouns implements a fork of the Compound governance framework (Section 2.6.1, Figure 2.1), with governance taking place on-chain and voting rights granted to Nouns NFT holders at one vote per Noun. The DAO regularly holds proposals on allocations from the treasury, and has funded a wide range of projects, from staking positions in Lido Finance to a global pizza party costing roughly USD\$42 000. Disagreement surrounding this kind of spending appears to have contributed to a group within the DAO initiating the first Nouns DAO fork [67].

The first fork saw 472 Nouns exit the original DAO to form a new instance, with the treasury split proportionally and the corresponding share of ether transferred to the new DAO. The forked DAO did not rebuild a working community; instead, a new *rage quit* mechanism within it allowed Noun owners to liquidate their NFTs for a proportional share of the new treasury. In effect, this gave those who forked a route to exit the Nouns ecosystem entirely. Nouns DAO has since undergone two further forks.

We selected Nouns DAO as the case study for mapping partisan communities for two reasons. First, forking a DAO is an extreme form of conflict resolution, and its presence in Nouns DAO is itself indicative of partisan divisions within the membership. Second, the forks provide a ground truth: they identify, after the fact, which addresses ultimately separated from the DAO, allowing us to examine the relationship between an address's community alignment and its future likelihood of joining a fork.

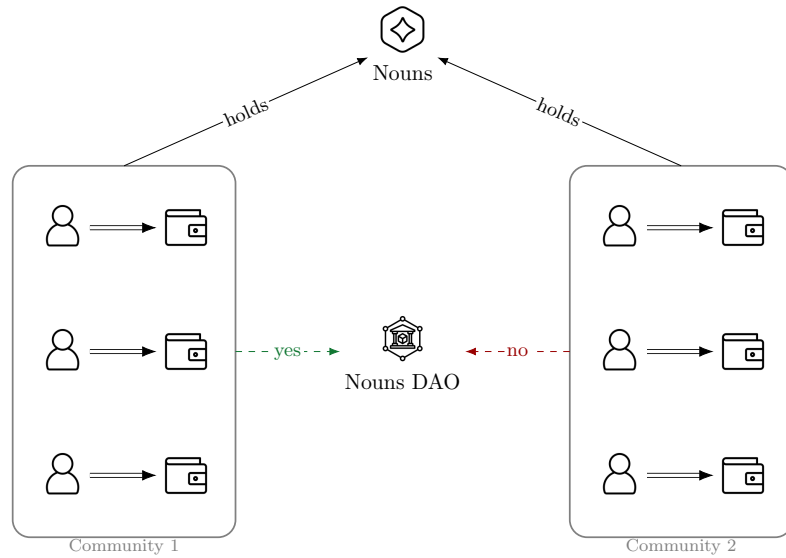


Fig. 4.3 Internal community governance in Nouns DAO. Two communities hold the Nouns token and vote in opposing directions in Nouns DAO. Community 1 exercises yes votes and Community 2 exercises no votes.

Figure 4.3 illustrates the Nouns DAO governance scenario using the visual grammar introduced in Section 2.9. Two communities of actors each hold the Nouns governance token and vote in Nouns DAO. Community 1 exercises yes votes and Community 2 exercises no votes. The diagram corresponds to the community-level analysis described in Section 2.9.2: detecting these opposing communities from on-chain voting data *before* the fork occurs is the goal of the method presented in this chapter.

## 4.4 Method

To identify partisan communities within DAOs, we use a multi-stage method that bridges on-chain data analysis with spatial and statistical validation. The following subsections describe each stage in turn (Sections 4.4.1–4.4.7).

The method begins with *blockchain data acquisition*: we gather raw transaction logs and event data from EVM-based DAOs through RPC calls. The data is structured into a *voter matrix* that encodes voting behaviour while accounting for abstentions and non-participation. We then perform a *community friction assessment* to quantify overall voter discord across proposals. This acts as a sieve, prioritising DAOs with significant partisan dynamics over those with consistently unanimous voting. Next we identify *active voters* by selecting addresses with a sufficient level of participation within a sliding proposal window; this limits subsequent analysis to engaged participants and reduces noise from sporadic voters. The core of the method lies in *pairwise dissimilarity computation*, where



| Voter Address                              | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 |
|--------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 0xe0e7ee036a7ed4a7699f8e137b5de5eff15e1801 | 1   | 1   | 0   | 1   | 1   | 1   | 1   | 1   | 1   |
| 0x1c85a8cd74bca05a993d62a42fd229a8495064ad | -1  | 1   | -1  | -1  | -1  | -1  | -1  | -1  | -1  |
| 0x4bc5bff1de9bf2d1bbcf0d0c5b128524b77e94ea | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  |
| 0x83782a5ec557be21ac356e3673e2fd5897ea8fad | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  |
| 0xb8f9f5a98ae392063b3ab0563c568a68e82eb5ad | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  | -1  |
| 0x70ece45cdc64a768f936ccb8ac851578251363e3 | -1  | 1   | 0   | -1  | -1  | 1   | 1   | 0   | 1   |
| 0x8becd7ebf3f910090f9264dfd0bb221ea04af8c3 | 1   | 1   | 0   | 1   | 1   | 1   | 1   | -1  | 1   |
| 0x0c66d954d9cb1ebdf4e37a000262323c83655e70 | 1   | 1   | -1  | -1  | 1   | 1   | 1   | -1  | 1   |
| 0x86030dbbce1c771ff6622c20455cd3619aa93c05 | 1   | 0   | 0   | 1   | -1  | 1   | 1   | 1   | 1   |
| 0xb8a9ed8d008fa1f161809f15a11f4161fb8b717e | 1   | 0   | 0   | 1   | 0   | 1   | 1   | 1   | 1   |

Fig. 4.4 A sample from the voter matrix for Nouns DAO with ten voting addresses (the rows) and nine proposals (the columns). The cell values represent either “Yes” votes (1s in green), “No” votes (0s in red), or all other cases (−1s in grey).

we measure voting divergence between addresses by comparing their shared voting history. This produces a dissimilarity matrix per proposal, capturing how often two voters opposed each other relative to their co-participation. We visualise these matrices using *multidimensional scaling* (MDS), mapping addresses into 2D space where proximity reflects voting alignment. Finally, we identify partisan communities through *dynamic clustering*, selecting cluster counts via *silhouette scoring* and testing robustness by comparing genuine voting data against randomised voting patterns.

The method is governance-framework agnostic and can be applied broadly across DAOs. We explain it using Nouns DAO as a running example: voting power in Nouns DAO is derived from NFT ownership, and its documented history of forks provides a ground truth against which we can compare our results.

#### 4.4.1 Blockchain Data Acquisition

We collected on-chain transaction and event logs from EVM blockchains (Ethereum and Arbitrum) through RPC calls, using a self-hosted Ethereum node and Infura as described in Section 2.3.2. We identified governance smart contracts for six DAOs: Lido, Compound, Tornado Cash, Nouns, Uniswap, and Arbitrum. For each DAO, we determined the relevant event signature by analysing historical transactions and filtering on the voting events emitted by the governance contract. The events contain proposal identifiers, voter addresses, and the direction of each vote (support, opposition, or abstention). We collected event logs across the entire operational history of each DAO, from the block in which the governance contract was deployed up to and including block 22 575 000 (May 2025). The Nouns DAO governance contract<sup>1</sup> was deployed in block 12985453 (August 2021).

### 4.4.2 Voter Matrix Construction

After parsing, each vote log entry contained the voting address, the proposal ID, and a support value. We transformed this data into a voter matrix, illustrated in Figure 4.4, in which rows correspond to voter addresses, columns to proposal IDs, and entries denote support: 1 for “Yes”, 0 for “No”, and  $-1$  for all other cases. The raw data distinguishes five states: 1 (Yes), 0 (No),  $-1$  (no voting power and no vote), 2 (had voting power and abstained), and 99 (had voting power but did not vote). We subsumed the latter three states ( $-1$ , 2, and 99) into a single  $-1$  value to simplify the analysis; each represents an absence of a yes or no vote, and the binary distinction is what our subsequent analysis requires. Determining voting eligibility precisely is non-trivial: an address might have held tokens at the time of a proposal but only become eligible retroactively due to delayed vesting schedules, delegation mechanics, or other temporal constraints, and accurate eligibility tagging requires checking historical token balances at specific block heights. To construct the per-proposal voter lists used in later stages, we extracted addresses with non-negative support values (0 or 1) for each proposal. Although DAOs often use a support value of 2 to denote abstentions that contribute to quorum, we adopted a binary classification (for/against) so that the voter matrix aligns with standard voting-analysis frameworks.

More formally, if  $A = \{a_1, a_2, \dots, a_n\}$  is the set of addresses that participated in a DAO and  $P = \{p_1, p_2, \dots, p_m\}$  is the set of proposals voted on by the DAO, then the voter matrix is defined such that  $v_{ij} \in \{0, 1, -1\}$  indicates the action taken by address  $a_i$  on proposal  $p_j$ . The matrix  $V$  has  $n$  rows and  $m$  columns:

$$V = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1m} \\ v_{21} & v_{22} & \cdots & v_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nm} \end{bmatrix}$$

For Nouns DAO, we constructed a matrix with 629 unique voter addresses (rows) and 330 proposals (columns), spanning the period from the deployment of the governance smart contract to the first Nouns DAO fork in block 18 144 239 (September 2023). Proposal IDs were numbered from 1 to 362 inclusive; we excluded 32 cancelled proposals that had no voting activity.

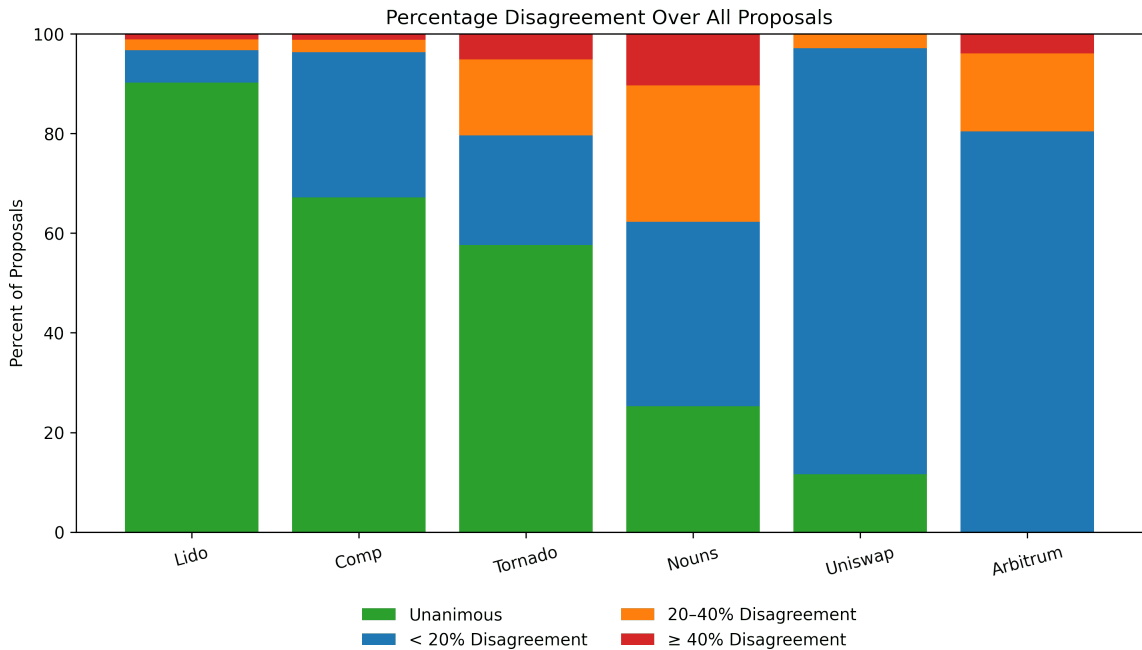


Fig. 4.5 A percent stacked bar chart showing disagreement across all proposals for six DAOs at the voter address level. Lido DAO voters are unanimous (0% disagreement) or exhibit low disagreement (in the  $(0, 20)\%$  range) for almost all of their proposals, whereas Nouns DAO voters exhibit medium disagreement (in the  $[20, 40)\%$  range) and high disagreement (in the  $[40, 50]\%$  range) for over 30% of their proposals.

### 4.4.3 Community Friction Assessment

To prioritise DAOs with meaningful partisan dynamics, we quantified community-level discord using two metrics derived directly from the voter matrix (Section 4.4.2): static disagreement percentages per proposal and rolling average disagreement percentages over sequential proposals.

For static disagreement analysis, we categorise proposals by the proportion of voters opposing the winning outcome. The thresholds are unanimous (0% disagreement), low  $((0, 20)\%)$ , medium  $([20, 40)\%)$ , and high  $([40, 50]\%)$ . These align with standard benchmarks for social consensus in collective decision-making studies. As shown in Figure 4.5, Nouns DAO exhibits medium-to-high disagreement in 30% of proposals, a stark contrast to Lido DAO’s near-unanimous voting patterns. This quantitative disparity motivates our selection of Nouns DAO as a high-friction case study, particularly given its documented fork in September 2023.

The rolling average metric (Figure 4.6) reveals temporal trends in partisan behaviour. By calculating the mean disagreement percentage across a 10-proposal window (moving forward one proposal at a time), we detect persistent voting divisions in Nouns DAO,

<sup>1</sup>0x6f3E6272A167e8AcCb32072d08E0957F9c79223d

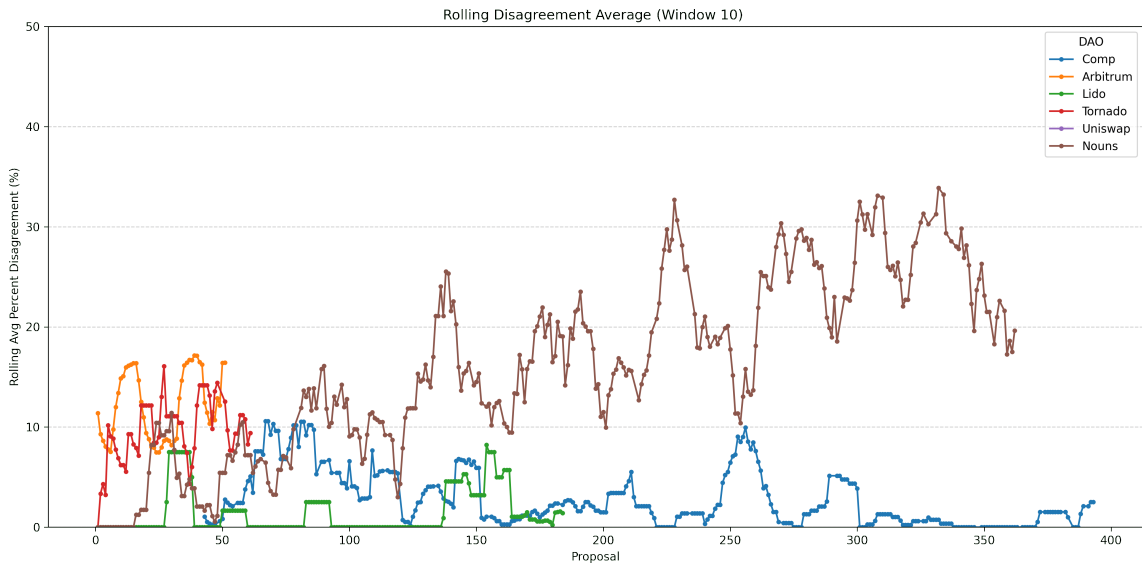


Fig. 4.6 A plot of the rolling average of disagreement for six DAOs using a 10-proposal sliding window. The  $x$ -axis contains the proposal IDs, which serve as a proxy for time. The  $y$ -axis shows the level of disagreement. The unanimity within Lido DAO and the conflict within Nouns DAO leading up to their first fork are apparent.

where disagreement spikes and sustained volatility preceded its fork. This metric serves two purposes: (1) identifying DAOs with persistent rather than transient conflict, and (2) pinpointing temporal clusters of contentious governance that warrant further analysis (see Section 4.4.7).

These friction metrics inform our subsequent dynamic clustering analysis. DAOs with medium/high disagreement exceeding 20% (Figure 4.5) and rolling averages exceeding 15% (Figure 4.6) are flagged for further analysis, while low-friction DAOs (e.g., Lido) are excluded. This filtering ensures our clustering experiments (Section 4.4.7) focus on cases where conflict is evident.

#### 4.4.4 Active Voter Identification

To address the sparsity of the voter matrix caused by fluctuating participation in DAOs, we compute a voter participation list. This distinguishes DAO voting from parliamentary voting, where persistent cohorts dominate. In DAOs, voting eligibility depends on dynamic factors such as token ownership, delegation, and time-based vesting, leading to variable active-voter sets across proposals. As a result, many addresses have minimal or no voting activity, introducing noise into community detection.

Our method calculates participation frequency using a sliding proposal window. For each proposal, we define a window containing the proposal and a fixed number of preceding proposals. For Nouns DAO, we used a window of size 10. We chose a sliding window of

10 as it captures a specific moment in time within the DAO. It provides enough historical proposal context where community formation can be observed, while also responding to changes once communities have formed. Larger sliding windows would delay observations of the community evolution, both in alignment and misalignment. Therefore, for proposal #20, the window included proposals 11 to 20 inclusive. For each voter address in each window, we compute their participation percentage:

$$\text{Participation \%} = \frac{\# \text{ of proposals in which the address voted}}{\# \text{ of proposals in the window}}$$

Addresses with a participation percentage below a fixed threshold (40% for Nouns DAO) were dropped from the filtered voter matrix. We chose 40% as this ensures that voters had participated in a sufficient number of proposals for the later pairwise dissimilarity comparison. This allows for enough overlap for meaningful comparisons and reduces noise of pairs of voters who have never voted with one another, while still allowing cases of no shared voting history to be represented in the data. This ensures subsequent analysis focuses on voters with sustained engagement. Early proposals (proposals #1–#9 for Nouns DAO) used shorter windows (e.g., the window for proposal #2 included only proposals #1 and #2), increasing voter counts until sufficient historical data was accumulated.

More formally, let  $w \in \mathbb{N}$  denote the proposal window size and  $\tau \in [0, 1]$  the participation threshold. For each proposal  $p_j \in P$ , we define its sliding window  $W_j \subseteq P$  as:

$$W_j = \{p_{\max(1, j-w+1)}, p_{\max(1, j-w+2)}, \dots, p_j\}.$$

For each address  $a_i \in A$ , we define its participation percentage  $\pi_{ij}$  in  $W_j$  as:

$$\pi_{ij} = \frac{1}{|W_j|} \sum_{p_k \in W_j} \mathbf{1}_{\{v_{ik} \geq 0\}},$$

where  $\mathbf{1}$  is the indicator function. We use  $|W_j|$  rather than  $w$  in the denominator so that the shorter windows used at the start of the dataset (e.g.  $|W_1| = 1$ ) are normalised correctly. The filtered voter matrix  $V'$  retains only rows  $a_i$  with  $\pi_{ij} \geq \tau$  and the columns corresponding to the proposals in  $W_j$ .

The filtered voter matrix concentrates the analysis on the addresses with sustained engagement, for which partisan alignment is meaningful to measure, and serves as the input to the pairwise dissimilarity computation (Section 4.4.5).

#### 4.4.5 Pairwise Dissimilarity Computation

The core of the method is the pairwise dissimilarity computation, which measures the voting divergence between voter addresses. It quantifies how often two addresses vote in opposition relative to their shared participation across proposals, producing a dissimilarity matrix per proposal that captures the voting alignment within the DAO. We have chosen vote-based pairwise dissimilarity as the metric to quantify voting dissimilarity between addresses based on their shared voting history, as it normalises disagreement by the number of proposals on which each pair of addresses cast valid votes.

For each proposal  $p_j \in P$ , we compute pairwise dissimilarity scores between all active voter addresses within the proposal's sliding window  $W_j$  (as defined in Section 4.4.4). Let  $A_j \subseteq A$  denote the set of active addresses for proposal  $p_j$ , filtered by the participation threshold  $\tau$ . For any two addresses  $a_i, a_k \in A_j$ , their dissimilarity score  $d_{ik}^j$  is calculated as:

$$d_{ik}^j = \begin{cases} \frac{\sum_{p_\ell \in W_j} \mathbf{1}_{\{v_{i\ell}, v_{k\ell} \geq 0 \wedge v_{i\ell} \neq v_{k\ell}\}}}{\sum_{p_\ell \in W_j} \mathbf{1}_{\{v_{i\ell}, v_{k\ell} \geq 0\}}} & \text{if } \exists p_\ell \in W_j : v_{i\ell}, v_{k\ell} \geq 0 \\ 1 & \text{otherwise} \end{cases}$$

The numerator counts the number of proposals within  $W_j$  where both addresses cast valid votes (i.e.,  $v_{i\ell}, v_{k\ell} \geq 0$ ) but voted in opposition to each other. The denominator represents the total number of proposals where both addresses participated with valid votes. When two addresses have no overlapping voting history within the window, we assign the maximum dissimilarity score of 1, as pairs with no shared votes cannot show voter alignment; this occurred in less than 0.6% of all pairs. This formulation ensures that dissimilarity scores are normalised to the range  $[0, 1]$ , where 0 indicates complete alignment (addresses always vote together when both participate) and 1 indicates complete opposition (addresses never agree when both participate).

Algorithm 1 outlines the computational procedure. For each proposal  $p_j$ , we construct the sliding window  $W_j$  and identify the set of active addresses  $A_j$  that meet the participation threshold. We then extract the relevant submatrix  $V'$  from the voter matrix  $V$  and compute pairwise dissimilarity scores.

The resulting dissimilarity matrix  $D^j$  for proposal  $p_j$  is an  $n \times n$  symmetric matrix where  $n = |A_j|$ . Each entry  $d_{ik}^j$  represents the dissimilarity between addresses  $a_i$  and  $a_k$ , with the diagonal entries being zero (complete self-similarity). Figure 4.7 shows a sample from the dissimilarity matrix for Nouns DAO proposal #300.

For Nouns DAO, this process generated 329 dissimilarity matrices (one per proposal with valid voting data), each capturing the voting alignment structure within the DAO at that specific point in time. These matrices serve as the foundation for subsequent

**Algorithm 1** COMPUTEDISSIMILARITY generates pairwise dissimilarity matrices for all proposals.

---

```

1: function COMPUTEDISSIMILARITY( $V, A, w, \tau$ )
2:    $R \leftarrow \{\}$  ▷ Dictionary to store results
3:   for all  $p_j \in P$  do
4:      $W_j \leftarrow \{p_{\max(1, j-w+1)}, \dots, p_j\}$  ▷ Sliding window
5:      $A_j \leftarrow \{a_i \in A : \pi_{ij} \geq \tau\}$  ▷ Active addresses
6:      $V' \leftarrow V[A_j, W_j]$  ▷ Filtered voter submatrix
7:      $n \leftarrow |A_j|$ 
8:      $D^j \leftarrow \mathbf{0}_{n \times n}$  ▷ Initialise dissimilarity matrix
9:     for  $i \leftarrow 1$  to  $n - 1$  do
10:      for  $k \leftarrow i + 1$  to  $n$  do
11:         $\text{valid} \leftarrow \{p_\ell \in W_j : v_{i\ell}, v_{k\ell} \geq 0\}$ 
12:         $\text{shared} \leftarrow |\text{valid}|$ 
13:        if  $\text{shared} > 0$  then
14:           $\text{opposing} \leftarrow |\{p_\ell \in \text{valid} : v_{i\ell} \neq v_{k\ell}\}|$ 
15:           $d_{ik}^j \leftarrow \text{opposing}/\text{shared}$ 
16:        else
17:           $d_{ik}^j \leftarrow 1$  ▷ No shared voting history
18:        end if
19:           $d_{ki}^j \leftarrow d_{ik}^j$  ▷ Symmetric assignment
20:      end for
21:    end for
22:     $R[p_j] \leftarrow \{\text{addresses} : A_j, \text{matrix} : D^j\}$ 
23:  end for
24:  return  $R$ 
25: end function

```

---

| Address                                     | 0x008    | 0x91d | 0xe08    | 0x475    | 0xc9eb | 0xe8d    | 0xcc2    | 0xa86 | 0xf6b | 0xdcdb |
|---------------------------------------------|----------|-------|----------|----------|--------|----------|----------|-------|-------|--------|
| 0x008c84421da5527f462886cec43d2717b686a7e4  | 0        |       |          |          |        |          |          |       |       |        |
| 0x91dcaa260cc4616e1a6e6b693db7207c5e42937   | 1        | 0     |          |          |        |          |          |       |       |        |
| 0xe085327c5ad2f77147f10973fed45fb19d734f7e  | 0.285714 | 0     | 0        |          |        |          |          |       |       |        |
| 0x4754b7e3dede42d71d6c92978f25f306176ec7e9  | 0.4      | 0.5   | 0        | 0        |        |          |          |       |       |        |
| 0xc9eb78abfad166c11e350156fd671b385265a6c87 | 0        | 0     | 0        | 1        | 0      |          |          |       |       |        |
| 0xe8d24be3abf6eb27b800f752a969d3c1e84fd401  | 0.6      | 0     | 0        | 1        | 0      | 0        |          |       |       |        |
| 0xcc2688350d29623e2a0844cc8885f9050f0f6ed5  | 0.5      | 0     | 0        | 0.666667 | 1      | 0        | 0        |       |       |        |
| 0xa86882277e69fbf0a51805cdc8b0a3a113079e63  | 0.333333 | 0.25  | 0.166667 | 0.4      | 0      | 0        | 0        | 0     |       |        |
| 0xf6b6f07862a02c85628b3a9688beae07fea9c863  | 1        | 0     | 0.2      | 0.333333 | 1      | 0.5      | 0.333333 | 0.25  | 0     |        |
| 0xdcdb4117e3a00632efcac3c169e0b23959f555e5e | 0.571429 | 0.5   | 0.142857 | 0.6      | 0      | 0.333333 | 0.25     | 0     | 0.6   | 0      |

Fig. 4.7 A sample from the dissimilarity matrix for Nouns DAO proposal #300. The rows are the addresses and the columns are the same addresses (abbreviated). A score of 1 represents complete dissimilarity; a score of 0 represents complete similarity.

multidimensional scaling visualisation (Section 4.4.6) and dynamic clustering analysis (Section 4.4.7).

#### 4.4.6 Spatial Visualisation via MDS

To visualise the alignment structure captured by our dissimilarity matrices, we use multidimensional scaling (MDS) to project the high-dimensional voting relationships into a two-dimensional space. MDS finds an embedding in which the Euclidean distances approximate the pairwise dissimilarities, so addresses with similar voting behaviour appear close together. As demonstrated in Section 4.2, MDS can be used to visualise political alignments. Therefore, we apply this method to the pairwise dissimilarity matrix to identify and visualise alignment between voting addresses.

For each proposal  $p_j$  with dissimilarity matrix  $D^j$ , we apply classical MDS to obtain a 2D embedding  $\mathbf{X}^j \in \mathbb{R}^{n \times 2}$  where each row  $\mathbf{x}_i^j$  represents the coordinates of address  $a_i$  in the visualisation space. The MDS algorithm minimises the stress function:

$$\text{Stress} = \sqrt{\frac{\sum_{i < k} (d_{ik}^j - \|\mathbf{x}_i^j - \mathbf{x}_k^j\|_2)^2}{\sum_{i < k} (d_{ik}^j)^2}}$$

where  $\|\cdot\|_2$  denotes the Euclidean distance in the 2D embedding space. This ensures that addresses with low dissimilarity scores appear close together in the visualisation, while highly dissimilar addresses are positioned far apart.

To maintain temporal continuity across proposals, we use the embedding coordinates from proposal  $p_{j-1}$  as initialisation for the MDS algorithm at proposal  $p_j$ . This prevents arbitrary rotations and reflections between consecutive visualisations, enabling coherent tracking of voter movement over time. We performed MDS using the `scikit-learn` library with the following parameters: metric MDS with Euclidean distance, maximum of 300 iterations, and convergence tolerance of  $10^{-6}$ . The embedding positions  $\mathbf{X}^j$  were stored for subsequent clustering analysis (Section 4.4.7).

Nouns DAO provides a natural ground truth for validating our visualisation approach. The DAO has undergone three forks since deployment, each initiated when a subset of NFT holders (Noun owners) signalled their intent to establish a new DAO instance via on-chain transactions. We identified fork participants by analysing NFT transfers to the new fork smart contracts, creating a binary classification of addresses as either “forkers” or “stayers”.

Figure 4.8 shows the MDS visualisation for proposal #334, where red vertices represent addresses that participated in a subsequent fork and blue vertices represent those that remained. The spatial clustering of the vertices shows that our dissimilarity-based approach captures behavioural divisions that manifest in real-world governance actions. This ground



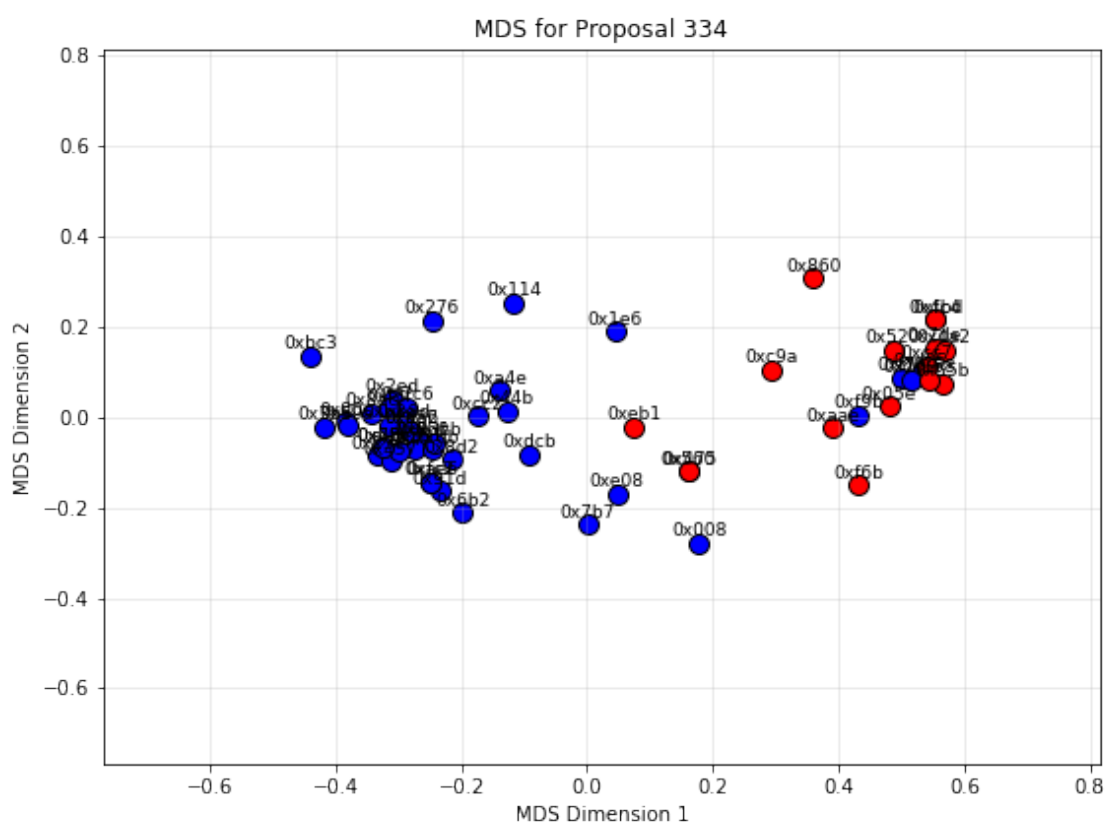


Fig. 4.8 MDS visualisation for Nouns DAO proposal #334 with ground-truth colouring. Each vertex represents a voter address, with positions determined by pairwise dissimilarity scores. Red vertices indicate addresses that participated in a subsequent fork, while blue vertices represent addresses that remained in the original DAO.

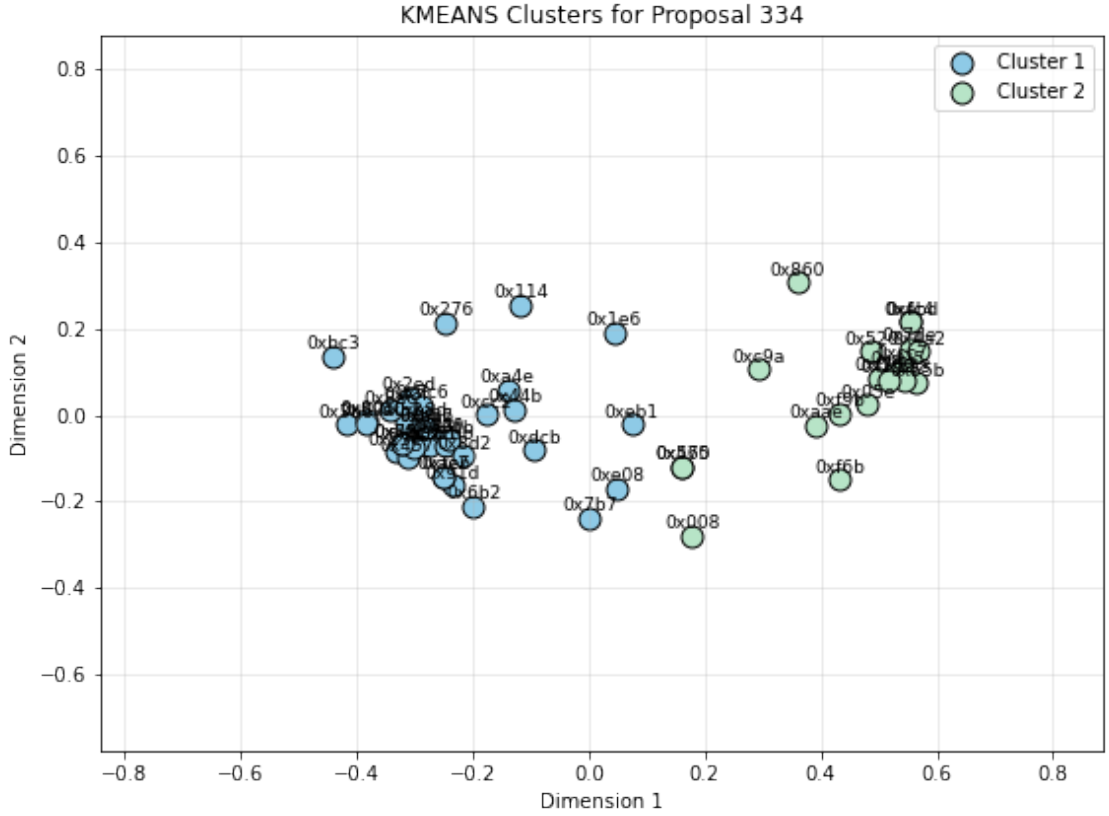


Fig. 4.9 MDS visualisation for Nouns DAO proposal #334 with  $k$ -means clustering. Note the overlap with the ground-truth clusters in Figure 4.8.

truth demonstrates that MDS translates abstract voting patterns into interpretable spatial relationships that can be used for automated partisan community detection.

#### 4.4.7 Dynamic Cluster Analysis

Once each proposal has an MDS embedding, we cluster the embeddings to identify partisan groups. This stage transforms the spatial patterns visible in our visualisations into quantifiable community structures that can be analysed and validated. We chose  $k$ -means clustering as it groups addresses that are in close proximity within the MDS visualisation, identifying communities with similar voting behaviours and allowing us to quantify changes within communities over time.

For each proposal  $p_j$ , we perform  $k$ -means clustering on the 2D coordinates  $\mathbf{X}^j$  obtained using MDS (see Section 4.4.6). Since the number of active voters varies across proposals and partisan structures evolve over time, we dynamically determine the optimal number of clusters rather than use a fixed value. We use the *silhouette score* to select the optimal cluster count  $k^*$  for each proposal. The silhouette score  $s(i)$  for voter  $i$  is calculated as:

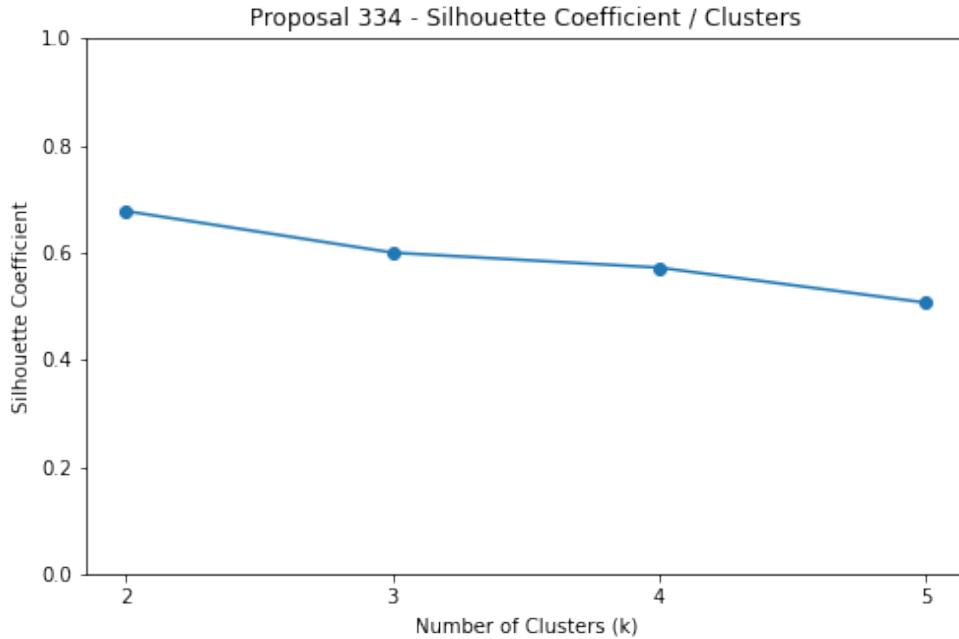


Fig. 4.10 Silhouette coefficient scores for cluster counts  $k = 2$  to  $k = 5$  on Nouns DAO proposal #334.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where  $a(i)$  is the average distance from voter  $i$  to all other voters in the same cluster, and  $b(i)$  is the average distance from voter  $i$  to all voters in the nearest neighbouring cluster. The overall silhouette score for a clustering configuration is the mean of the individual silhouette scores across all voters. We implemented silhouette scoring to allow for the dynamic nature of community structures. This allowed us to determine the most appropriate number of communities within each proposal window, enabling us to observe when voters consolidated into a small number of cohesive groups or fragmented across multiple communities.

We evaluate cluster counts from  $k = 2$  to  $k = 5$  inclusive and select the configuration with the highest silhouette score, as shown in Figure 4.10. We restrict the search to this range because partisan structures in DAOs typically present as a small number of aligned groups rather than as highly fragmented communities. In supplementary tests up to  $k = 10$ , 22% of clusterings exceeded five groups overall, but fewer than 3% did so after proposal #319. Hence, restricting to  $k \leq 5$  has a negligible impact on detecting partisan structures.

As a preliminary validation, we can use the ground truth provided by Nouns DAO’s documented forks. Figure 4.9 shows the clustering results for proposal #334, where  $k^* = 2$ . The clusters closely match the ground truth division shown in Figure 4.8, with forking addresses predominantly clustered together and distinct from addresses that remained in the original DAO.

#### 4.4.8 Limitations

The methodology we presented in this chapter includes several design choices across multiple steps that may influence the resulting community structures. Our choices of a sliding window of 10 proposals, a voter threshold of 40%, vote-based pairwise dissimilarity, MDS,  $k$ -means clustering, and silhouette scoring over cluster counts up to 5 were based on their appropriateness in the context of this study. However, alternative parameter values and clustering approaches were not evaluated in this research.

This is a limitation of this study. In future research, we will perform a sensitivity analysis to evaluate the robustness of the communities detected in the study and implement alternative methods for determining distance and clustering the resulting data.

## 4.5 Results

Our analysis of Nouns DAO governance data spans from the deployment of its governance contract at block 12985453 (8 August 2021) through proposal #362 at block 18050498 (2 September 2023). This time frame captures the complete pre-fork governance history, providing insight into how partisan communities emerged and solidified before the DAO’s fragmentation.

### 4.5.1 Visualisation of Community Structure

The MDS visualisation in Figure 4.8 shows a spatial representation for proposal #334, which we selected for detailed analysis due to its high concentration of participating fork addresses. The graph maps 2D positions of governing addresses based on their voting alignment, where proximity indicates similar voting behaviour and distance reflects increasing voting dissimilarity.

A clear bimodal structure emerges, with two distinct clusters separated along the horizontal axis. The rightmost cluster contains addresses that subsequently participated in one of Nouns DAO’s three forks (in red), while the leftmost cluster contains addresses that remained in the original DAO (in blue). This spatial segregation demonstrates that our dissimilarity-based approach captures the underlying meaningful differences in observed

voting behaviour that align with the eventual fragmentation of the DAO. The left-right clustering pattern persists across all subsequent proposals, suggesting that once partisan identities formed, they remained through the pre-fork period.

## 4.5.2 Quantitative Clustering Analysis

Applying *k*-means clustering with silhouette score optimisation to the MDS coordinates confirms our visual observations. For proposal #334, the algorithm identified two as the optimal number of clusters, in agreement with the binary fork/no-fork ground truth. Fourteen of the fifteen fork addresses were assigned to one cluster, with only a single fork address placed in the other. The clustering algorithm, operating solely on geometric proximity without knowledge of future fork participation, reconstructed the partisan divide that would later lead to the DAO's fragmentation.

## 4.5.3 Temporal Evolution of Partisan Behaviour

Our analysis shows a shift in fork addresses' governance engagement over time. During the early period (proposals #1 to #256), fork addresses participated sporadically, with an average of only 1.86 addresses voting per proposal. From proposal #257 (22 March 2023) onwards, their participation increased. In the final 91 proposals of our dataset, at least four fork addresses participated in every vote, with average participation jumping to 8.27 addresses per proposal.

This suggests that addresses that would later fork the DAO increased their governance involvement as observable divisions deepened. While non-forking addresses also increased their participation (from 28.86 to 39.93 average participants), the proportional increase was much smaller, indicating that engagement among the addresses that would later fork increased disproportionately as the fork approached.

## 4.5.4 Consolidation into Partisan Communities

Figure 4.11 shows the temporal consolidation of fork addresses into a unified partisan community. The stacked area chart tracks cluster assignments across proposals #257 through #362, revealing how initially fragmented fork addresses gradually coalesced into a single, dominant cluster as the DAO approached its breaking point. This process shows that partisan communities do not emerge suddenly but form through sustained governance participation. Fork addresses become increasingly unified in their cluster assignment, suggesting that their observed voting behaviour became progressively more aligned over time through repeated participation in governance.

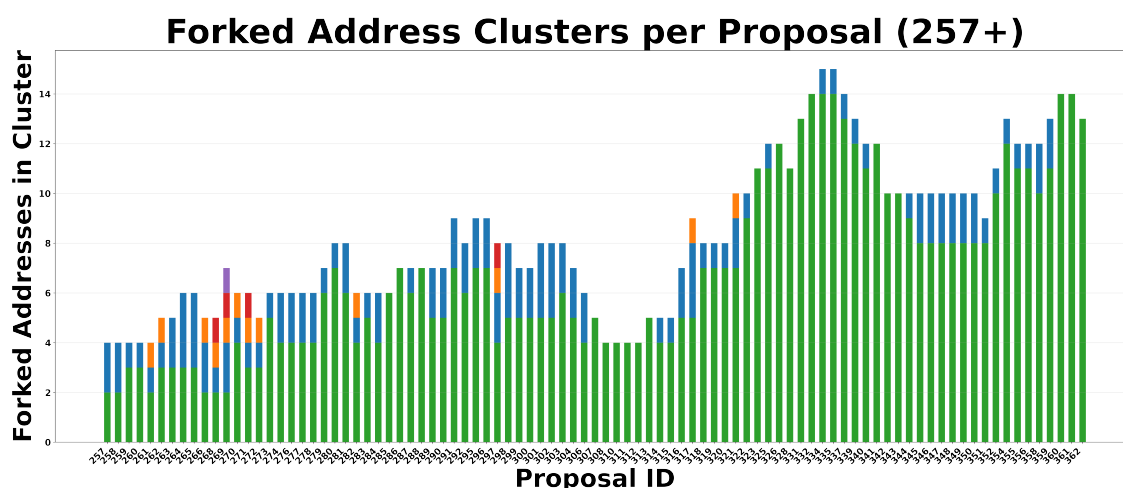


Fig. 4.11 A stacked area chart showing the assignment of forked addresses in Nouns DAO to  $k$ -means clusters across proposals #257 through #362. The addresses gradually coalesce into a single cluster as the fork approached.

### 4.5.5 Validation Against Randomised Data

Table 4.1 compares genuine voting data with 100 randomised iterations in which participants and outcomes were preserved but vote assignments shuffled (seeds 0–99), isolating the effect of genuine alignment from random behaviour.

The results are as follows. First, genuine voter data produces fewer clusters per proposal (2.77 average) than randomised data (3.36 average), indicating that genuine voting behaviour exhibits more coherent community structure than would occur by chance. This gap widens as we focus on periods closer to the fork, with the final 14 proposals showing just 2.08 clusters on average versus 3.87 in randomised data.

Second, the percentage of fork addresses concentrated within a single cluster differs sharply between genuine and randomised data. Across all proposals, 72.20% of fork addresses cluster together in genuine data, compared with only 62.64% in randomised data. This differential becomes more pronounced in the pre-fork period: in the final 44 proposals, 90.96% of fork addresses cluster together in genuine data, against just 47.36% in randomised data.

Our results show that the partisan communities that ultimately fragmented Nouns DAO could be detected, in retrospect, before the fork occurred. Addresses that ultimately participated in forks exhibited progressively more aligned voting behaviour months before the decision to fork was enacted. These findings suggest that governance data may contain early indicators of emerging partisan divisions. However, this evidence is retrospective; a prospective prediction or monitoring system would require further validation across additional DAOs and false-positive cases.

Table 4.1 Forked cluster dominance and average clustering comparison. For each proposal range, the table reports the average number of  $k$ -means clusters and the percentage of forked addresses assigned to the dominant cluster, alongside the minimum, maximum, and mean values obtained from a shuffled baseline.

| Range                | Metric              | Value | Shuff.<br>Min. | Shuff.<br>Max. | Shuff.<br>Avg. |
|----------------------|---------------------|-------|----------------|----------------|----------------|
| Proposals<br>1–362   | Avg # Clusters      | 2.77  | 3.14           | 3.57           | 3.36           |
|                      | % in Forked Cluster | 72.20 | 57.70          | 67.64          | 62.64          |
| Proposals<br>257–362 | Avg # Clusters      | 2.48  | 3.23           | 4.05           | 3.66           |
|                      | % in Forked Cluster | 79.36 | 44.60          | 59.31          | 50.40          |
| Proposals<br>319–362 | Avg # Clusters      | 2.14  | 3.06           | 4.14           | 3.62           |
|                      | % in Forked Cluster | 90.96 | 41.66          | 59.69          | 47.36          |
| Proposals<br>349–362 | Avg # Clusters      | 2.08  | 2.50           | 4.67           | 3.87           |
|                      | % in Forked Cluster | 90.28 | 34.82          | 63.24          | 46.99          |

## 4.6 Conclusions

In this chapter, we have identified partisan community formation within Nouns DAO and retrospectively analysed the behavioural voting patterns that preceded the DAO’s known fragmentation. The alignment between the detected communities and the forking outcome provides behavioural validation of the method against a documented fork within a single DAO, rather than universal validation across all DAOs. The fork provides an external label for fragmentation, but it does not reveal private motives, ideology, or deliberate coordination.

We presented a method to map partisan divisions by analysing on-chain governance data. The method begins with a voter matrix representing a DAO’s governance activity across its proposals and voter addresses. We measured the overall friction within the community based on the outcomes of governance proposals, and explored this friction at the address level by filtering the data to include only active voting participants. We segmented proposals into a rolling window of ten proposals to capture how voting behaviour evolved throughout the DAO’s lifecycle, and considered active voters to be those who had participated in 40% of proposals within the governance window. This filter removed addresses with minimal engagement and reduced noise in the data. We then conducted pairwise dissimilarity analysis on the active voting addresses within each proposal window, comparing the binary votes of each pair of addresses. The dissimilarity scores were used to visualise the alignment of addresses through multidimensional scaling. Finally,

once the addresses were projected into two dimensions, we identified clusters using  $k$ -means clustering and selected the optimal configuration using silhouette scoring. This framework transformed individual voting records into a coherent visual representation of voter alignment for each proposal in the DAO's history.

We used Nouns DAO as the case study for our analysis, as its past documented forks provided a ground truth for validating the method. We showed that addresses that later participated in a fork are increasingly grouped into a single cluster, with over 90% aligned just before the split. Moreover, the average number of clusters falls to roughly two by this stage, indicating two partisan communities. We validate these results by randomising voter assignments while preserving participation and outcomes; across 100 iterations of randomised voter assignments, on average 47% of forked addresses clustered together, and the average number of clusters rose above four.

Beyond confirming that internal communities exist within DAOs, we have shown how our method can automate community detection from raw blockchain log events. The degree to which communities disagree can be quantified and visualised, exposing partisan divisions within DAOs. We found that addresses belonging to a particular cluster had an increased likelihood of participating in subsequent DAO forks. This is significant because analysis at the community level can provide insight into the subsequent behaviour of individual addresses. The insights are particularly important in the context of forking, which can destabilise and delegitimise a DAO's governance system. In answering SRQ3, this chapter provides the retrospective evidence for RQ 3 and, through the detection of internal communities, contributes to RQ 1. It establishes the internal strand of the thesis-level argument: viewing voting records at the community level reveals partisan structure that address-level analysis cannot see.



# Chapter 5

## Metagovernance

### 5.1 Introduction

In Chapter 4, we investigated internal DAO dynamics. As Section 3.1.2 discussed, smart contracts on a blockchain are composable: any contract can call functions on any other contract in the same network. DAO governance contracts are no different, and external contracts can build on top of them just as freely. Whereas this composability is largely seen as a positive in decentralised finance, its consequences for decentralised governance are less clear.

We defined *metagovernance* in Section 2.8 and reviewed external DAO governance dynamics in Section 3.3. Metagovernance is known to occur within blockchain networks, but its frequency and impact remain understudied. In this chapter, we present a method to automate the detection of metagovernance on the Ethereum blockchain by tracing the tokens of prominent DAOs through to the smart contracts and external organisations that interact with them. We investigate the mechanics through which metagovernance operates, measure its prevalence between DAOs, and characterise how external communities exercise influence through three case studies.

Specifically, this chapter addresses RQ 1 by detecting external communities that participate in DAO governance on the Ethereum network. This is achieved through a semi-automated methodology that takes a DAO governance token as input and identifies metagovernance entities that participate as external communities within that DAO. While this is demonstrated through a curated token list, this framework could be applied to other tokens outside of this study, and reused with alternative indexing and labelling sources.

We begin from an initial dataset of governance tokens. We refine prior methodologies by tracing each token to the governance contracts that consult it; this approach establishes a connection between the governance contract and its voting power source, and handles a wide variety of DAO frameworks and voting schemes. Once the token-to-DAO rela-

relationship has been identified, we gather and process the corresponding voting data. This produces a list of potential metagovernance relationships, which we label and visualise as a network graph. Our network graph visualises observable governance influence found in the governance data, but it does not contextualise the internal governance process within an external community. The transactions within a 2-of-3 multi-signature wallet can show us the outcome of an external governance process, but they do not reveal the internal decision-making process taking place off-chain. The same applies to external communities voting through Snapshot spaces, whose internal deliberations may remain opaque. Further analysis of the voting data, complemented by case studies, reveals three metagovernance dynamics. We use the descriptive labels strategic, decisive, and nexus to describe these observed behaviours; they are illustrative examples rather than a complete taxonomy.

The remainder of this chapter is organised as follows. Section 5.2 describes our method for identifying governance smart contracts and extracting contract account voters from both on-chain and off-chain DAOs. Section 5.3 reports our results, including the metagovernance network we constructed across 61 DAOs and three case studies that illustrate strategic, decisive, and nexus metagovernance. Section 5.4 concludes.

## 5.2 Method

To investigate DAO-to-DAO voting mechanisms, we collected on-chain and off-chain data. We extracted data from Ethereum mainnet using an Ethereum archive node, including event logs, trace calls, and historical contract calldata, from block height 0 to 18 299 999 (October 2023) inclusive. We collected off-chain data from a variety of sources, including voting data from the Snapshot GraphQL API, labelled event signatures from Etherface [42], smart contract Application Binary Interfaces (ABIs) and labelled contract data from Etherscan, and a DAO token list from CoinMarketCap [26].

The DAO token list was a seed dataset of twenty tokens with the highest market capitalisation, labelled with ‘Ethereum’ and ‘DAO’ by CoinMarketCap. We refined the list to sixteen tokens primarily governed via the Ethereum blockchain. We then cross-referenced the entities between the different sources, matching each token on the DAO token list with its corresponding governance smart contract on Ethereum mainnet and its corresponding Space on Snapshot. We detail this process in the following subsections.

### 5.2.1 Governance Smart Contracts

Our first task was to match the set of DAO governance tokens with their governance smart contracts. This is typically a manual process that requires reading a protocol’s documentation and GitHub repository and inspecting its on-chain deployments. We

created an automated process that combines on-chain and off-chain data to produce similar results. `IDENTIFYGOVERNANCESMARTCONTRACT` identifies the governance smart contract associated with a contract account address (Algorithm 2). The input contract account address ( $A$ ) can point to a governance token, or to any source of voting power. The output is the governance contract account address that wields voting power over  $A$ .

---

**Algorithm 2** The Governance Smart Contract Identification Heuristic (`IDENTIFYGOVERNANCESMARTCONTRACT`) takes a contract account address ( $A$ ) as input and produces a governance contract account address as output.

---

```

1: function IDENTIFYGOVERNANCESMARTCONTRACT( $A$ )
2:    $B \leftarrow \text{CONTRACTINVOCATORS}(A)$ 
3:    $C \leftarrow \{\}$ 
4:   for all  $b \in B$  do
5:     if HASEETHERSCANABI?( $b$ ) then
6:        $ES \leftarrow \text{HASHEDEVENTSIGNATURES}(b)$ 
7:       if ISGOVERNANCECONTRACT?( $ES$ ) then
8:          $C \leftarrow C \cup \{b\}$ 
9:       end if
10:    end if
11:  end for
12:  return MOSTFREQUENTLYINVOCATED( $C, A$ )
13: end function

```

---

The algorithm first identifies the set of contract accounts that call functions on the input contract address (Line 2). These are obtained from the historical trace data, specifically the `to` field of contract calls directed at the input contract. Governance systems based on token voting typically separate the governance contract, which manages proposals and votes, from the voting power contract, which records token balances. When a vote is cast, the governance contract calls the voting power contract to determine the caller’s voting weight. In one-token-one-vote frameworks, this is most commonly done via the `balanceOf` function, which returns the number of tokens held by a given address. By starting from the voting power contract and following the trace data backwards, we surface the governance contracts that consult it.

Next, we filter the set of contract addresses to those that appear to be governance contracts (Lines 3–11). For each contract on our list, we retrieved its ABI from the Etherscan API (Line 5). The ABI gives us complete access to the contract’s functions and events, and we hashed the events linked to each contract (Line 6). We then matched these hashes against a local database of known governance event signatures, built from Etherface

and 4byte [1], restricted to signatures whose names contain governance-related terms such as “vote” or “proposal”. This produces a set of candidate governance contracts ( $C$ ).

Finally, we select the candidate that most frequently invoked functions on the input contract address as the governance contract (Line 12).

---

**Algorithm 3** The Multi-Signature Contract Account Identification Heuristic (IDENTIFYMULTISIGVOTERS) takes a governance contract account ( $C$ ) as input and produces a set of multi-signature contract accounts as output ( $W$ ).

---

```

1: function IDENTIFYMULTISIGVOTERS( $C$ )
2:    $ES \leftarrow \text{HASHEDVOTINGSIGNATURE}(C)$ 
3:    $V \leftarrow \text{CONTRACTVOTERS}(C, ES)$ 
4:    $W \leftarrow \{\}$ 
5:   for all  $v \in V$  do
6:     if HASMULTISIG?( $v$ ) then
7:        $W \leftarrow W \cup \text{GETMULTISIG}(v)$ 
8:     end if
9:   end for
10:  return  $W$ 
11: end function

```

---

Our second task was to navigate from a governance contract account to the multi-signature contract accounts that voted in its governance process. We created an automated process to achieve this. IDENTIFYMULTISIGVOTERS identifies the contract account voters for a single governance contract account address (Algorithm 3). The input governance contract account address ( $C$ ) is the output of the previous algorithm. The output is a set of contract account voters ( $W$ ) that may be engaged in metagovernance.

We first collect all event logs emitted by the contract and order them by frequency, with the most frequent first. We assume that the most frequently occurring event log on a governance contract is related to voting (Line 2), since governance systems generally produce more vote actions than proposal submissions. We then focus on events whose topic hash matches our criteria for voting events. We look for log entries labelled “voter” that contain an Ethereum address; if no such entries exist, we examine the data section of each log for Ethereum addresses. Once we have identified all addresses associated with voting, we filter to those that are contract accounts (Line 3), leaving a list of contract accounts that participated in the governance process.

### 5.2.2 Delegation Data

We identify delegation for on-chain contract accounts using a procedure similar to that of `IDENTIFYGOVERNANCESMARTCONTRACT` (Lines 5–6), and resolve the delegation with a function called `GetDelegateSignature`. The voting power address serves as input, and the function performs signature matching against the keyword “delegate”. Unlike `IDENTIFYGOVERNANCESMARTCONTRACT`, which returns a single contract, `GetDelegateSignature` returns all delegation-related signatures. We then parse the delegate logs to extract contract account delegations.

### 5.2.3 Snapshot Spaces

To detect Snapshot Spaces from a source of voting power, we indexed all Spaces and Strategies using the Snapshot GraphQL API. This allowed us to look up a source of voting power, such as a governance token address, and search for any Snapshot Space that referenced it as part of its Strategy. We restricted the list of Snapshot Spaces to those with at least 50 followers, since anyone can create a Space that references any governance token. We then retrieved all historical voting data from the Snapshot API, focusing exclusively on voting addresses that were contract accounts.

### 5.2.4 Account Labelling

We labelled the list of contract account voter addresses using Etherscan’s public name tags. Public name tags attach a human-readable identifier to a smart contract address, including identifiers for multi-signature accounts, treasuries, and even known exploiters.

## 5.3 Results

In our analysis of 16 DAOs, we found 12 that use an on-chain governance framework. Of these, we identified 9 correctly and one incorrectly. Our method was unable to recognise MakerDAO because the structure of its event logs differs from the conventional pattern, producing a mismatch during the event log signature step. Although we identified Yearn’s governance structure from our Snapshot data, we erroneously tagged the Yearn locker as the governance contract because the locker received a disproportionately large number of trace calls. We also encountered difficulty parsing voting data for UMA, which uses an unconventional vote-revelation scheme that is not easily detected through log counts. Half of the 16 DAOs draw voting power from sources other than the token contract (for example, Curve’s `veCRV`), which required manual verification of the voting power source. In total, we identified 663 contract accounts participating in governance operations. Of

these, we labelled 42 using Etherscan’s tag system, and we labelled five additional DAOs via Snapshot Spaces.

We expanded the dataset by applying the same algorithm to the tagged DAOs with governance tokens, iterating to a depth of three and excluding any DAOs not engaged in governance on Ethereum mainnet. This process correctly identified two further on-chain addresses and incorrectly identified four. For Convex, for example, our system flagged the Pirex CVX contract as a governance contract: Pirex CVX is a token wrapper for v1CVX that interacts directly with the cvxLocker and emits vote logs relevant to Votium, which our heuristic misinterpreted as a governance role. We also found 11 additional Snapshot addresses, of which only two contained no contract account voting.

We identified the corresponding Snapshot Space for every DAO in the original 16. This was expected: the voting power contract for each had been manually derived from its documentation, and the follower-count filter (Section 5.2.3) discarded Spaces that referenced a token without a real community behind them. Because our dataset focused on DAOs with significant market capitalisation, the algorithm might face difficulty with lesser-known DAOs.

Our success rate in detecting on-chain governance was 61%: of the 18 DAOs we attempted to process end-to-end, we automated governance contract detection, log parsing, and contract account voter retrieval for 11. The remaining 39% (7 DAOs) divides into two groups. Two were on-chain misidentifications caused by inaccuracies in our algorithm’s trace ordering and signature matching, where wrapper contracts such as the Yearn locker and the Pirex CVX wrapper were flagged in error. The other 28% (5 DAOs) failed at the Snapshot stage: although our pipeline returned a Space for each, no contract account voting could be extracted from those Spaces, leaving no metagovernance edges to add.

### 5.3.1 Metagovernance Network

Figure 5.1 visualises metagovernance as a network. Each vertex represents a DAO, and each directed edge from a source to a target represents the source DAO exerting influence on the target DAO. We identified 61 labelled smart contracts, each represented as a vertex. Convex Finance has the highest in-degree, receiving participation in its governance from 12 other metagovernors. Curve Finance and Balancer follow, each with 10.

The three DAOs most heavily involved in metagovernance all govern token emissions, which suggests that economic incentives tied to gauge proposal outcomes attract metagovernors. This concentration may also reflect a bias in our method: contract account voting on gauge contracts requires whitelisting, which may produce more thorough labelling of metagovernors on Etherscan. Convex Finance complicates this picture: it does not require

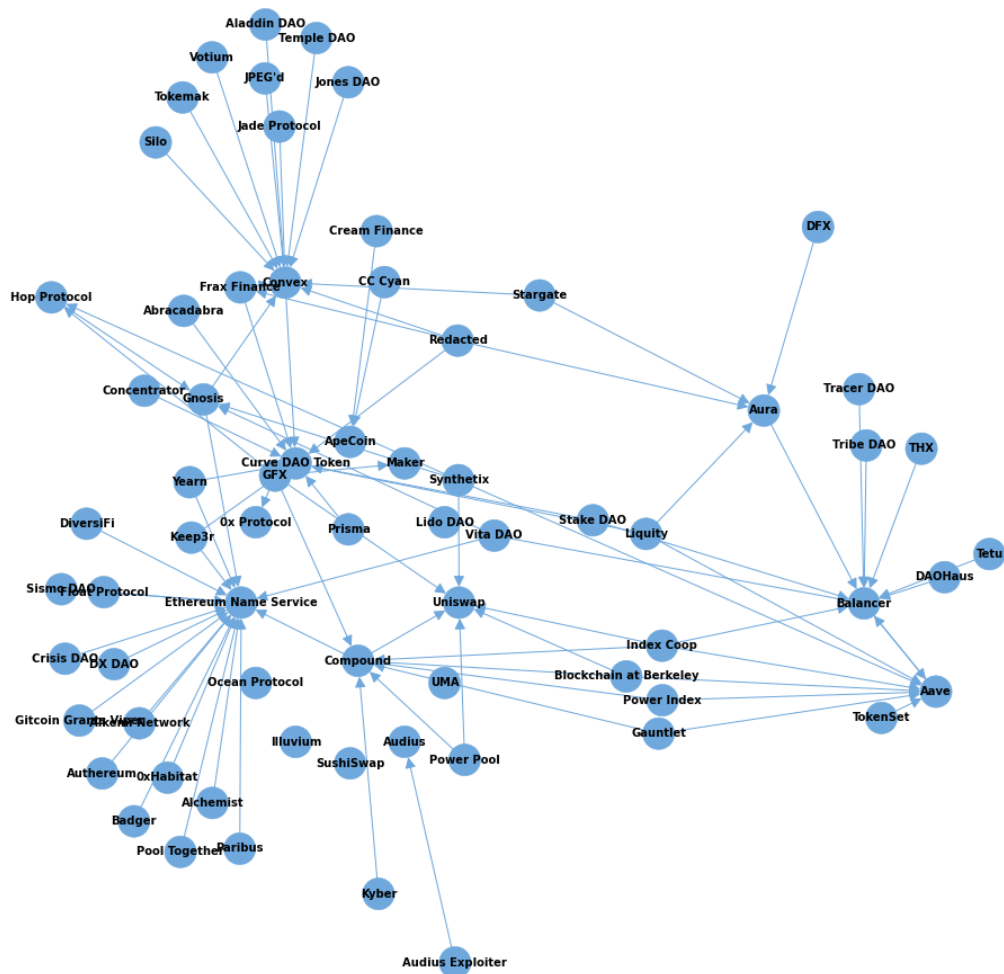


Fig. 5.1 The metagovernance relationship between DAOs can be visualised as a network: each vertex corresponds to a DAO and each directed edge connects a source to a target where the DAO corresponding to the source metagoverns, or influences the governance of, the DAO corresponding to the target.

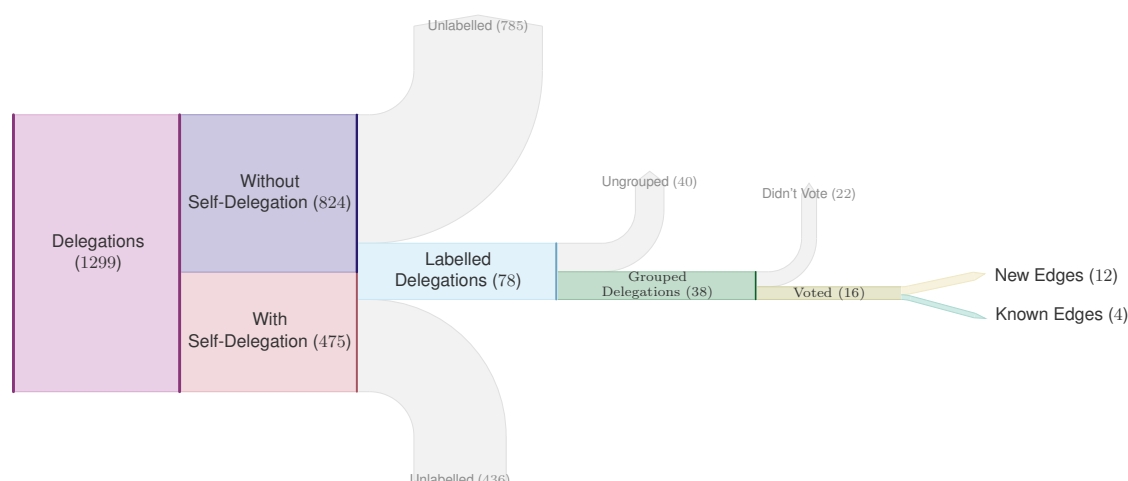


Fig. 5.2 DAOs delegate voting rights to third parties who vote on their behalf. Although our dataset included 1299 instances of contract accounts delegating their voting rights, this only added 12 additional edges to the metagovernance network in Figure 5.1. At each fork, the two branches have widths proportional to the square root of their delegate counts, normalised so that they sum to the width of the parent. This scaling, applied recursively from root to leaves, preserves the ordering and approximate relative magnitude at each split while keeping the smallest leaves (e.g. “Known Edges” with 4 delegates) legible alongside the initial flow of 1299.

whitelisting for participation in its own governance, yet has accumulated a central role in Curve Finance’s governance through an off-chain framework.

The vertex with the highest out-degree is GFX Labs, a blockchain research and development organisation that participates in DAO governance, receives substantial delegations, and frequently creates proposals. Three DAOs (Synthetix, Index Coop, and Redacted) share second position in out-degree. Synthetix operates a governance council, and Redacted, a liquidity aggregator, uses its liquidity to acquire governance positions. Index Coop is the subject of a case study in Section 5.3.3.

### 5.3.2 Delegation Results

Figure 5.2 visualises the filtering of on-chain delegations into new metagovernance edges, read from left to right. Our analysis identified six on-chain DAOs that use delegation, across two distinct frameworks. The first requires token holders to delegate to themselves (*self-delegation*) before they receive governance rights. The second distributes voting rights without the need for self-delegation. Of the 1299 contract account delegations we observed, 475 were self-delegations and 824 were not.

The two flows merge at a labelling step. Using Etherscan’s tag system, we labelled 78 of the 1299 delegating addresses and discarded the remaining 1221 as unlabelled.



Grouping the labelled addresses by their parent DAO yielded 38 distinct DAO-to-DAO delegations; the 40 addresses that did not aggregate into a DAO are excluded. Inspection of the voting logs showed that only 16 of the 38 delegates had cast votes, with the other 22 dropped. Of the 16 active delegates, four were already edges in our metagovernance network and the remaining 12 were added as new edges (Figure 5.1).

### 5.3.3 Case Studies

We analysed three case studies drawn from our metagovernance network, each chosen to illustrate a different metagovernance dynamic. As in Section 5.1, the labels *strategic*, *decisive*, and *nexus* describe the observed behaviours rather than a complete taxonomy. The first illustrates *strategic metagovernance*: a single high-stakes vote in which the metagovernor’s voting power exceeded 15%. The second illustrates *decisive metagovernance*: a vote that directly determined a proposal’s outcome. The third illustrates *nexus metagovernance*: a DAO that acts as a governance hub for multiple other DAOs, evidenced by a high out-degree in the metagovernance network and frequent references to the governed DAOs in its own proposals.

#### Liquity and Aave

Liquity is a decentralised lending protocol and the issuer of the LUSD stablecoin [71]. Aave is a decentralised lending protocol for both stable and variable on-chain assets [2]. Liquity cast a single vote with 100 000 AAVE tokens, or 17% of the total voting power, on Aave DAO’s Proposal #95. The proposal was first raised on 3rd June 2022 in the Aave governance forum, with the title “ARC: Add support for LUSD (Liquity)”. ARC is a permissioned version of the Aave protocol designed for anti-money-laundering (AML) and know-your-customer (KYC) compliance, so tokens must be whitelisted to appear on the platform; new assets are listed only after a successful governance proposal.

The proposal transitioned to a Snapshot governance vote on 5th July and concluded on 10th July with 94% approval. Liquity’s direct involvement with the on-chain proposal began with a delegation of AAVE tokens to its “Liquity: Bounties” contract account. On 16th August, an EOA delegated 100 000 AAVE tokens to the Liquity contract; nine days later, the contract submitted Proposal #95 to the governance contract and cast a “yes” vote later the same day.<sup>1</sup> The proposal passed.

The proposal had direct benefits for Liquity, as it expanded the utility of LUSD. However, the rapid accumulation of voting power to influence the token’s listing raises concerns characteristic of strategic metagovernance.

<sup>1</sup><https://archive.today/sOL8o>

### Blockchain at Berkeley and Compound Finance

Blockchain at Berkeley was founded in 2016. They describe themselves as “driving innovation in the blockchain industry by building an ecosystem that empowers students to make an impact through practical education, consulting for enterprise companies, and conducting open source research” [21]. Within our metagovernance network (Figure 5.1), Blockchain at Berkeley participates in the governance of Aave, Compound, and Uniswap. They have voted 70 times across these three DAOs, accounting for almost 6% of the total votes cast per proposal on average.

Compound Finance [27] is a decentralised lending protocol. On 15th April 2022, Proposal #100,<sup>2</sup> titled “Rewards Adjustments: Kickstart Rewards: Step Two”, was created within Compound’s DAO. The proposal sought to end the COMP rewards programme and replace it with a new programme focused on bootstrapping new markets. Voting ran from 18th to 21st April. The proposal was rejected, with 492 678 votes in favour and 499 849 against. Blockchain at Berkeley cast the deciding vote of 50 007 against.<sup>3</sup>

By focusing on votes cast by contract addresses that were decisive in proposal outcomes, our analysis reveals the influence that a single external community can exert within a DAO. Blockchain at Berkeley operates through a multi-signature wallet<sup>4</sup> that requires authorisation from two of its eleven owners; the off-chain process by which those signers reach consensus is not publicly documented.

Figure 5.3 illustrates this scenario using the visual grammar introduced in Section 2.9. Three actors within the Blockchain at Berkeley community boundary each control an EOA. The community controls a 2-of-11 multi-signature wallet that holds the COMP governance token and casts a no vote in Compound DAO. The diagram shows how a single external community, operating through a multi-signature wallet, cast the deciding vote on Proposal #100.

### Index Coop

Index Coop simplifies access to complex DeFi strategies through ERC-20 tokens [58], offering products such as tokenised index funds (DPI). A key distinction between traditional financial stewardship and Index Coop’s metagovernance model lies in who makes the governance decisions. In traditional finance, decision-makers are portfolio managers, boards, and executives; in Index Coop, governance rights are embedded in the INDEX token, which is separate from the underlying assets such as DPI.

<sup>2</sup>See <https://compound.finance/governance/proposals/100>.

<sup>3</sup>See <https://archive.today/wUtjJ>.

<sup>4</sup>See <https://archive.today/Tfzqh>.

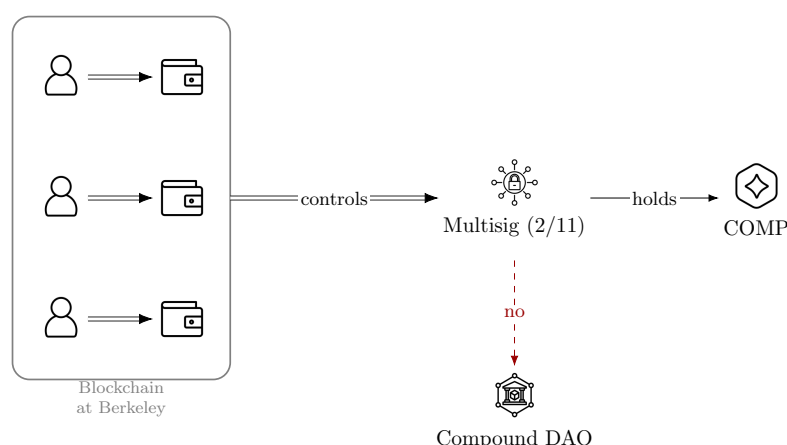


Fig. 5.3 Blockchain at Berkeley’s participation in Compound governance. Three actors within the Blockchain at Berkeley community control a 2-of-11 multi-signature wallet that holds COMP tokens and casts a no vote in Compound DAO.

We identified Index Coop’s active participation in the governance of four DAOs by counting the number of Index Coop proposals that referenced each: Aave (669 proposals), Compound (179), Uniswap (114), and Balancer (33). Index Coop also voted on proposals in those DAOs directly: 120 times on Aave, 32 times on Compound, twice on Uniswap, and four times on Balancer. Several factors account for the discrepancy between Index Coop’s internal proposals and its direct votes. First, Index Coop mandates a quorum for governance actions, so a referenced proposal that fails to reach quorum produces no corresponding vote. Second, the referenced proposals may relate to different stages of governance: Uniswap’s on-chain proposals, for example, follow “temperature check” Snapshot votes. Third, our heuristic may include false positives, where mentions of the four DAOs were not direct metagovernance decisions.

Index Coop’s involvement across four DAOs makes it a clear instance of nexus metagovernance.

## 5.4 Conclusions

In this chapter, we have presented a method for identifying external community governance interactions, characterising the mechanics of inter-DAO governance through multi-signature smart contracts, the forms in which metagovernance communities manifest, and the influence external communities exert on the DAOs they govern.

The detection method has two parts:

1. *On-chain governance contract identification*: starting from a voting power contract, we sort all contract accounts that interact with it by frequency and cross-reference their event signatures against a database of voting and proposal signatures.
2. *Contract account voter extraction*: from the identified governance contract, we collect the voting event log and extract all voter addresses. We retain only those addresses that contain executable code (i.e., contract accounts), since these are the candidates for external community participation.

For off-chain metagovernance, we matched a DAO's on-chain voting power contracts to Snapshot Spaces using a database constructed from Snapshot's GraphQL API. When multiple Spaces referenced the same voting power contract, we selected the most-followed Space, excluding any with fewer than 50 followers. We then parsed the voting data from the Snapshot API and retained only voters that were contract accounts.

We labelled contract accounts using Etherscan's public name tags, supplemented by names and comments in the contract source code. This labelling allowed us to identify the entities behind metagovernance interactions and to expand the dataset by following labelled edges. The result is a metagovernance network that visualises the metagovernance relationships between DAOs.

In Chapter 4, we identified internal communities within a DAO's governance from its governing addresses. In this chapter, we have shown that through multi-signature smart contracts and delegations, those governing addresses may themselves represent entire external communities. We investigated a subset of DAOs on the Ethereum network: across 61 established DAOs, we identified 72 external community relationships. These relationships are a detected subset rather than a complete census of metagovernance on Ethereum: our heuristics did not capture every DAO framework, as shown by the detection misses reported in Section 5.3, and further relationships may exist both within and beyond the DAOs we examined. We selected three of these for closer examination, and they exhibit very different forms: from a university group operating a multi-signature wallet (Blockchain at Berkeley), through a single protocol acquiring delegated voting power to list its own token (Liquity), to an entire DAO with its own token holders that replicates the underlying DAO's proposals on its own governance platform (Index Coop).

The impact of these external communities also varies across the three cases. Blockchain at Berkeley cast the deciding vote on a Compound proposal that would otherwise have passed, demonstrating that a single external community's vote can determine the outcome. Liquity proposed and voted to list its own token on the underlying DAO's lending platform, using delegated voting power to advance its own protocol's interests. Index Coop relocated the governance process of the underlying DAOs by replicating their proposals on its

own governance platform, where votes are cast using the INDEX token rather than the underlying DAO's token.

Although this is a small subset of the relationships in our dataset, the three cases together suggest three concerns. First, external communities can be decisive in the governance outcomes of the DAOs they govern, indicating that they hold real power within those DAOs. Second, they may use that voting power to directly benefit their own community, introducing potential conflicts of interest with the underlying community. Third, external community governance can relocate the locus of voting from the underlying DAO's governance contract to another smart contract operating under a different governance framework, reducing transparency, which is fundamental to DAO governance.

Active external DAO communities exist throughout Ethereum's prominent DAOs. They are not uniform: they range from a handful of signers controlling a multi-signature contract to entire protocols with hundreds of token holders. The interconnected DAO governance ecosystem currently operates without major disruption, but it creates conditions for inter-DAO friction that the underlying DAOs neither authorised nor control.

DAOs govern protocol upgrades, treasury management, and other consequential matters. Token holders within a single DAO typically share aligned incentives for its success; metagovernance introduces external parties whose incentives may not be aligned with the underlying DAO. Because blockchain governance lacks the regulatory and social checks present in corporate governance, the outcomes of metagovernance relationships are difficult to anticipate. Tools that identify and contextualise metagovernance, of the kind we have begun to develop in this chapter, are a necessary step toward understanding the governance dynamics that emerge across DAO boundaries. In this way, the chapter answers SRQ1 and contributes the external-community evidence for RQ 1, advancing the thesis-level argument by showing that a single governing address can represent an entire community.

# Chapter 6

## The veToken Model

### 6.1 Introduction

In Chapter 4, we used historical governance records to show that communities within DAOs can be detected, and that friction between them can have severe consequences, including the fragmentation of the membership through a DAO fork. In Chapter 5, we identified a range of external DAO communities that vote through multi-signature addresses, producing a network of interconnected DAOs. We examined the dynamics between these external communities and the underlying DAOs they participate in, and found that external communities can exert substantial influence over governance outcomes. Friction at this level is also harder to detect and resolve than its internal counterpart.

In this chapter, we investigate a DAO with multi-layered external communities tied to economic incentives, governed by a token model that addresses several of the limitations of token-voting designs reviewed in Section 3.2.1: the *Vote-Escrowed Token* (veToken) model. The model requires stakeholders to *lock* their tokens for a fixed period in exchange for voting weight. Locked tokens cannot be transferred, so voters who acquire weight commit to the protocol for the duration of their lock, aligning their preferences with the long-term prospects of the decentralised organisation. We explain the model and analyse its emergent outcomes. We describe its implementation by Curve Finance, a popular automated market maker, and the ecosystem of protocols that has arisen on top of it: yield aggregators such as Convex Finance, voting markets such as Votium, and significant participants such as Frax Finance, a stablecoin issuer that is itself governed by a DAO.

This chapter contributes primarily to RQ 2 and secondarily to RQ 1 by characterising tokenised external communities within the Curve ecosystem. We examine the dual nature of the veToken model. The model avoids the shortcomings of earlier token-voting designs by aligning stakeholder and organisational goals. However, as Section 3.1 discussed, the smart contracts that make up governance protocols do not exist in isolation, and the

emergence of higher-level protocols built on top of them creates a dynamic in which influence is disproportionately exercised at upper tiers of the ecosystem, where the rules of engagement have been distorted so as not to match those of the underlying DAO. In this context, distortion does not refer to the modification of the veToken contract, as the contract is immutable, but to the alteration of the intended incentive structure, lockup logic, and distribution of governance influence through composable smart contracts that enable a higher tier of governance. This is not necessarily a violation of the underlying protocol's rules.

We demonstrate this by examining the impact of Votium's incentivisation mechanisms on the outcomes of Convex's fortnightly proposals. We then examine Frax, which locks relatively few tokens directly with Curve but locks substantially more through Convex and acquires further voting weight through Votium bribes.

## 6.2 The veToken Model

The veToken model was proposed and implemented by Egorov as part of Curve Finance, an automated market maker that specialises in stablecoin trading [39, 30]. The model was implemented using smart contracts that extend Aragon [29], a DAO governance framework. The critical difference between it and previous systems was that it replaced the one-token-one-vote model with a voting weight proportional to lock time.

Users need to lock or escrow tokens for a fixed period in exchange for voting weight. The longer the period, the more voting weight is granted. For Curve Finance, the token is known as CRV and the corresponding locked tokens or voting weight is known as veCRV. During the lockup period, the locked tokens are non-transferable, i.e., CRV is transferable but veCRV is not.

Voting weight,  $w$ , for a user can be calculated using:

$$w = a \cdot \frac{t}{t_{\max}}$$

where  $a$  is the number of tokens (e.g., CRV),  $t$  is the desired lockup period, and  $t_{\max}$  is the maximum lockup period [39]. In Curve Finance, the minimum lockup period is one week and the maximum is four years. Consequently, a user who locks up their tokens for one week would need to lock up 208 times the number of tokens compared to a user who locks for the maximum lockup period of four years to get the same voting weight. The intention is to align voters' potentially short-term interests with the protocol's long-term goals by tying voting weight to the duration of the lockup.

The veToken model also addresses voter apathy. Decentralised organisations typically suffer from poor voter turnout [48], and voting weight alone might not be incentive enough

for users to lock tokens for extended periods. The model therefore includes an economic incentive: the allocation of new tokens (CRV tokens in the case of Curve Finance) is directed via voting using a mechanism known as *gauges*.

### 6.2.1 Gauges

Curve Finance is a platform for stablecoin trading. It uses gauges to measure the amount of liquidity provided by users to the various liquidity pools. Each gauge is assigned a weight and those weights determine the daily emission of new CRV tokens. The weights are set every week at Thursday midnight UTC after a gauge proposal vote. Therefore, users with voting weight decide on the allocation of new CRV tokens to liquidity providers. In fact, the voters may be liquidity providers and they can choose to direct the new tokens to themselves. This incentivises the lockup of tokens for extended periods.

### 6.2.2 Higher-Level Protocols

Curve has spawned several higher-level protocols and strategies. For example, yield aggregators such as Convex Finance [28], Yearn Finance [132], and Stake DAO [104] provide smart contracts that hold and lock CRV on behalf of users. This allows users to lock tokens for the maximum lockup period, to trade tokenised claims on those locked tokens, and to amortise gas costs. Voting markets such as Votium [120] allow anyone to incentivise particular Curve gauges by rewarding votes with bribes. Frax Finance [47] is a stablecoin issuer that interacts with many of these protocols. For example, it uses Curve to lock CRV for veCRV, it holds tokenised veCRV via Convex, and it incentivises Curve gauges using Votium.

Figure 6.1 illustrates the Curve/Convex governance ecosystem using the visual grammar introduced in Section 2.9. Two communities hold the v1CVX token and vote in Convex, which controls a voter address that holds veCRV and casts votes in Curve DAO. Two tokenisation chains capture the token relationships: CRV is tokenised by veCRV and CVX is tokenised by v1CVX. The diagram shows how governance decisions at the Convex level propagate down to Curve through the controls and holds edges that connect the two tiers.

## 6.3 Method

We collected on-chain and off-chain data from three primary sources within the Curve ecosystem. Firstly, we gathered transaction data from the Ethereum blockchain relating to Curve: transactions involving the CRV token contract, the veCRV token contract, and the Gauge Controller contract. Curve executes its governance entirely on-chain. Our data for



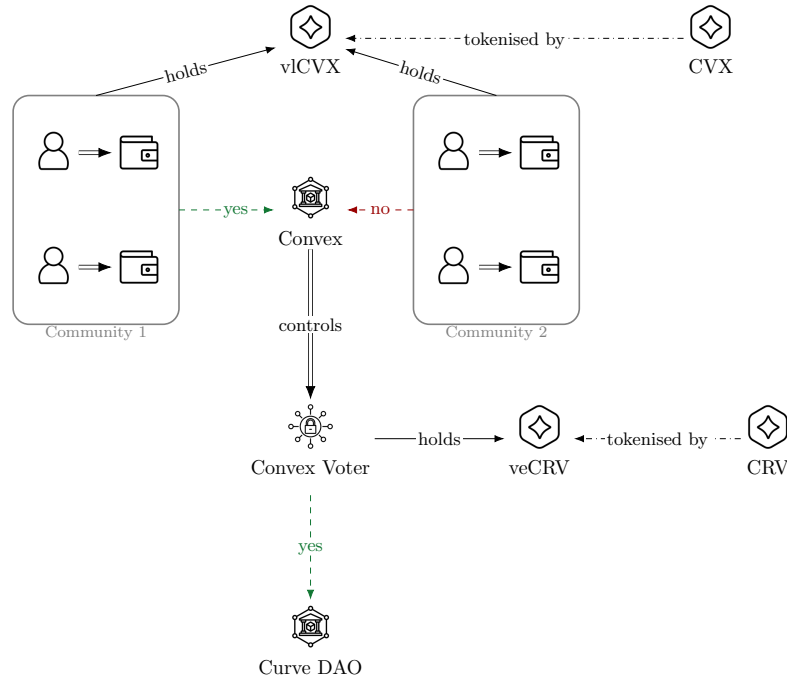


Fig. 6.1 Tokenised metagovernance in the Curve/Convex ecosystem. Two communities hold vlCVX and vote in Convex. Convex controls a voter address that holds veCRV and votes in Curve DAO. CRV is tokenised by veCRV and CVX is tokenised by vlCVX, creating two layers of governance above Curve.

Curve covers the period from the deployment of the CRV token on 12th August 2020 until 17th March 2023. Secondly, we gathered transaction data from the Ethereum blockchain relating to Convex: transactions involving the CVX token contract, the vlCVX token contract, and the cvxCRV token contract. CVX is Convex’s governance token; vlCVX is a locked version of CVX that works in a manner similar to CRV and veCRV, except that the maximum lockup period is much shorter (16 weeks) and the voting weight per vlCVX is determined by the amount of veCRV the Convex protocol owns. Users that lock CVX to get vlCVX receive voting weight on Convex, and in turn, voting weight from the veCRV held by Convex. Our data for Convex covers the period from the deployment of the CVX token on 17th May 2021 until 17th March 2023. We also collected the details of all Convex proposals via Snapshot [102] using its GraphQL API. Thirdly, we gathered transaction data from the Ethereum blockchain relating to Votium: transactions involving the bribing contracts for the gauges. Our data for Votium covers the period from the deployment of the Votium bribing contracts on 12th September 2021 until 17th March 2023. We mapped the bribes in the Votium contracts to their associated gauge in the Curve Gauge Controller contract.

## 6.4 Results

We begin with results relating to Curve’s implementation of the veToken model. These can be considered first-order results, in that they relate directly to CRV and veCRV. In the subsequent subsections (Sections 6.4.1 and 6.4.2), we consider two significant second-order results that relate to higher-level protocols, i.e., Convex, Votium, and Frax, that build upon the Curve ecosystem.

Firstly, we analysed participation in the weekly gauge proposals and the non-gauge proposals. The former are economically incentivised through bribes and CRV emissions whereas the latter are not. We found 9551 unique addresses that lock their CRV tokens to get veCRV, with 2555 (27%) voting in the fortnightly gauge proposals. Participation in the non-gauge proposals is significantly lower. For example, ‘ownership proposals’, which seek the community’s decision on adding new gauges, receive votes from an average of 24 unique addresses. This mirrors the low participation found in other decentralised organisations [48] and shows that the better turnout of economically incentivised gauge proposals does not carry over to non-gauge proposals.

Secondly, we considered the quantity of CRV tokens that are locked for veCRV: there are 644 million CRV (46%) locked for veCRV, leaving 770 million CRV in circulation. The tokens are locked with an average remaining lockup period of 181 weeks, or 3.5 years. This duration indicates a high level of commitment by the token holders to the ecosystem and a long-term outlook on the value of the protocol. These numbers show the veToken model is working as expected: users are incentivised to lock up their CRV tokens for extended periods to gain voting weight, and they use that weight to vote on gauge proposals and direct CRV emissions to liquidity providers. This enthusiasm does not extend to non-gauge proposals, where average participation remains at the 24 addresses noted above.

### 6.4.1 Bribes and Voting Outcomes

We turn our attention to the higher-level protocols. Convex participates in the Curve ecosystem as one of nine whitelisted contract accounts that can lock CRV for veCRV. In fact, 290 million (45%) of all locked CRV belongs to Convex, making it the entity with the largest voting weight in Curve. However, Convex’s voting weight is itself governed by its own token, CVX. There are 71 million CVX, of which 78% is locked for vlCVX to gain voting weight. This is higher than the equivalent figure for Curve. Voting on Convex is conducted off-chain via Snapshot, with the results relating to Curve proposals posted on-chain.

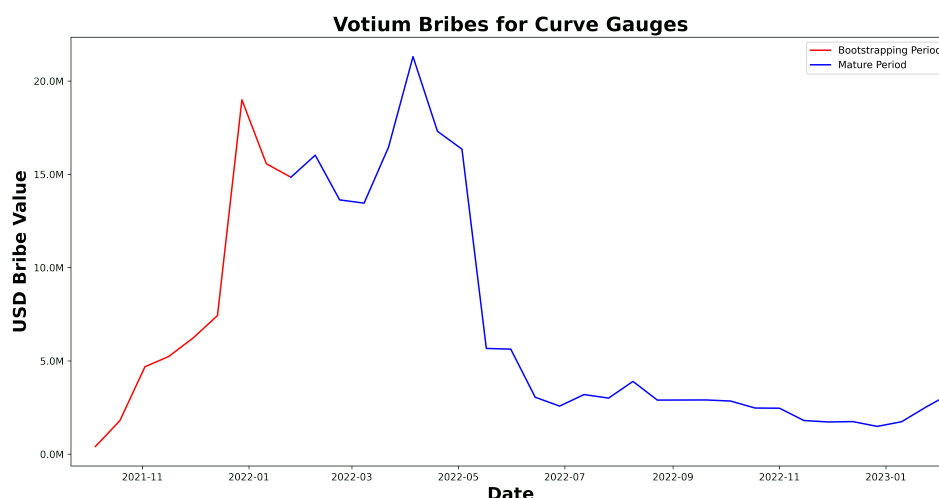


Fig. 6.2 Votium incentivises holders of veCRV and vICVX to vote for particular Curve gauges through bribes. The chart shows the total USD value of those bribes for each fortnightly vote. The USD value is highly dependent on broader market conditions. We divide the data into a bootstrapping phase (red) and a mature phase (blue).

As with Curve, the fortnightly gauge proposals attract the most interest: 1124 unique addresses and 95% of all vICVX vote in the gauge proposals. The number of unique addresses might be higher except for delegation, where a single address can vote using voting weight delegated to it from many other addresses. There is less interest in non-gauge proposals, which receive votes from an average of 77 unique addresses. All of this voting activity appears as a single address when analysing Curve in isolation.

Votium operates at a higher level than Convex. It gathers and distributes bribes to holders of veCRV and vICVX in return for them voting a particular way in the gauge proposals. Figure 6.2 shows the total USD value of those bribes for each fortnightly vote. The USD value is highly dependent on broader market conditions: the peaks in late 2021 and early 2022 coincide with peaks across cryptocurrency markets. At the time of writing, the total value of Votium bribes to date is USD\$248 million, and the value for the most recent fortnightly period is over USD\$3 million.

We can divide the data in Figure 6.2 into a bootstrapping phase and a mature phase. The bootstrapping phase includes the first eight gauge proposals; the mature phase includes all subsequent gauge proposals and continues to the present day. Figure 6.3 illustrates the change in voting behaviour between the two phases. Each dot represents a single instance of a fortnightly vote for a gauge proposal. The  $x$ -axis shows the percentage of the total USD value of Votium bribes directed to each gauge. The  $y$ -axis shows the percentage of the total vote received by each gauge. In the bootstrapping phase (red dots) the correlation

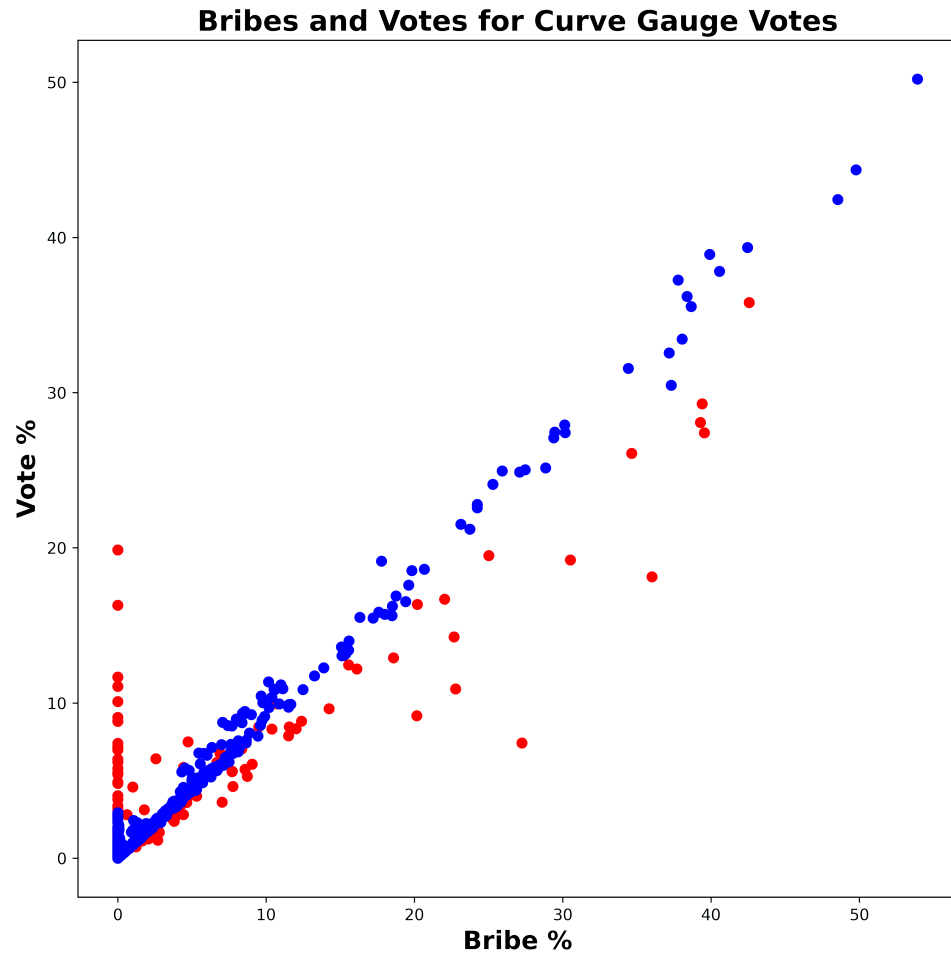


Fig. 6.3 Each dot represents a single instance of a fortnightly vote for a Curve gauge. The  $x$ -axis shows the percentage of the total USD value of Votium bribes directed to each gauge. The  $y$ -axis shows the percentage of the total vote received by each gauge. The red dots correspond to the bootstrapping phase (corr. coeff. = 0.88); the blue dots correspond to the mature phase (corr. coeff. = 0.99).

coefficient between the percentage of bribes attracted and the percentage of votes received is 0.88. In the mature phase (blue dots) the correlation coefficient is 0.99.

There were few outliers in the mature phase. Of the 691 dots in Figure 6.3, only one represented a gauge that received a percentage of votes less than 0.8 times its relative bribe in a given fortnight. In other words, voting outcomes are strongly associated with bribes. Conversely, there were 91 instances where a gauge received a percentage of votes more than 1.2 times its relative bribe. Two overlapping factors account for these. Firstly, in 71 of these instances the gauge received a vote that was less than 1% of the total number of votes for a given fortnight; the bribes and votes associated with these gauges were too small to be meaningful. Secondly, 79 of these instances involved Frax gauges: since Frax owns vlCVX and is a major briber to Votium (see Section 6.4.2), it is to be expected that Frax would vote for its own lesser-bribed gauges.

A correlation of 0.99 demonstrates a strong association between the distribution of Votium bribes and voting outcomes; it does not, by itself, establish that bribes directly cause those outcomes. Votium is specifically designed to incentivise those with voting power to direct their voting weight towards particular gauges or delegate their voting power to the Votium address, where it will “choose the best incentives on every proposal” [120]. This incentivisation mechanism, combined with the observed correlation, makes an influence on voting behaviour plausible; however, other factors may also contribute to the observed association. For example, an address with voting power may have exposure to that gauge and be a beneficiary of emissions from that gauge, posing an alternative economic incentive, and bribes may anticipate votes that would have been cast regardless.

This analysis shows a strong association between the monetary value of the distribution of bribes and outcome of gauge proposals within Convex, where a greater value of bribes on a given gauge generally receives a greater proportion of votes, suggesting that bribes shape gauge proposal outcomes. Figure 6.6 illustrates this on a per gauge/gauge proposal basis. In the heat map, the non-white cells show the difference in percentage points between the bribes directed to and the votes received by a Curve gauge for a single instance of a fortnightly vote. The white cells indicate no bribes or votes. All of the percentage-point differences fall between  $-6\%$  and  $2\%$ . As Charlie Munger once said, “Show me the incentive, and I will show you the outcome.”

### 6.4.2 The Cost of Votes

Frax is a central player in the Curve ecosystem. It participates at the three levels already mentioned: it locks CRV for veCRV, it locks CVX for vlCVX, and it makes bribes via Votium. The Frax/USDC pool has the second-highest total value locked (TVL) across all

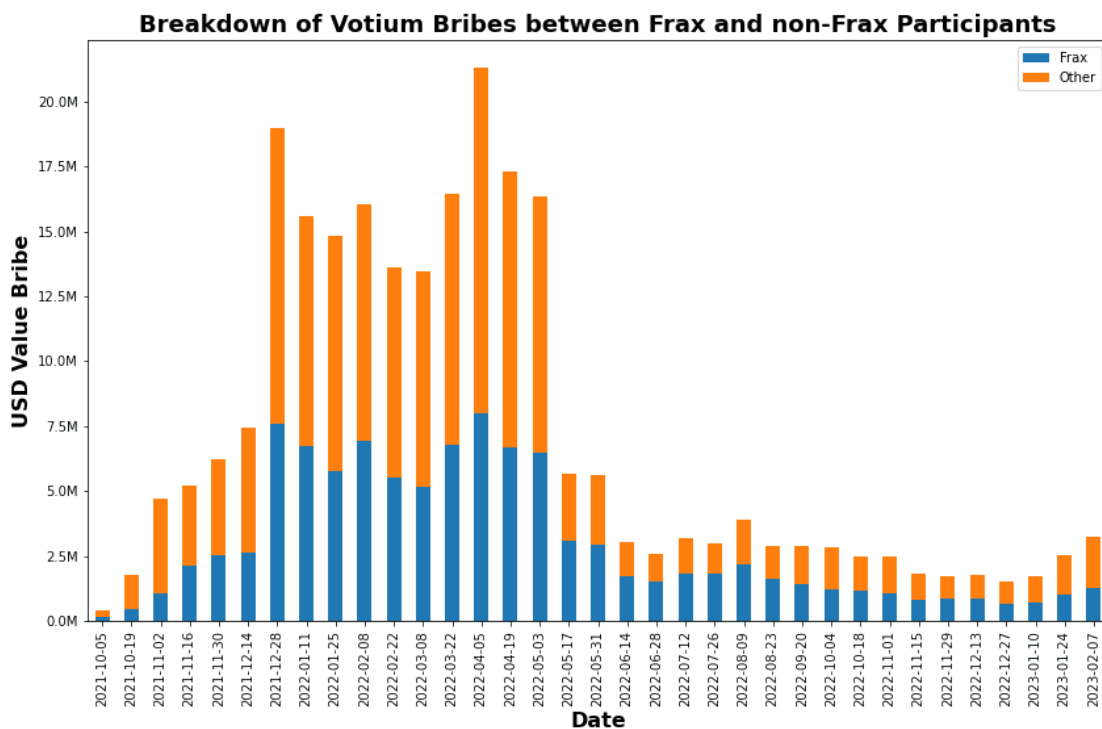


Fig. 6.4 Frax incentivises holders of veCRV and vlCVX to vote for Frax pools through Votium. It contributes almost half of the bribes to Votium.

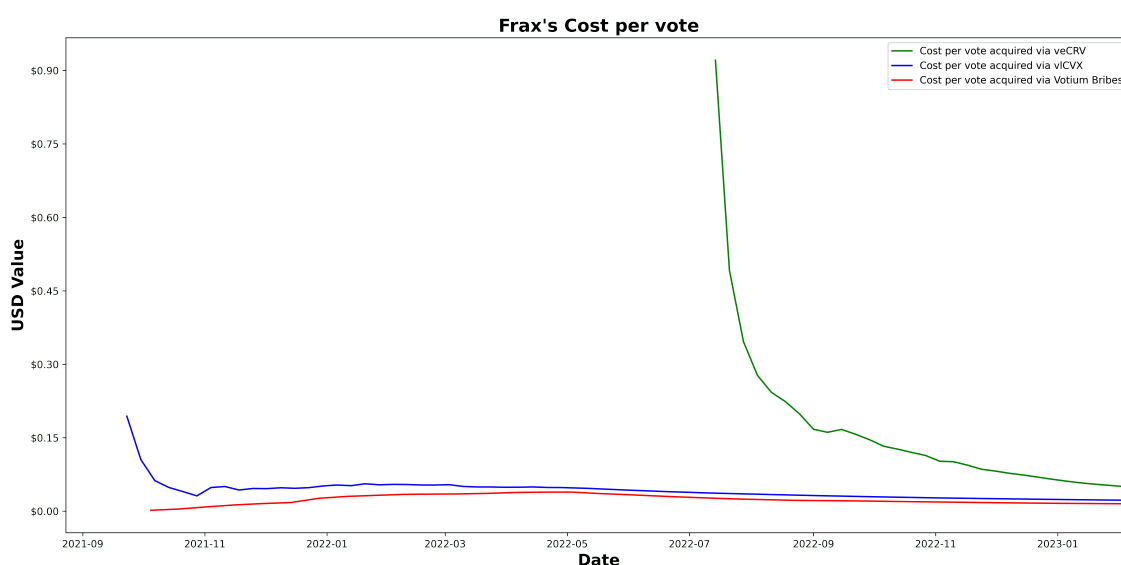


Fig. 6.5 Frax acquires votes for Frax pools through at least three distinct avenues: by locking CRV for veCRV, by locking CVX for vlCVX, and by paying Votium bribes. The cost per vote varies depending on the avenue, and the cost changes over time.

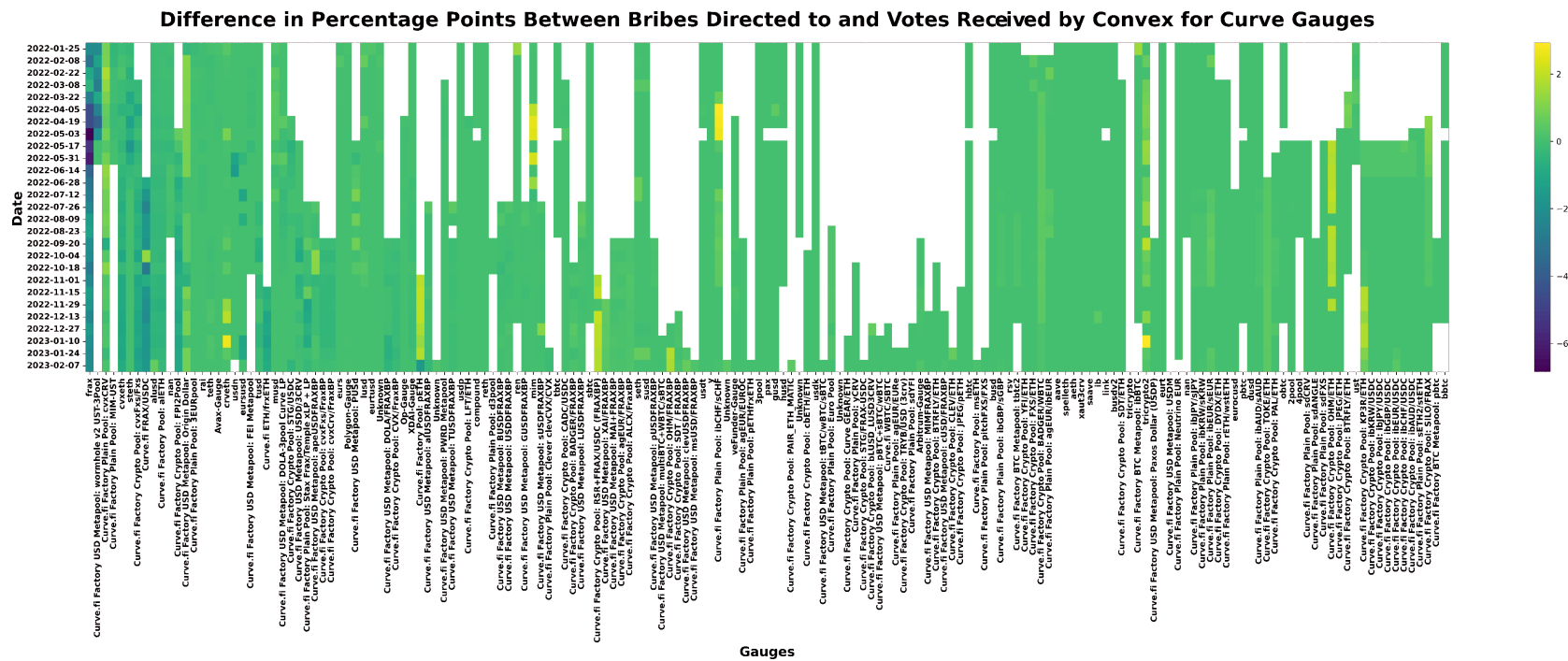


Fig. 6.6 The non-white cells in the heatmap show the difference in percentage points between the bribes directed to and the votes received by a Curve gauge for a single instance of a fortnightly vote. The white cells indicate no bribes or votes. The gauges are arranged left-to-right based on the total bribe amount per gauge, starting with the largest.

Curve pools, behind only the ETH/staked ETH pool. We considered Frax's involvement in the Curve ecosystem at each level.

At the Curve level, Frax began locking CRV for veCRV in July 2022 after a successful governance proposal whitelisted the Frax staking contract. As of February 2023, Frax has locked 697 114 CRV for veCRV, or just 0.1% of all locked veCRV. We determined the USD value of the locked CRV by the price at which it traded when it was locked. This resulted in a total locked value of USD\$618 830. Frax used this voting weight to exercise a total of 17.3 million votes. This yielded a cost per vote of USD\$0.051. The cost per vote decreases over time as the upfront cost of the token is amortised (see the green line in Figure 6.5). These tokens also have the potential to be traded once the lockup period has expired.

At the Convex level, Frax plays a more influential role. Using the same metric as above, we found that, as of February 2023, Frax has locked USD\$64.74 million worth of CVX for vlCVX. To determine its voting weight at the Curve level, we examined the amount of veCRV tokens held by Convex during each fortnightly period and the percentage of voting weight held by Frax. This showed that Frax exercised a total of 2.88 billion votes, which yields a cost per vote of USD\$0.022 (see the blue line in Figure 6.5).

Finally, we analysed Frax's bribes via Votium. Frax is the largest contributor to bribes, accounting for 41% of all bribes to vlCVX, or USD\$103.69 million. Based on the bribe-vote correlation reported in Section 6.4.1, we estimate that this contribution corresponds to 6.72 billion votes at the Curve level. As of February 2023, Votium bribes for Frax cost USD\$0.015 per vote (see the red line in Figure 6.5), making it the most cost-efficient option for Frax. Unlike the previously described levels, the value of bribes is realised immediately, whereas locking tokens still provides Frax with a claim to the asset once the lock reaches maturity.

The veToken model creates a system where voting weight can be acquired through three distinct mechanisms. Most directly, users lock tokens (say, CRV for veCRV) and receive the corresponding voting weight. Through an intermediary, users hold a claim to locked tokens and the voting weight they confer (say, CVX locked for vlCVX). At a further remove, users bribe the holders of locked tokens (veCRV or vlCVX holders) to vote a particular way. We can calculate a cost per vote at each of these levels.

## 6.5 Conclusions

In this chapter, we described the veToken model and its implementation within the Curve ecosystem. We analysed its usage by looking at locking rates, lockup durations, voting participation, and financial incentives. The veToken model is intended to align voter and



protocol incentives by tying voting weight to long lockups. However, through multiple layers of external governance, the mechanics of the model can be distorted. At higher tiers, the rules of engagement change: voters can be directed through bribes from higher-level protocols, and the distribution of those bribes is strongly associated with the outcome of gauge proposals. This effectively replaces four-year-long lockup periods with bi-weekly gauge proposals in which the strong association between bribes and voting outcomes is consistent with economic influence on voter behaviour. We also showed that the cost per vote depends on how the vote is acquired: directly by locking at the base level, indirectly through yield aggregators that lock on behalf of depositors, or through bribes paid to voting markets.

Our findings show a marked difference in participation between gauge and non-gauge proposals. For non-gauge proposals, voter participation is low, mirroring the behaviour of many other DAOs as discussed in Section 3.2. For gauge proposals, participation is significantly higher. This affirms the design of the veToken model: users commit to locking tokens for extended durations in exchange for voting weight, and in turn participate when that weight has economic consequence.

Participation in gauge proposals is economically incentivised. Voters use their voting weight to direct emissions of CRV. Higher-level protocols, including yield aggregators such as Convex and voting markets such as Votium, have emerged to maximise the return on this weight for users. Convex's 45% share of all locked CRV and the USD\$248 million in bribes routed through Votium are associated with substantial voting power and economically incentivised voting within the ecosystem: since January 2022, when Votium began to gain recognition, we observed a 0.99 correlation between voting percentages and the share of bribes distributed to each gauge. This contrasts with prior accounts of DAO governance in which voting coalitions and influential leaders play the dominant role [106]. Communities operating at higher tiers of the governance ecosystem can therefore alter the intended mechanisms of an underlying DAO. By changing the rules of engagement at upper tiers, external communities create more economically incentivised governance, which in turn accrues voting power at those tiers.

One entity that bribes through Votium and exerts influence at every level of the Curve ecosystem is Frax, a stablecoin issuer. Frax holds less than 0.1% of all veCRV, yet wields significant influence at higher levels through Convex and through more than USD\$100 million in Votium bribes. In this way, Frax can seek to influence gauge proposal outcomes by incentivising voters without locking tokens for extended periods. The maximum lockup for Convex is just 16 weeks, against four years for Curve, and Votium imposes no lockup at all. One might expect such votes to carry a premium, since they avoid the illiquidity of a long lock. The opposite holds: the cost per vote acquired through Convex is currently lower than the cost through Curve, and the cost through Votium is lower still.

While our analysis in this chapter is specific to the Curve-Convex-Votium ecosystem of layered metagovernance, our findings hold broader implications for veToken models that use gauge emissions. Indeed, similar governance ecosystems exist among DAOs, with protocols such as Balancer, Velodrome and even Frax using veToken governance models with economically incentivised governance bribes. While the empirical findings presented here are specific to Curve, the governance mechanisms identified apply to other veToken ecosystems that employ vote-escrow governance, gauge emissions and economically incentivised voting. Moreover, the tokenisation and composition of governance rights extends beyond veToken models. For example, protocols such as Index Coop use composability to wrap governance tokens, creating additional governance layers. However, we would expect the dynamics within these protocols to differ as they do not employ gauge-based emissions or external vote-incentive markets that economically reward particular governance outcomes.

In summary, our study shows the dynamic interplay between the levels of the Curve ecosystem and, more generally, the veToken model. The token-locking mechanism is straightforward to understand, but the higher-level protocols built on top of it produce complex outcomes. The model was developed to address the limitations of the one-token-one-vote design. In doing so it has spawned additional protocols and external communities that capitalise on its behaviour. In decentralised finance, this composability is viewed as a positive (the “money legos” metaphor). For governance, it means that higher-level protocols can have surprising and undue influence on lower-level ones. These findings answer SRQ2 and provide the evidence for RQ 2, while the characterisation of tokenised external communities contributes to RQ 1. Within the thesis-level argument, this chapter supplies the distortion strand: layered communities can alter the effective operation of an underlying DAO’s governance.

# Chapter 7

## Token Composition

### 7.1 Introduction

In Chapter 6, we examined the veToken model and the layered ecosystem that has emerged on top of it. Convex Finance composes Curve’s vote-escrowed CRV into its own staking and bribe markets; Frax, Yearn, and other protocols compose Convex in turn. Each layer wraps the layer below in a new token, and each accrues governance influence by virtue of holding what is wrapped. This compositional pattern is not unique to Curve. In this chapter, we step back from any single ecosystem and characterise token composition across the Ethereum blockchain.

Within Chapter 5 we identified metagovernance relationships within DAO voting data using multi-signature addresses to identify potential metagovernors, thereby identifying external governance communities. Chapters 4, 5, and 6 collectively provide direct evidence of internal, external, and tokenised governance communities, respectively.

In this chapter we take a different perspective, by providing structural and infrastructural evidence of the token composition patterns that make layered governance possible. This structural evidence contributes to RQ 1 by characterising the infrastructure through which tokenised external communities can operate. By identifying token wrapping relationships created through smart contract composition, the token graph reveals how tokens can be composed through successive layers. Where the underlying tokens confer voting rights, these relationships represent potential pathways for layered governance, specifically tokenised metagovernance. However, these compositional relationships do not in themselves demonstrate that governance influence has been exercised through them, or reveal the presence or behaviour of a governance community. Nevertheless, token composition provides an underlying mechanism through which tokenised metagovernance can occur, as we observed in Chapter 6.

Ethereum hosts hundreds of thousands of ERC-20 tokens. Many are simple, in that they are not composed of other tokens. Others are composed: a liquidity pool token, for example, represents a share of a collection of other tokens, and DEX Screener [32], a popular pool tracker, lists over one hundred thousand of them. Tokens are also being composed in more elaborate ways: PT-weETH-25APR2024 (0xb18c87) is a token issued by Pendle Finance [88] on the Arbitrum blockchain that transitively depends on many other tokens. At the time of writing, this token has a market capitalisation of over one billion US dollars. Examining token composition matters for both technical and financial risk: if an investor purchases PT-weETH-25APR2024, what tokens does the investment depend upon? In this chapter, we present a method for examining token composition at both the macro- and micro-level and apply it to the Ethereum blockchain.

Our method extracts *meta-events* from EVM logs. Contracts emit low-level events; the ERC-20 standard, for example, specifies a Transfer event whenever a token is transferred [117]. A meta-event is a heuristic-defined sequence of one or more low-level events emitted by a single transaction. A meta-event could signify a token being tokenised by another token, i.e., a deposit of an underlying token with a contract and the minting of a new share, or the burning of a share and the withdrawal of an underlying token from a contract. This meta-event, which we call a *tokenising meta-event*, can be identified from multiple Transfer events within a single transaction. The tokenising meta-events can be represented as a *token graph*: each vertex represents a token and each directed edge represents the token corresponding to the source vertex being tokenised by the token corresponding to the target vertex. We apply various forms of graph analysis to the token graph and visualise the structure of token compositions.

The remainder of this chapter is organised as follows. Section 7.2 introduces token composition, the token graph, and our data sources, building on the smart contract composition literature reviewed in Section 3.1.2. Section 7.3 presents our analysis. Section 7.4 concludes.

## 7.2 Token Composition

Tokens are central to blockchain-based protocols and applications [119]. Token composition is a technique where one or more tokens can be combined to create new tokens. The entitlements of the input tokens are appropriated by the new token. We extract data from EVM logs (Section 7.2.1) and use it to construct a token graph (Section 7.2.2), a representation of the tokenisation process. In Section 7.3 we analyse the graph and relate it to the tokenisation process.

Table 7.1 Each tokenising meta-event contains the address of the source token, the address of the target token, one of two possible pairs of actions (deposit & mint or withdraw & burn), the amount of the source token that was deposited or withdrawn, the amount of the target token that was minted or burned, and a transaction hash. The table includes four sample entries from the full set of 4032033 tokenising meta-events: the earliest and latest tokenising meta-events that have deposit & mint and withdraw & burn actions.

| Source Token       | Target Token         | Actions         | Source Amount | Target Amount | Tx Hash  |
|--------------------|----------------------|-----------------|---------------|---------------|----------|
| ARC<br>(0xac709f)  | SWT<br>(0xb12a3c)    | deposit & mint  | <i>dust</i>   | <i>dust</i>   | 0x549a12 |
| DGZ<br>(0x84178d)  | preDGZ<br>(0x18aa6e) | withdraw & burn | 1371          | 150           | 0x2da232 |
| BONE<br>(0x981303) | tBONE<br>(0xf7a038)  | withdraw & burn | 5183          | 5160          | 0x5dbe32 |
| WETH<br>(0xc02aaa) | aWETH<br>(0x030ba8)  | deposit & mint  | 25            | 25            | 0xb4281a |

### 7.2.1 EVM Logs, Events and Meta-Events

EVM logs record specific occurrences or outputs generated during the execution of contract code. They enable off-chain applications to react to on-chain events. A popular event is the Transfer event emitted by ERC-20 tokens [117]:

```
event Transfer(address indexed from,
               address indexed to,
               uint256 value);
```

Listing 7.1 The ERC-20 Transfer event specifies three parameters: from, to, and value.

The event has two special cases. If the from address is the zero address (0x0) then the contract mints new tokens. If the to address is the zero address then it burns existing tokens.

A single transaction can emit multiple events. We define a *meta-event* to be a sequence of events that match some pattern and are emitted by a single transaction. We define a *tokenising meta-event* to be a meta-event where the pattern is two Transfer events: one must indicate a transfer of tokens to the contract and another must indicate a new token being minted (*deposit & mint*), or one must indicate a token being burned and another must indicate a transfer of an existing token from the contract (*withdraw & burn*). We want to identify instances where a token is tokenised by another. In the terminology of ERC-4626 [97], a tokenising meta-event corresponds to either a Deposit event or a Withdraw event. However, our tokenising meta-event does not require the contract to follow ERC-4626 and emit those precise events.

We extracted all Transfer events from Ethereum mainnet from block height 0 to 16685101 (February 2023) inclusive using Geth’s `eth_getLogs` RPC method [51]. From the Transfer events, we identified 4032033 tokenising meta-events using the pattern described above. Table 7.1 shows a sample of the data. The first row is the earliest deposit & mint and the second is the earliest withdraw & burn; the third is the latest withdraw & burn and the fourth is the latest deposit & mint.

For example, the first row indicates a transaction that deposited a dust amount of ARC [10] to a contract in exchange for newly minted SWT [107] in January 2017. The third row indicates a transaction that withdrew 5183 BONE [101] from a contract in exchange for burning 5160 tBONE in February 2023. We do not consider the individual utility or value of the tokens, only whether Token *X* was deposited with a contract to mint Token *Y*, and/or Token *Y* was burned by a contract to withdraw Token *X*.

We filter the tokenising meta-events to include only those meta-events that involve two tokens, Token *X* and Token *Y*, such that there is at least one instance of Token *X* being deposited with a contract to mint Token *Y*, and at least one instance of Token *Y* being burned by a contract to withdraw Token *X*. In other words, the “and/or” conjunction in the previous paragraph is replaced by “and”. This excludes *one-way token upgrades* where Token *X* can be deposited with a contract to mint Token *Y* but Token *X* cannot be withdrawn from the contract, and *one-way token burns* where Token *Y* can be burned by a contract to withdraw Token *X* but Token *X* cannot be deposited with the contract to mint Token *Y*. Of the 4032033 tokenising meta-events, 3461723 meet the additional requirement. We refer to the *unfiltered* and *filtered tokenising meta-events* in Section 7.2.2.

We also incorporate market capitalisation data from CoinGecko [25] and liquidity pool data from DEX Screener [32]. CoinGecko aggregates token data including market price, exchange volume, and market capitalisation. DEX Screener stores, parses, and analyses blockchain data to produce a token screener with charts.

### 7.2.2 The Token Graph

We can construct a directed graph from the tokenising meta-events as follows. Each vertex corresponds to a token. Each directed edge from a source vertex to a target vertex corresponds to a set of tokenising meta-events that deposits the source token and mints the target token, and/or withdraws the source token and burns the target token. In the terminology of ERC-4626, the source token is the *asset* and the target token is the *share*. Figure 7.1 shows an example token graph.

In the unfiltered case, the directed graph has 23687 vertices (distinct tokens) and 23549 edges; each edge represents a pair of tokens where the second tokenises the first with either a deposit & mint or a withdraw & burn action. Filtering to retain only edges where both

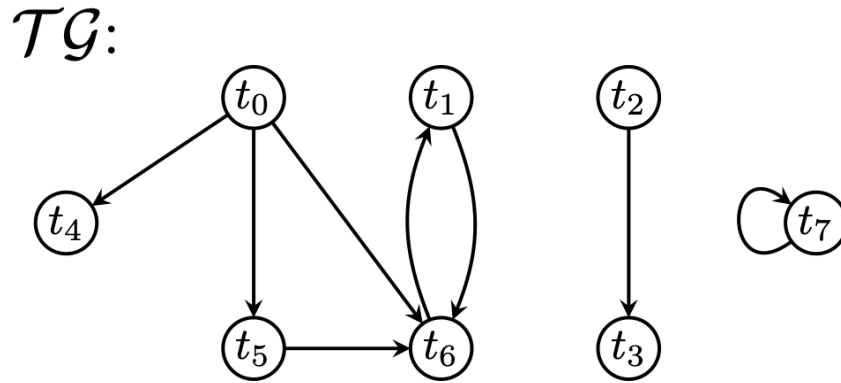


Fig. 7.1 A token graph,  $TG$ , with eight vertices,  $t_0, t_1, \dots, t_7$ , and eight edges. Each edge corresponds to a token tokenising another token. For example, the token represented by  $t_5$  tokenises the token represented by  $t_0$ .

actions are observed yields 7536 edges connecting 8424 of the original vertices. We refer to the *unfiltered* and *filtered token graphs* in the remainder of this chapter.

The filtered graph is therefore a higher-confidence subset of the unfiltered token graph and not a complete representation of token composition within Ethereum. While the filtered graph requires both the tokenisation operation and its inverse operation, reducing potential false positives within the data set, it may also remove legitimate composition relationships whose inverse operations were not observed in the dataset. This results in the unfiltered graph providing greater overall coverage of token composition relationships, while the filtered graph provides greater confidence in the relationships detected.

### 7.2.3 Data Limitations

Our input data, namely, EVM logs, CoinGecko market data, and DEX Screener liquidity pool data, have limitations. Firstly, EVM logs are unauthenticated: a contract can emit an event of its choosing. There is no guarantee that, say, an ERC-20 Transfer event accurately reflects an actual transfer [53]. Furthermore, the first special case highlighted in Section 7.2.1 (mint) is recommended by ERC-20<sup>1</sup> but the second (burn) is not. However, EVM logs for the Ethereum blockchain are generally accurate and malicious contracts can be easily excluded. For an aggregated analysis, such as ours, we believe the impact of these limitations is minimal.

Secondly, the data from CoinGecko and DEX Screener are snapshots that were gathered in April 2024 whereas the EVM logs have a temporal component. It is possible that a token had an entry on CoinGecko in the past, but, at the time the data was gathered, the entry no longer existed. It is also possible that a token was tracked by DEX Screener in the past, but,

<sup>1</sup>“A token contract which creates new tokens SHOULD trigger a Transfer event with the `_from` address set to 0x0 when tokens are created.” [117]

Table 7.2 Each edge in a token graph represents a set of tokenising meta-events. The top five most significant edges in terms of the number of tokenising meta-events they contain are shown. The results are the same for both the unfiltered and filtered token graphs, i.e., the same five edges are present in both.

| Source Token     | Target Token      | # Tokenising Meta-Events |
|------------------|-------------------|--------------------------|
| SHIB (0x95ad61)  | xSHIB (0xb4a812)  | 402 186                  |
| BONE (0x981303)  | tBONE (0xf7a038)  | 203 734                  |
| SUSHI (0x6b3595) | xSUSHI (0x879824) | 120 221                  |
| LEASH (0x27c70c) | xLEASH (0xa57d31) | 75 180                   |
| USDC (0xa0b869)  | aUSDC (0xbcca60)  | 69 373                   |

at the time the data was gathered, it was no longer being tracked. It is also possible that the token coverage of CoinGecko or DEX Screener is incomplete or inaccurate. However, as a high-level measure of token popularity, we believe the impact of this potential mismatch is minimal.

Thirdly, tokenising meta-events are a heuristic for identifying instances where one token is tokenised by another. False positives create edges in the token graph where the token corresponding to the source is not tokenised by the token corresponding to the target; false negatives are pairs of tokens where one is tokenised by the other but there is no corresponding edge in the token graph. We discuss both cases in Section 7.3 and Section 7.4.

## 7.3 Analysis

In this section we examine the macro-topological structure of the unfiltered and filtered token graphs including degree distributions, connected component structure, and cyclic structure. We also examine the micro-topological structure of some individual tokens. We visualise the composition of tokens and identify their direct and transitive dependencies.

Each edge in a token graph represents a set of tokenising meta-events. Before examining the structure of the graphs, we identify the most significant edges in terms of the number of tokenising meta-events they contain. Table 7.2 shows the results for both the unfiltered and filtered token graphs. Three of the five edges represent the staking of memecoins ( $\text{SHIB} \rightarrow \text{xSHIB}$ ,  $\text{BONE} \rightarrow \text{tBONE}$ , and  $\text{LEASH} \rightarrow \text{xLEASH}$ ), one represents the staking of a governance token for a decentralised exchange ( $\text{SUSHI} \rightarrow \text{xSUSHI}$ ), and one represents the supply of a stablecoin to a decentralised lending market ( $\text{USDC} \rightarrow \text{aUSDC}$ ). Significance



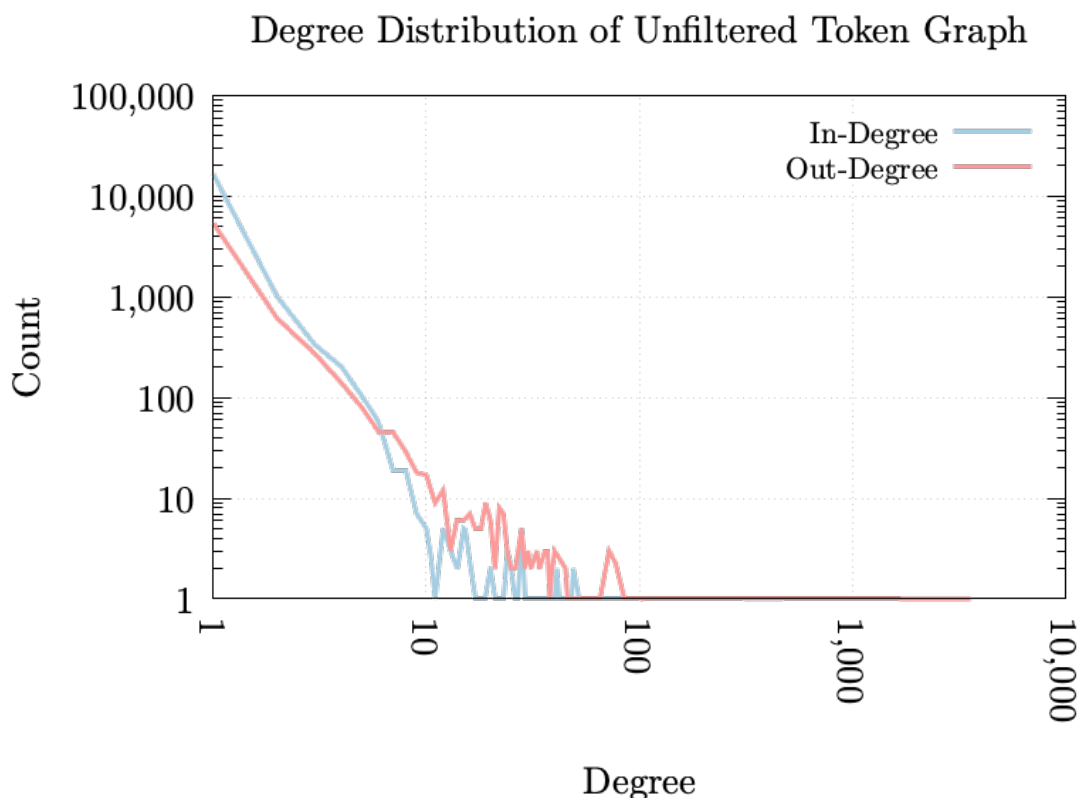


Fig. 7.2 The in- and out-degree distributions of the unfiltered token graph show an inverse relationship between the degree of a vertex and the number of vertices with that degree. Few vertices have high degree and many have low degree.

could also be measured by the volume of tokens transacted or the present USD value of the locked tokens, among other criteria.

### 7.3.1 Degree Distributions

The in- and out-degree distributions of the unfiltered and filtered token graphs show an inverse relationship between the degree of a vertex and the number of vertices with that degree (see Figure 7.2 and Figure 7.3). Table 7.3 shows the top five vertices in the unfiltered and filtered token graphs by in-degree and out-degree. The out-degree entries are easy to explain: they are tokens that are deposited with contracts to mint many other types of tokens. They include stablecoins (USDC, DAI, USDT and sUSD), wrapped ether (WETH), and wrapped bitcoin (WBTC). Figure 7.4 illustrates this using the *tokenised by* edge from the visual grammar introduced in Section 2.9. WETH is tokenised by three derivative tokens (aWETH, stETH, and pETH), with stETH itself tokenised by wstETH. USDC is tokenised by aUSDC and cUSDC. Each *tokenised by* edge in the diagram corresponds to an edge

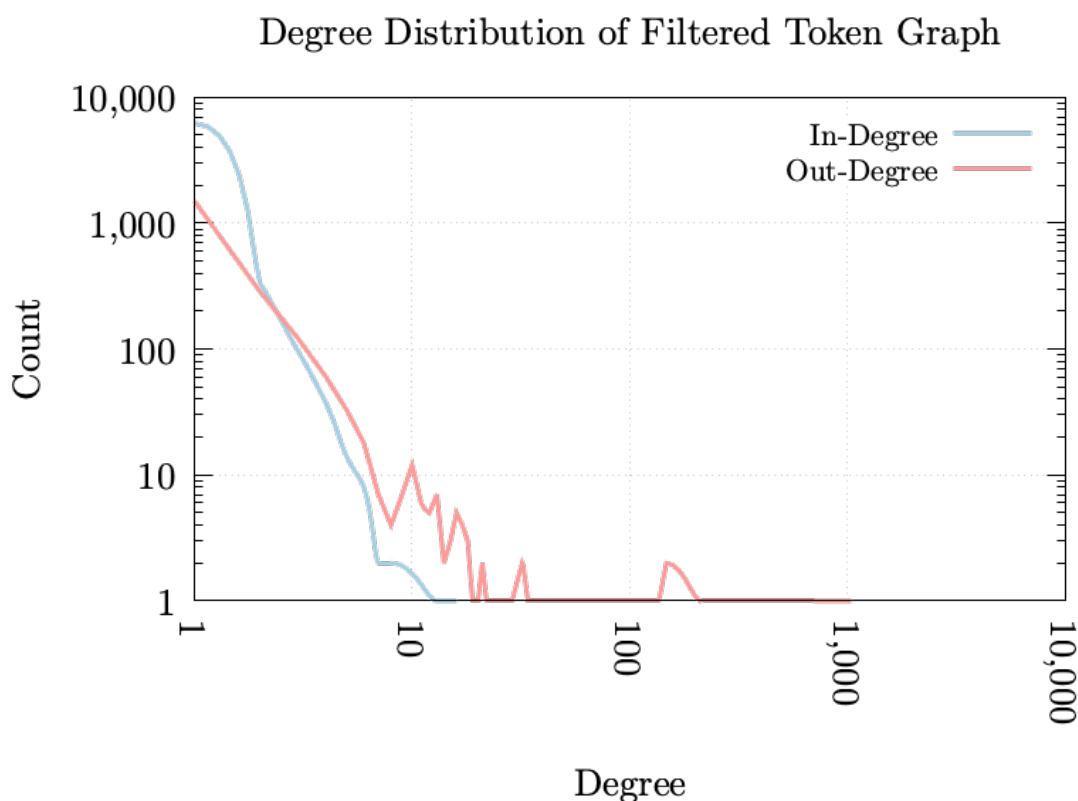


Fig. 7.3 The in- and out-degree distributions of the filtered token graph show a similar inverse relationship as in Figure 7.2.

Table 7.3 The top five vertices (tokens) in the unfiltered and filtered token graphs by in-degree and out-degree.

|            | Top Five by In-Degree        |        | Top Five by Out-Degree |        |
|------------|------------------------------|--------|------------------------|--------|
|            | Token                        | Degree | Token                  | Degree |
| Unfiltered | CHI (0x000000 <sup>2</sup> ) | 471    | USDC (0xa0b869)        | 3587   |
|            | USDP (0x145668)              | 117    | DAI (0x6b1754)         | 1923   |
|            | aUSDC (0xbcca60)             | 84     | USDT (0xdac17f)        | 1175   |
|            | aWETH (0x030ba8)             | 63     | WETH (0xc02aaa)        | 951    |
|            | aDAI (0x028171)              | 54     | sUSD (0x57ab1e)        | 548    |
| Filtered   | XDP2 (0xe68c1d)              | 16     | USDC (0xa0b869)        | 1037   |
|            | XDP1 (0x134fc6)              | 15     | DAI (0x6b1754)         | 752    |
|            | cyUSD (0x1d0914)             | 14     | USDT (0xdac17f)        | 396    |
|            | iDOL (0x7591a3)              | 13     | WETH (0xc02aaa)        | 281    |
|            | agEUR (0x1a7e4e)             | 8      | WBTC (0x2260fa)        | 211    |

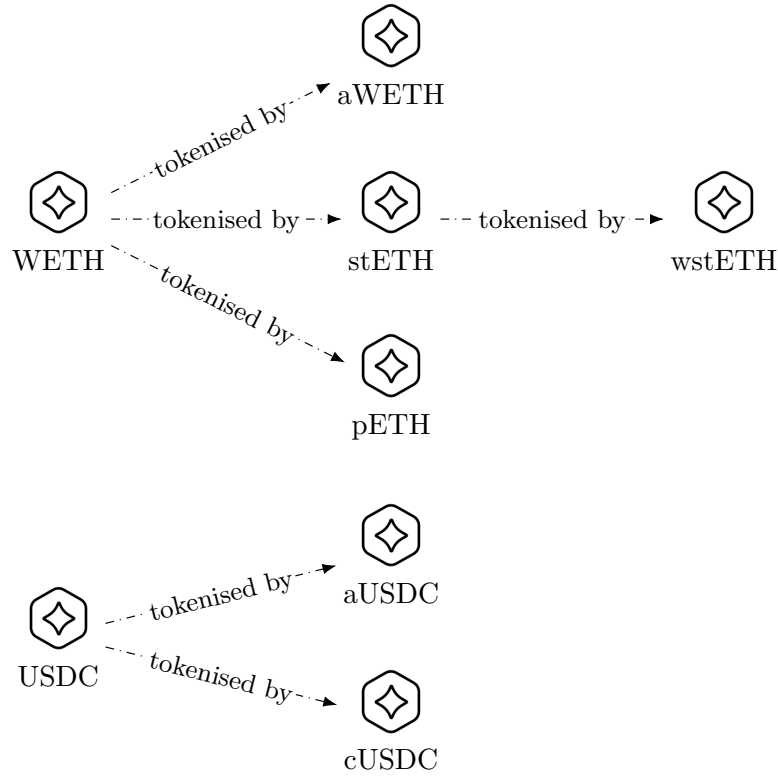


Fig. 7.4 Token composition for WETH and USDC. WETH is tokenised by aWETH, stETH, and pETH; stETH is tokenised by wstETH. USDC is tokenised by aUSDC and cUSDC. Each *tokenised by* edge corresponds to a tokenisation relationship in the token graph.

in the token graph, and the high out-degree of these base tokens reflects the number of protocols that tokenise them.

The in-degree entries are more complex and have multiple explanations. For example, CHI [111] is a gas token created by 1inch, a decentralised exchange aggregator, that is burned to obtain a reduction in transaction fees; in some transactions the burning of CHI is combined with the withdrawal of another token. This is a false positive generated by our heuristic since CHI does not tokenise a token. The remaining in-degree entries in the unfiltered category are due to token swaps performed during a deposit. For example, aDAI [2] is a yield-bearing token issued by AAVE, a decentralised lending market, in exchange for the stablecoin DAI. However, in some transactions, other tokens are supplied and swapped to DAI. These are also false positives since only DAI is tokenised by aDAI. In the filtered category, the in-degree entries are more reliable. For example, the iDOL token is minted when a user deposits various forms of the SBT token (e.g., SBT09180200, SBT09250200, etc.) according to the Lien Protocol [70]. Similarly, the agEUR token is a stablecoin issued by the Angle Protocol [8] that accepts a variety of tokens as collateral.

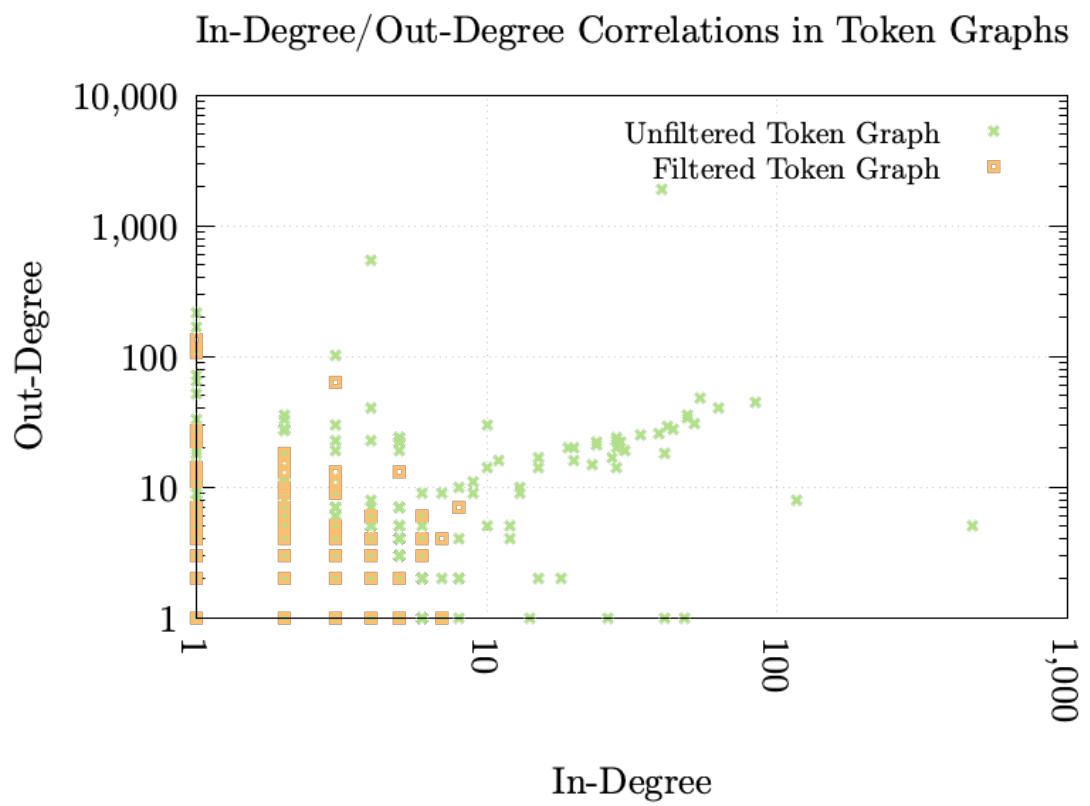


Fig. 7.5 In the unfiltered and filtered token graphs, we can identify vertices with both high in-degree and high out-degree.

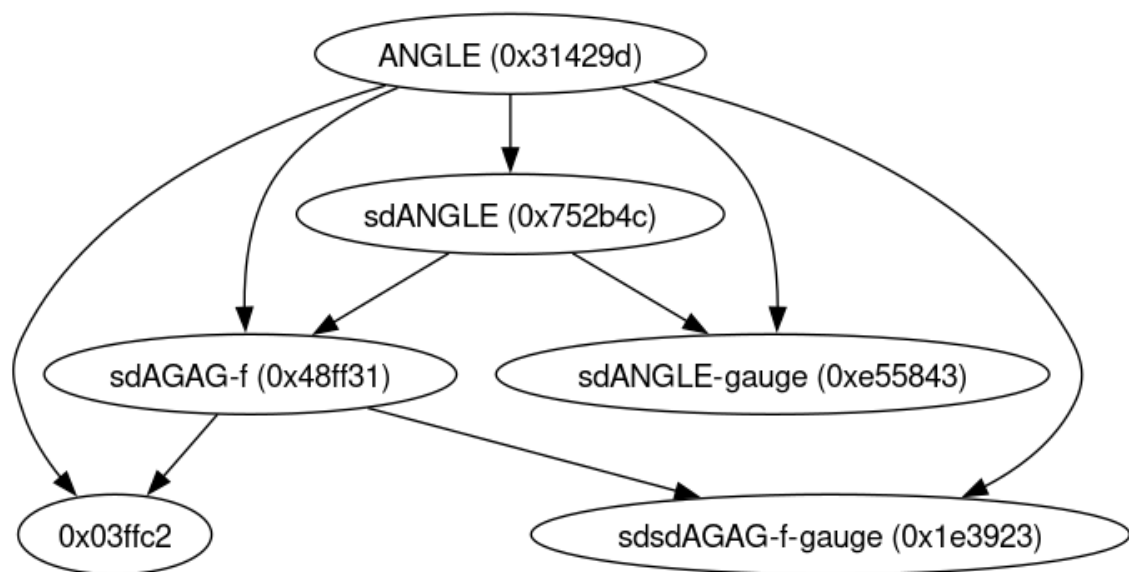


Fig. 7.6 The unfiltered token graph has a weakly connected component whose vertices correspond to tokens related to the Angle Protocol and Stake DAO.

Figure 7.5 plots in-degrees against out-degrees in the unfiltered and filtered token graphs. The tokens whose corresponding vertices have both high in-degree and high out-degree are tokens that tokenise many other tokens, and are themselves tokenised by many other tokens. Examples include mUSD, a stablecoin issued by the mStable protocol [86] and the agEUR token. Stablecoins can be minted from various forms of collateral and themselves used as collateral to mint other tokens, which accounts for both their high in- and out-degrees.

### 7.3.2 Connected Component Structure

The example token graph in Figure 7.1 contains three weakly connected components:  $\{t_0, t_1, t_4, t_5, t_6\}$ ,  $\{t_2, t_3\}$ , and  $\{t_7\}$ . The unfiltered graph has 4082 weakly connected components. A giant component contains 13 794 vertices (approximately 58%) and 17 711 edges (approximately 75%). There are 3336 components (approximately 82% of the total) with only two connected vertices. These vertices correspond to pairs of tokens where at least one tokenises the other, but neither tokenises, or is tokenised by, a third. All tokens represented by the vertices in these components have zero market capitalisation according to CoinGecko, and none are traded in any liquidity pool according to DEX Screener. Their lack of popularity reflects their isolation in the token graph.

There are 418 components, other than the giant component, with more than two connected vertices. Figure 7.6 is an example from this set. It shows the tokenising relationships between tokens related to the Angle Protocol [8] and Stake DAO [104]. Stake

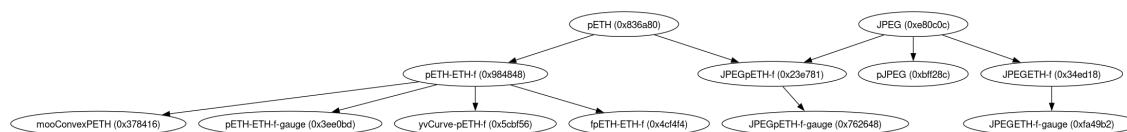


Fig. 7.7 The filtered token graph contains a weakly connected component whose vertices correspond to tokens related to the JPEG’d Protocol.

DAO implements investment strategies based on other decentralised protocols. Their “Liquid Lockers” generate liquidity, voting power, and yield from lockable tokens. ANGLE is Angle’s governance token. It can be deposited with Stake DAO to mint sdANGLE. It is not possible to burn sdANGLE and withdraw ANGLE: the edge between those vertices represents a one-way operation and is not present in the filtered token graph. ANGLE and sdANGLE can be deposited in a gauge (liquidity pool) to mint sdANGLE-gauge. We use this visualisation to identify the direct and transitive dependencies of tokenising tokens such as sdANGLE and sdANGLE-gauge. Many of the other components in the set produce similar insights relating to other tokens and protocols.

The filtered graph has 1491 weakly connected components. A giant component contains 4648 vertices (approximately 55%) and 5247 edges (approximately 70%). There are 1162 components (approximately 78% of the total) with only two connected vertices. There are 196 components, other than the giant component, with more than two connected vertices. Figure 7.7 is an example from this set that relates to the JPEG’d Protocol [60]. JPEG’d is a decentralised lending market where users can supply non-fungible tokens (NFTs) as collateral and obtain loans in pETH, a synthetic token that tracks the price of ether. JPEG is the protocol’s governance token. The figure shows the various ways in which pETH and JPEG can be tokenised by liquidity pools such as Curve’s pETH/WETH pool [30]. Unlike the operations represented by edges in Figure 7.6, all operations represented by edges in Figure 7.7 can be reversed by either depositing the underlying and minting the share, or burning the share and withdrawing the underlying.

Both the unfiltered and filtered graphs contain giant weakly connected components. Exploring such structures requires an interactive user interface with navigational aids such as selection, panning, and zooming. As an illustrative example of the structure in the giant components, we present the longest directed path in the filtered token graph. It comprises nine tokens in sequence: renBTC → sBTC → crvRenWSBTC → tbtct/sbtctCrv → tbtct/sbtctCrv → ibBTC → wibBTC → ibbtct/sbtctCRV-f → bibbtct/sbtctCRV-f. Each token can be deposited with a contract to mint the next, and the reverse operations can also be performed (withdraw & burn). The chain involves tokens from several different protocols. It is an example of “composition in the wild”, groups of interacting smart contracts that span multiple protocols.

### 7.3.3 Cyclic Structure

The example token graph in Figure 7.1 contains an undirected cycle  $(t_0, t_5, t_6)$ , a directed cycle  $(t_1, t_6)$ , and a loop  $(t_7)$ . Both the unfiltered and filtered token graphs contain undirected cycles. For example, Figure 7.6 contains (ANGLE, sdANGLE, sdANGLE-gauge). A priori, it was not apparent if the graphs would contain directed cycles or loops. The filtered token graph contains neither directed cycles nor loops. That is, there is no sequence of  $n$  tokens represented by vertices  $t_0, t_1, \dots, t_{n-1}$  such that  $t_i$  is tokenised by  $t_{(i+1) \bmod n}$ ,  $0 \leq i < n$ . Is this due to a technical limitation in our tokenising meta-event heuristic, or is it the case that no such contracts have been deployed during the time period covered by the data? We believe the answer is the latter since it *is* possible to deploy a contract that creates tokenising meta-events that produce a directed cycle in the filtered token graph. We have created such a contract<sup>3</sup> to demonstrate this. It can produce directed cycles of arbitrary length (including loops) in the filtered token graph.

The unfiltered token graph does not contain any loops. However, it does contain a small number of directed cycles. The graph has 50 non-trivial strongly connected components, that is, strongly connected components with more than one vertex. We manually investigated the tokens involved in these directed cycles. Many are “test tokens” or tokens from defunct protocols, e.g., TST (0x50e508) and TST2 (0x70b34d), TSH (0x46bada) and TCH (0x2fe3e4), etc. All have zero market capitalisation according to CoinGecko, and only two (USDx (0x2f6081) and xBond (0xa8f8dc)) are traded in liquidity pools according to DEX Screener. Although directed cycles and loops can occur in a token graph, they are not commonplace.

## 7.4 Conclusions

In this chapter, we presented a graph representation of token composition on Ethereum. We extracted tokenising meta-events from EVM logs, defined the token graph, and analysed its macro- and micro-topological structure. The unfiltered graph contains 23 687 tokens and 23 549 composition edges; the filtered graph, which retains only pairs where both deposit & mint and withdraw & burn actions are observed, contains 8 424 tokens and 7 536 edges. Both have giant weakly connected components. Base assets such as USDC, DAI, and WETH have very high out-degree. Stablecoins occupy a distinctive position with high in- and out-degrees, since they can be minted from various forms of collateral and themselves used as collateral.

<sup>3</sup>The contract is published at <https://github.com/harrigan/tokenised-tokens-contracts>: The repository contains a smart contract (ERC20ExchangeWrapper) that can tokenise any token (IERC20 underlyingToken) with any other (IERC20Wrapper overlyingToken).

Composition rarely stops at one layer. `stkcvcxcrvRenWBTC-abra` (`0xb65ede`), for example, represents a staked deposit of a share of a liquidity pool for synthetic and wrapped versions of Bitcoin’s native token; its base assets include `renBTC` (`0xeb4c27`) and `WBTC` (`0x2260fa`). The longest directed path in the filtered graph spans nine tokens drawn from several protocols, an example of “composition in the wild” where dependencies traverse organisational boundaries. Chapter 6 observed this layering within a single ecosystem; the token graph generalises the picture to the network as a whole.

The tokenising meta-event heuristic produces some false positives, most notably from gas tokens such as `CHI` and from token swaps performed during a deposit. Refining the heuristic to remove these is a direction for future work.

The token graph records compositional dependencies between tokens and, where those tokens confer voting rights, the potential pathways along which governance influence can travel. It does not by itself establish that any particular relationship is used to exercise governance influence, nor does it reveal the community behaviour behind it. Figure 7.7 shows a weakly connected component containing `JPEG`, the governance token of the `JPEG’d` protocol, together with the liquidity pool tokens that wrap it. The component shows how the voting rights attached to `JPEG` can be composed through successive wrappings; establishing whether they were exercised, and by whom, would require additional analysis.

The token graph therefore contributes structural evidence to RQ 1 and completes the thesis-level argument: the layered, community-driven governance documented in Chapters 4 to 6 rests on a composability that is a network-wide property of Ethereum.



## Chapter 8

# Discussion and Conclusions

This chapter synthesises the findings from Chapters 4–7 in relation to the three research questions posed in Chapter 1. Prior work on DAO governance has overwhelmingly adopted the individual address as the unit of analysis [48, 121]. This thesis takes a different approach: we analyse governance through the lens of communities. We distinguish internal communities, groups of addresses whose voting records show sustained alignment (Chapter 4), from external communities, organisations that participate in governance through smart contract accounts (Chapter 5), and from tokenised external communities, protocols that wrap a DAO’s governance token and operate their own governance process on top (Chapter 6). The token composition graph (Chapter 7) provides system-level evidence of the composability that enables these layered structures.

Table 8.1 summarises each research question, the problem it addresses, our approach, the key finding, and its implications. However, the findings within this thesis should be distinguished from their broader implications. In Section 1.3 we define the general research questions within this thesis, in addition to the associated specific research questions. The specific research questions (Section 1.3) define the scope of the empirical studies conducted in this thesis and described in Chapters 4–7. The studies undertaken to answer the SRQs are a first step towards answering the broader research questions (RQ1–RQ3). Within this scope, the thesis directly demonstrates the identification of groups through historical voting behaviour, detected metagovernance relationships through multi-signature identification on Ethereum, and widespread token composition across the Ethereum network. These results are specific to the DAOs investigated: partisan internal community groups were detected within Nouns DAO, and external communities of metagovernors were identified through multi-signature addresses across a curated list of DAOs. In answering the specific research questions, we have provided evidence towards RQ 1. While we found that higher tiers of governance change the rules of engagement and a strong association between bribes and voting outcomes within the Curve ecosystem, providing evidence towards

RQ 2, our research was limited to a single veToken system. While we discovered the formation of communities and fork-related voting alignment within Nouns DAO, providing evidence towards RQ 3, our findings were based on a retrospective ground truth inferred from voting behaviour. These findings do not establish that their broader implications occur universally across DAOs. Generalised prediction of DAO fragmentation, governance distortion across all composable governance systems, and complete identification of governance communities therefore require further empirical validation.

The sections that follow discuss each question in turn, then draw the findings together into a layered perspective on DAO governance, before addressing limitations, future work, and conclusions.

## 8.1 Detecting and Characterising DAO Communities

Communities can be detected at multiple levels of DAO governance using complementary methods, and the difficulty of detection varies by community type and governance venue.

For internal communities, Chapter 4 showed that pairwise dissimilarity analysis of voting records, combined with multidimensional scaling and cluster analysis, can identify groups of addresses with sustained alignment. The method operates on a voter matrix that encodes participation and support across proposals. This representation is governance-framework agnostic: any DAO with recorded votes can be transformed into a voter matrix, making the approach applicable beyond the Nouns DAO case study in which it was validated. The preliminary analysis of UK parliamentary voting data confirmed that the method correctly separates known political parties, providing an independent check on the dissimilarity and clustering pipeline before applying it to pseudonymous blockchain data.

For external communities, Chapter 5 developed heuristics to identify governance hubs and extract smart contract voters. The results revealed an asymmetry between on-chain and off-chain detection. Off-chain detection through Snapshot was straightforward: we indexed all Snapshot spaces and matched voting power sources to governance tokens, successfully identifying all relevant spaces. On-chain detection proved more difficult, achieving a 61% success rate. The heterogeneity of DAO governance frameworks, even among prominent DAOs, means that no single heuristic reliably identifies all governance contracts from trace data alone. Labelling the extracted contract accounts required manual inspection of Etherscan name tags, a step that limits scalability. Despite these constraints, we identified 72 metagovernance relationships across 61 DAOs, suggesting that external community governance is widespread on Ethereum.

For tokenised external communities, Chapter 6 characterised the Curve/Convex/Votium ecosystem, showing how Convex locks CRV on behalf of depositors to accumulate 45%

| Research Question                                                                                                                          | Problem                                                                                                                                                                                         | Our Approach                                                                                                                                                                                                  | Key Finding                                                                                                                                                                                                               | Implications                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RQ 1: How can communities be detected and characterised from Ethereum governance data?</b>                                              | Prior analysis treats addresses as independent actors, ignoring that some may represent coordinated groups. No current method exists to identify governance relationships between DAOs.         | Heuristics to identify governance hubs and extract smart contract voters from on-chain and off-chain DAOs. Pairwise dissimilarity analysis and clustering to detect internal communities from voting records. | Governance hub detection is difficult on-chain but straightforward off-chain. DAOs are heavily interconnected through token ownership and delegation. Internal communities can be inferred from Nouns DAO voting records. | Interconnected DAO communities are prevalent on Ethereum. Identifying these relationships is a prerequisite for monitoring governance behaviour. |
| <b>RQ 2: How do communities distort the intended governance mechanisms?</b>                                                                | Governance mechanisms encode specific incentive structures but are composable, allowing new protocols to build on top. It is unclear whether composability benefits governance as it does DeFi. | Analysis of three governance layers within the Curve ecosystem (Curve, Convex, Votium), examining the relationship between bribing incentives and voting outcomes.                                            | Curve's four-year veToken lockups can be distorted through metagovernance into fortnightly bribed governance proposals.                                                                                                   | External communities can create governance arrangements at higher layers where bribes are strongly associated with voting outcomes.              |
| <b>RQ 3: How do community voting patterns provide early-warning signals of partisan division and organisational fragmentation in DAOs?</b> | Individual-level analysis obscures group behavioural patterns. High voter churn and low participation make it difficult to detect alignments.                                                   | Voter matrices from raw governance data, pairwise dissimilarity and clustering to identify partisan communities, validated against randomised baselines.                                                      | Community-level analysis of historical voting records can provide insight into governance behaviour.                                                                                                                      | Forks split both the membership and the treasury. Early warning signs can help prevent costly fragmentation.                                     |

Table 8.1 Summary of research questions, problems, approaches, key findings, and implications. Each row corresponds to a research question and draws on the chapters that address it.

of all locked CRV and operates its own governance process on top of Curve. This form of detection was guided by domain knowledge of the Curve ecosystem rather than by a general-purpose heuristic, but it revealed the most consequential community structures in the study.

Chapter 7 provided system-level evidence that the token wrapping observed in the Curve ecosystem is not an isolated pattern. The token graph constructed from EVM logs contains over 8400 tokens connected by more than 7500 tokenisation edges, with a giant connected component spanning 55% of all vertices. Stablecoins and wrapped assets such as WETH and WBTC occupy the highest out-degree positions, confirming that the most widely held tokens are also the most frequently composed. These findings show that the conditions for tokenised metagovernance exist wherever governance tokens are composable, which on Ethereum is nearly everywhere.

A cross-cutting insight emerges from considering these detection methods together. An address that appears as a single voter in the internal community analysis of Chapter 4 may represent an entire external community identified in Chapter 5, built on a tokenisation chain visible in Chapter 7. Detection at one layer informs analysis at others: identifying a multi-signature voter in a DAO's records is the first step towards understanding the community behind it, and the token graph reveals the compositional dependencies that connect governance layers.

## 8.2 Distortion of Governance Mechanisms

External communities can distort the intended governance mechanisms of an underlying DAO, and the degree of distortion increases at higher tiers of governance.

At the level of direct participation, Chapter 5 documented three case studies that illustrate different forms of distortion. Blockchain at Berkeley cast the deciding vote on Compound Proposal #100 through a 2-of-11 multi-signature wallet, defeating the proposal by a margin of fewer than 8000 votes. The governance decision of a single external community, reached through an opaque off-chain process, determined the outcome. Liquity proposed and voted to list its own stablecoin on Aave, using 100000 delegated AAVE tokens to do so. The governance power was accumulated rapidly and deployed in the community's direct economic interest. Index Coop replicated the governance proposals of four underlying DAOs within its own governance process, using its own INDEX token for voting, effectively relocating governance decisions to a different token and a different set of voters. These cases represent different points on a spectrum of distortion: from exercising disproportionate influence within the existing rules, to redirecting governance in self-interest, to relocating the governance process entirely.

At the level of tokenised metagovernance, Chapter 6 showed that distortion can become systematic. Curve Finance designed the veToken model to align voter and protocol incentives through token locking periods of up to four years. Convex circumvents this design by locking CRV for the maximum four-year period on behalf of depositors, who receive liquid cvxCRV in return and can exit at any time. Users who wish to vote on how Convex exercises its voting weight lock CVX for vlCVX, with a maximum lock of just sixteen weeks. Convex also relocates Curve’s on-chain governance to the off-chain Snapshot protocol. At a further level, Votium enables anyone to incentivise gauge votes through bribes, requiring no token locking at all. The result is that 45% of all locked CRV is controlled by Convex, and the distribution of Votium bribes exhibits a 0.99 correlation with gauge voting outcomes in the mature phase. The intended mechanism of long-term commitment has been replaced, in practice, by a fortnightly bribe market.

Chapter 7 places this distortion in a broader context. The token graph shows that token wrapping is a structural feature of Ethereum, not an anomaly confined to the Curve ecosystem. Composability is widely regarded as a positive property of decentralised finance, often described as financial building blocks [99] or money legos [7]. For governance, however, composability has ambiguous consequences. When a new protocol wraps a governance token, it inherits the voting power but not necessarily the rules of engagement. The rules at higher tiers may contradict those of the underlying DAO, as the Curve ecosystem demonstrates. This means that any DAO whose governance token is composable is, in principle, exposed to governance distortion by external communities that it did not authorise and may be unable to prevent.

### 8.3 Early-Warning Signals of Division

Community voting patterns can contain early-warning signals of partisan division within a DAO before it manifests as a fork, as our retrospective analysis of Nouns DAO shows. Chapter 4 showed that addresses which later participated in a Nouns DAO fork increasingly clustered together in voting behaviour as the fork approached. Over 90% of forking addresses aligned in a single cluster in the final proposals before fragmentation, while the average number of clusters fell to just over two. In 100 iterations of randomised voting data over the same pre-fork period, only 47% of forking addresses clustered together on average, with the maximum across all iterations reaching 60%. The contrast between genuine and randomised data confirms that the clustering reflects real alignment.

These results show that community-level analysis reveals dynamics invisible at the individual address level: the gradual consolidation of a partisan bloc, the increase in participation by addresses that would later fork, and the reduction of governance to two

opposing blocs. A DAO that appears stable when each voter is examined in isolation may already contain a consolidating partisan community detectable through the methods of Chapter 4.

## 8.4 The Layered Perspective

The four contribution chapters examine DAO governance at increasing levels of complexity. Chapter 4 analyses the voting behaviour of individual addresses within a single DAO. Chapter 5 shows that some of those addresses are smart contracts representing external organisations. Chapter 6 reveals that some external organisations wrap the underlying DAO's token and build entirely new governance processes on top. Chapter 7 demonstrates that this tokenisation pattern is pervasive across Ethereum, not confined to any single ecosystem.

The visual grammar introduced in Chapter 2 (Section 2.9) formalises this progression. It begins with individual-level analysis: an actor controls an EOA that holds a token and votes in a DAO (Figure 2.3). It then introduces community boundaries that group actors with aligned behaviour (Figure 2.4), multi-signature addresses that represent multiple actors behind a single voting address (Figure 2.5), and tokenised-by edges that connect governance tokens across layers (Figure 2.6). The contribution chapters instantiate each level of this grammar with empirical data. The grammar captures the composable, directed nature of governance relationships that our analysis reveals.

This layered perspective exposes a recursive structure. An external community can itself be a DAO, which may have its own internal communities (detectable through the methods of Chapter 4) and may itself be subject to tokenised metagovernance from a further layer above. The Curve ecosystem illustrates this: Convex has its own token holders who form communities that vote on how Curve's voting power is exercised, and Votium adds a further layer where bribes direct those votes. At each level, the governance process appears as a single address when viewed from below, masking the community dynamics that produced the decision. This is why single-layer analysis is insufficient for understanding DAO governance.

A property of public blockchains that distinguishes DAOs from traditional governance is the permeability of governance boundaries. In corporate governance, the boundary between an organisation and its shareholders is legally defined and regulated. In DAO governance, any entity can build a new governance layer on top of an existing DAO without its permission, simply by acquiring or wrapping the governance token. Chapter 7 provides empirical evidence that this permeability is pervasive: the giant connected component of

the filtered token graph spans 55% of all tokens, and the longest directed path traverses nine tokens across multiple protocols.

## 8.5 Implications for Prior Literature

The findings of this thesis revise several positions in the existing literature on DAO governance.

Prior work on smart contract identification has used bytecode hashing, function signature matching, and behavioural clustering [49, 89, 57, 34]. These approaches are effective for well-defined classes of contract but offer limited support for governance research, where the relevant contracts are heterogeneous and often bespoke to a single DAO. The metagovernance pipeline of Chapter 5 introduces a complementary approach: starting from a curated list of governance tokens, we trace interactions and apply signature matching only to contracts that engage with those tokens. Grounding identification in a token list narrows the search space and links each contract to a known protocol, which is essential when the goal is to compare governance behaviour across DAOs rather than to classify contracts in the abstract.

Existing analyses of DAO centralisation, participation, and influence treat each address as an independent voter [48, 121, 106]. Our results challenge this framing in two directions. Internal community detection (Chapter 4) shows that addresses with sustained voting alignment can be aggregated into single governance entities, lowering the number of distinct decision-makers and raising measured centralisation. External community detection (Chapter 5) shows the converse: a single multi-signature address may represent a community of signers whose collective decision-making expands the effective set of governing actors. Centralisation calculations that ignore both effects misrepresent the distribution of influence.

Comparative analyses of DAO governance mechanisms typically evaluate one-token-one-vote designs against vote-escrowed and delegated alternatives in isolation [39, 109, 95]. Chapter 6 shows that in-isolation evaluation is incomplete. Curve’s veToken model, designed to bind voters to long-term commitment, is undermined in practice by Convex and Votium, which operate at higher tiers with shorter lockups, separate tokens, and bribe markets. A governance mechanism cannot be assessed without considering the higher-tier mechanisms that may be built upon it, because composability allows those higher tiers to override the original incentive structure.

Composability is widely framed as a virtue of decentralised finance, with metaphors such as money legos [7] and financial building blocks [99] reflecting the gains from open contract interfaces. Chapter 7 provides empirical support for the prevalence of

composability: more than half of all tokens lie in a single connected component of the token graph. For governance, the same property has costs that the composability literature does not address. Any DAO whose token can be wrapped is exposed to governance distortion by entities it has neither authorised nor anticipated.

Prior work on DAO forks treats them as discrete events analysed after the fact and emphasises the loss of resources and members that follows fragmentation [13, 94]. No quantitative method has been proposed for anticipating a fork from governance data alone. Chapter 4 addresses this gap: the consolidation of forking addresses into a single internal community before the fragmentation of Nouns DAO produced a measurable signal that distinguished real partisan alignment from randomised baselines. This positions internal community analysis as a candidate tool for governance monitoring, although prospective validation across additional DAOs remains future work.

## 8.6 Limitations

Several limitations constrain the scope and generalisability of our findings.

The internal community detection method (Chapter 4) was validated on Nouns DAO only. We selected Nouns DAO because its documented forks provided a ground truth for evaluating whether partisan communities could be detected before fragmentation. This choice was methodologically necessary, but it means that generalisation to other DAOs remains unverified. The method itself is governance-framework agnostic, as it operates on voter matrices that can be constructed for any DAO with recorded votes, but we have not demonstrated this in practice beyond the preliminary UK parliamentary analysis.

All four studies analyse the Ethereum blockchain. DAOs exist on other networks, and cross-chain governance, where a DAO on one blockchain participates in governance on another [90], is not captured by our methods.

Blockchain addresses are pseudonymous. A single actor may distribute tokens across many addresses to create an appearance of decentralisation (a Sybil attack), and a single multi-signature address may represent the collective decision of many actors. Our analysis cannot fully resolve the mapping from addresses to the actors behind them. The internal community detection in Chapter 4 infers alignment from observed behaviour rather than known identity, which is both a strength (it requires no identity infrastructure) and a limitation (it cannot distinguish genuine coordination from coincidental agreement).

The metagovernance detection pipeline (Chapter 5) depends on Etherscan name tags for labelling contract account voters. These tags may be incomplete or inaccurate, and the manual inspection step limits scalability. The on-chain governance hub heuristic



achieved a 61% success rate, meaning that roughly 40% of governance contracts were not automatically detected. This constrains the completeness of the metagovernance network.

Each study covers a specific time period. The internal community analysis (Chapter 4) collected data up to May 2025 but focused on the pre-fork period ending September 2023; the metagovernance analysis (Chapter 5) uses data up to October 2023; the veToken analysis (Chapter 6) covers data up to March 2023; the token graph (Chapter 7) covers data up to February 2023. DAO governance ecosystems evolve, and our findings represent the state of these systems at the time the data was collected.

External communities that deliberate off-chain, such as Blockchain at Berkeley’s multi-signature signers or Index Coop’s token holders voting on Snapshot, produce governance outcomes that are visible on-chain, but the deliberation process itself is not. Face-to-face discussions, messaging platforms, and forum debates that shape governance decisions are outside the scope of on-chain analysis. This thesis captures the outcomes of governance, not the full process.

## 8.7 Future Work

There are several directions in which the work presented in this thesis could be extended. The following subsections outline how the thesis could improve its robustness, broaden its application, and how it can address the wider implications.

### 8.7.1 Methodological Validation

Across this thesis, we have implemented new methodological approaches to analysing DAOs and the voting members within them. While our analysis of external communities included 61 DAOs, our methodological approach to investigating internal DAO communities was limited to Nouns DAO. Moreover, the parameters within our analysis of Nouns DAO, such as the rolling window and active voter threshold, could be varied to assess the robustness of the results. Therefore, the internal community detection method should be applied to additional DAOs beyond Nouns DAO, with different parameters to assess its generalisability.

Incorporating proposal categories would allow the method to determine if governing addresses differed across all of the governance proposals within a DAO or on specific issues within the DAO. This could provide insights into community formation within specific issues rather than the entirety of a DAO’s proposals. Due to the variety of proposal categories, some governance decisions may have greater consequences for a DAO than others, potentially highlighting more accurate DAO communities.

### 8.7.2 Broader Empirical Coverage

As stated in the previous section, within future studies the methods described in this thesis should be applied to a greater number of DAOs to cover more on-chain DAO frameworks and more off-chain DAO protocols, outside of the Snapshot protocol.

Future research should also investigate blockchains outside of the Ethereum network. The most obvious extension of analysis would be to EVM-based blockchain networks. While this analysis could reuse much of the methodology and codebase developed in this thesis, by adapting the data by connecting the existing tools to nodes operating on other EVM-compatible networks, the governance dynamics of DAOs outside of the Ethereum network may differ. This analysis could be extended further to blockchain networks that support smart contracts but operate independently of the EVM.

Extending the analysis of DAOs to other blockchains could also offer more insights into DAOs within Ethereum. Within this thesis, we examined DAOs within Ethereum, including the analysis of metagovernors. However, future work could extend this analysis to cross-chain metagovernance, where external governance relationships span multiple blockchain networks. With the increasing adoption of additional blockchain networks, including Layer 2 networks, examining these cross-chain relationships is the logical extension of the external governance and metagovernance analysis conducted in this thesis.

### 8.7.3 Governance-Monitoring Tools

The analyses conducted within this thesis were performed retrospectively using historical blockchain data such as past transactions and log emits. While this allowed us to identify and analyse internal communities, external governance relationships, and token composition, a governance-monitoring tool would require the methods described in our analysis to operate dynamically as new blocks are produced and governance activity takes place.

The internal community detection method we describe in Chapter 4 could be adapted for live governance monitoring. Our analysis of Nouns DAO was retrospective, as the forks had already taken place and served as our ground truth for community analysis. Future work could analyse DAOs prospectively, where changes in community structure are monitored as governance activity occurs. Quantitative fragmentation-risk scores could then be assigned to proposal windows based on observed changes in voting alignment, allowing patterns identified retrospectively in Nouns DAO to be monitored over time.

Further development would also be required to make the external governance analysis suitable for live monitoring. The on-chain metagovernance detection heuristic developed in Chapter 5 identified 61% of the known governance contracts within our dataset. Future work should refine this heuristic to improve its detection rate and reduce false positives. The heuristic could also be extended to identify and handle specific DAO frameworks.

The manual analysis required to identify external governance relationships should also be reduced. For example, the manual labelling of smart contracts could also be partially automated by using large language models (LLMs) to parse DAO documentation and associate smart contract addresses with named entities.

Once the issues identified in this section to move from a static analysis to a dynamic analysis are addressed, the internal community detection of Chapter 4, the metagovernance mapping of Chapter 5, and the token composition analysis of Chapter 7 could be combined into a unified governance-monitoring system.

### 8.7.4 Qualitative and Interpretive Work

The analysis conducted in this thesis is empirical quantitative analysis. However, external communities whose governance deliberations occur outside the blockchain require qualitative methods to complement on-chain analysis. Interviews with multi-signature signers and members of metagovernance organisations could reveal the decision-making processes that produce the on-chain votes we observe. Moreover, governance forums, proposal discussions, and other publicly available deliberative material could similarly be examined alongside voting records to provide greater context for the observed governance behaviour.

This type of analysis could contextualise inferred results found in this thesis. For example, it could expand on voter alignment context by determining the extent to which voting alignment identified quantitatively corresponds to deliberate coordination or coincidental agreement. We recognise that this type of analysis would present challenges due to the pseudonymous nature of blockchain addresses, which makes identifying and contacting the individuals or organisations responsible for observed governance activity difficult.

### 8.7.5 Ethical and Practical Implications

Within future work, it is important that researchers remain vigilant to the ethical implications of this research, as DAO governance makes decisions for organisations that control billions of dollars worth of crypto assets. Within our research, we have shown a strong association between bribes and voting outcomes, and Austgen et al. [12] show that governance rights can be traded covertly through trusted execution environments, enabling vote-buying without leaving an on-chain trace. Live monitoring systems of DAO governance, especially systems that can detect potential fragmentation, could be used to target DAOs and exploit vulnerabilities in times of volatility.

As DAOs themselves may be targeted by malicious actors, individual governing addresses may also become targets. Future qualitative research involving governance partici-

pants should therefore preserve participant privacy and pseudonymity wherever possible and avoid unnecessarily linking blockchain addresses to real-world identities. Moreover, approaches to labelling addresses using LLMs may incorrectly associate addresses with identities or motives (such as malicious actions) that are incorrect. Future work should clearly label when AI tools have been used to automate and infer the labels assigned.

## 8.8 Conclusions

This thesis investigated DAO governance through the community perspective rather than the individual address perspective that dominates prior work. We defined communities as cohesive groups of actors that collectively exert influence within a DAO's governance, distinguished internal communities from external and tokenised external communities, and developed methods to detect and characterise each type from Ethereum governance data.

For RQ 1, we showed that communities can be detected at multiple levels using complementary methods. Internal communities are identifiable through pairwise dissimilarity analysis of voting records, an approach that is governance-framework agnostic and requires no identity infrastructure. External communities are detectable through governance hub identification and smart contract voter extraction, a process that is straightforward for off-chain governance but achieves only a 61% success rate for on-chain governance due to the heterogeneity of DAO frameworks. Across 61 DAOs, we identified 72 metagovernance relationships, suggesting that external community governance is widespread on Ethereum.

For RQ 2, we showed that external communities can distort the intended governance mechanisms of an underlying DAO, with the degree of distortion increasing at higher tiers. In the Curve ecosystem, the veToken model's four-year locking requirement is circumvented by Convex, which locks CRV on behalf of depositors and offers its own governance token with a sixteen-week maximum lock, and by Votium, which requires no locking at all. Convex controls 45% of all locked CRV, and the distribution of Votium bribes exhibits a 0.99 correlation with gauge voting outcomes. The intended mechanism of long-term commitment has been replaced, in practice, by a fortnightly bribe market. The token composition graph shows that the composability enabling this distortion is a structural feature of Ethereum, meaning that any DAO whose governance token can be wrapped is, in principle, exposed to similar dynamics.

For RQ 3, we showed, retrospectively, that community voting patterns can contain early-warning signals of partisan division and organisational fragmentation. In Nouns DAO, over 90% of addresses that later participated in a fork aligned in a single cluster before the split, compared with an average of 47% in randomised data. Community-level

analysis reveals the gradual consolidation of a dissident bloc that is invisible when each voter is examined in isolation.

The overarching finding of this thesis is that DAO governance is shaped by communities operating at multiple layers, and the composability of smart contracts enables governance structures to emerge that the underlying DAO did not anticipate and may be unable to prevent. Viewing governance through the community perspective reveals dynamics that are invisible when each address is examined in isolation: the gradual consolidation of partisan blocs before a fork, the decisive influence of a single external community on a governance outcome, and the effective replacement of intended governance mechanisms by bribe markets at higher tiers. As DAOs manage increasing amounts of capital and make increasingly consequential decisions, understanding governance at the community level is necessary.

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# Appendix A

## DAO Governance Attacks

This appendix accompanies the timeline of governance attacks shown in Figure 3.2 of Chapter 3. Table A.1 lists each incident in chronological order, from the Steem/Tron takeover in March 2020 to the Compound proposal of July 2024, together with a short description and a citation to a primary report or post-mortem. The selection draws on the survey of Feichtinger et al. [45] and on subsequent public reporting. Both successful attacks and unsuccessful attempts are included. The incidents span the full range of attack vectors discussed in Section 3.4: token accumulation through open markets, exchange custody, or flash loans; bribery and the redirection of protocol emissions; malicious proposals disguised as routine upgrades; and disputes that escalated into forks or forced liquidations.

Table A.1 Governance attacks on DAOs, in chronological order.

| Incident                     | Date      | Description                                                                                                                            |
|------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| Steem/Tron [68]              | Mar. 2020 | Justin Sun used exchange-custodied STEEM tokens to install witnesses and seize control of the Steem blockchain's governance.           |
| MakerDAO [79]                | Oct. 2020 | Concerns arose that flash loans could borrow enough MKR to manipulate MakerDAO governance votes within a single transaction.           |
| True Seigniorage Dollar [24] | Mar. 2021 | An attacker accumulated over 33% of governance tokens, passed a malicious implementation update, and minted 11.8 billion TSD tokens.   |
| Venus VIP-42 [116]           | Sep. 2021 | A large token holder attempted to pass a proposal granting themselves administrative control over the Venus Protocol treasury.         |
| Yuan Finance [134]           | Sep. 2021 | An attacker exploited the governance mechanism to drain the Yuan Finance treasury, forcing the project to migrate.                     |
| Mochi Inu / Curve [14]       | Nov. 2021 | Mochi Protocol manipulated CRV gauge votes to direct Curve emissions to their own liquidity pool, extracting value from the ecosystem. |
| Justin Sun / Compound [113]  | Feb. 2022 | Justin Sun accumulated a large COMP position, raising concerns of a governance takeover of the Compound lending protocol.              |
| Build Finance [45]           | Feb. 2022 | An attacker acquired enough BUILD tokens to pass a proposal granting themselves control of the DAO's \$470,000 treasury.               |

*Continued on next page*



Table A.1 (continued)

| Incident                     | Date          | Description                                                                                                                                                |
|------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Beanstalk [36]               | Apr. 2022     | An attacker used a flash loan to acquire sufficient governance tokens to pass a malicious proposal, draining \$182 million from the protocol.              |
| Yam Finance treasury [103]   | Jul. 2022     | A governance attack attempted to drain Yam Finance's \$3.1 million treasury, which the community successfully defended.                                    |
| Audius [110]                 | Jul. 2022     | An attacker exploited a storage collision through the governance system to pass a malicious proposal, draining \$1.08 million.                             |
| Balancer / Humpy [62]        | May–Dec. 2022 | A whale accumulated 35% of veBAL voting power and directed protocol emissions to pools they controlled, extracting disproportionate value.                 |
| Binance / Uniswap [81]       | Oct. 2022     | Binance was accused of using UNI tokens held in custody on behalf of users to vote on Uniswap governance proposals.                                        |
| Mango Markets [100]          | Oct. 2022     | After exploiting Mango Markets via oracle manipulation, Avraham Eisenberg used stolen MNGO tokens to vote on a proposal allowing him to keep \$47 million. |
| Yam Finance redemption [131] | Jan. 2023     | Following the earlier treasury attack, Yam DAO voted to redeem remaining treasury funds and wind down the project.                                         |
| Swerve Finance [15]          | Mar. 2023     | An attacker attempted to pass governance proposals to transfer \$1.3 million from the defunct Swerve Finance protocol but failed to reach quorum.          |

*Continued on next page*

Table A.1 (continued)

| Incident                     | Date      | Description                                                                                                                                           |
|------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rook DAO [93]                | Apr. 2023 | Activist investors elected delegates on a dissolution platform and forced liquidation of the \$25 million Rook DAO treasury.                          |
| Aragon / RFV Raiders [9]     | May 2023  | Activist investors accumulated ANT tokens ahead of a planned \$200 million treasury transfer to DAO control, prompting Aragon to cancel the transfer. |
| Tornado Cash [16]            | May 2023  | An attacker disguised a malicious proposal as a routine update, gained control of Tornado Cash governance, and minted additional TORN tokens.         |
| Nouns DAO fork [56]          | Sep. 2023 | Disgruntled NFT holders used the newly implemented fork mechanism to exit the DAO with \$27 million from the treasury.                                |
| BigCap [20]                  | Sep. 2023 | A scam governance proposal was submitted to the BigCap DAO, which the community identified and rejected.                                              |
| Synthetify [66]              | Oct. 2023 | A malicious governance proposal on the Solana-based Synthetify protocol went unnoticed and passed, resulting in a \$230,000 loss.                     |
| Indexed DAO [3]              | Dec. 2023 | Multiple hijack attempts targeted the Indexed DAO treasury, which the community defeated before distributing remaining funds.                         |
| Compound / Golden Boys [112] | Jul. 2024 | A group passed a proposal allocating 499,000 COMP (\$24 million) to a yield vault they controlled, using votes sourced from a centralised exchange.   |